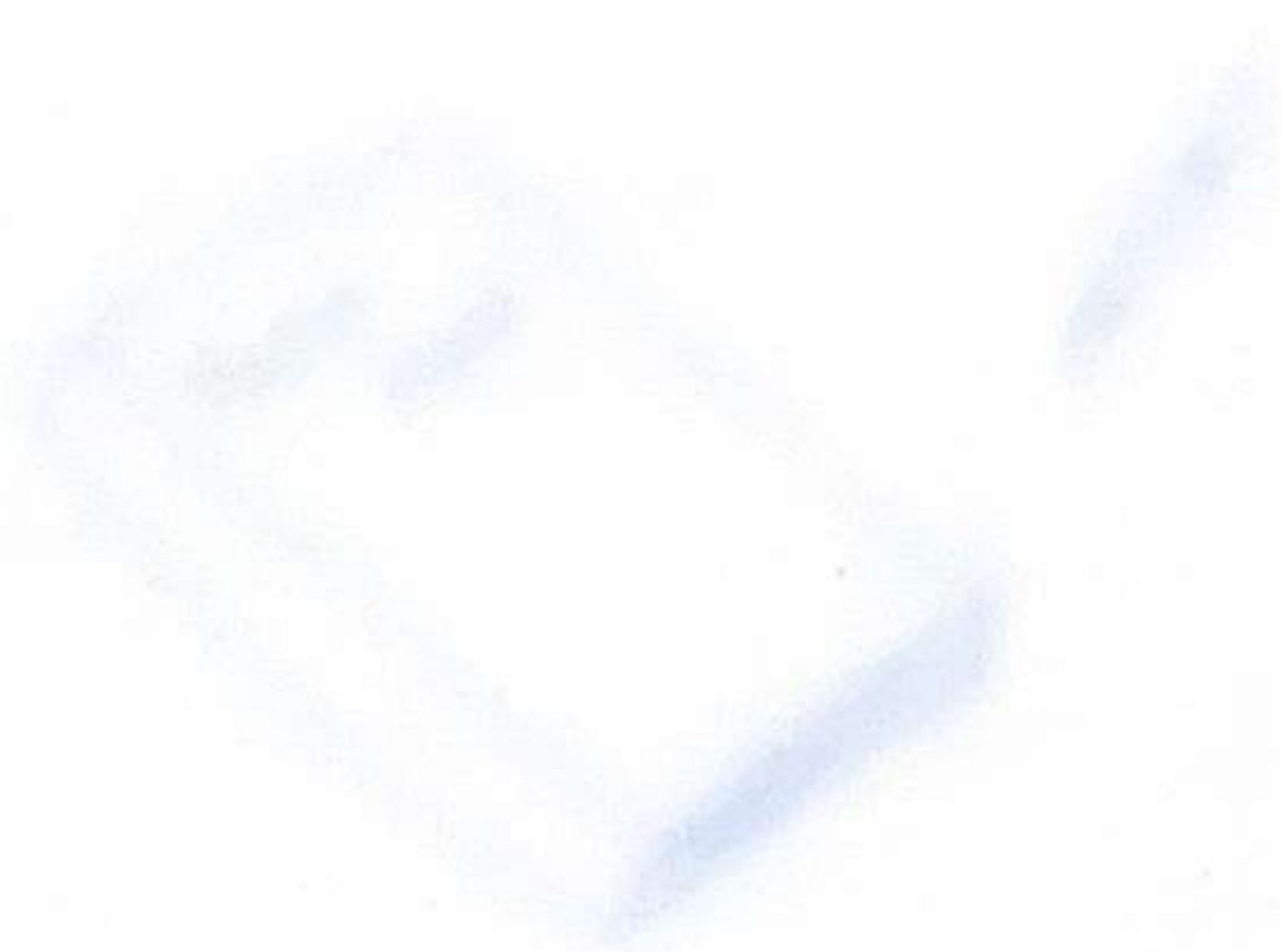


十一级 ^{specialist}
month
笔记本

Specialist

Mathematics



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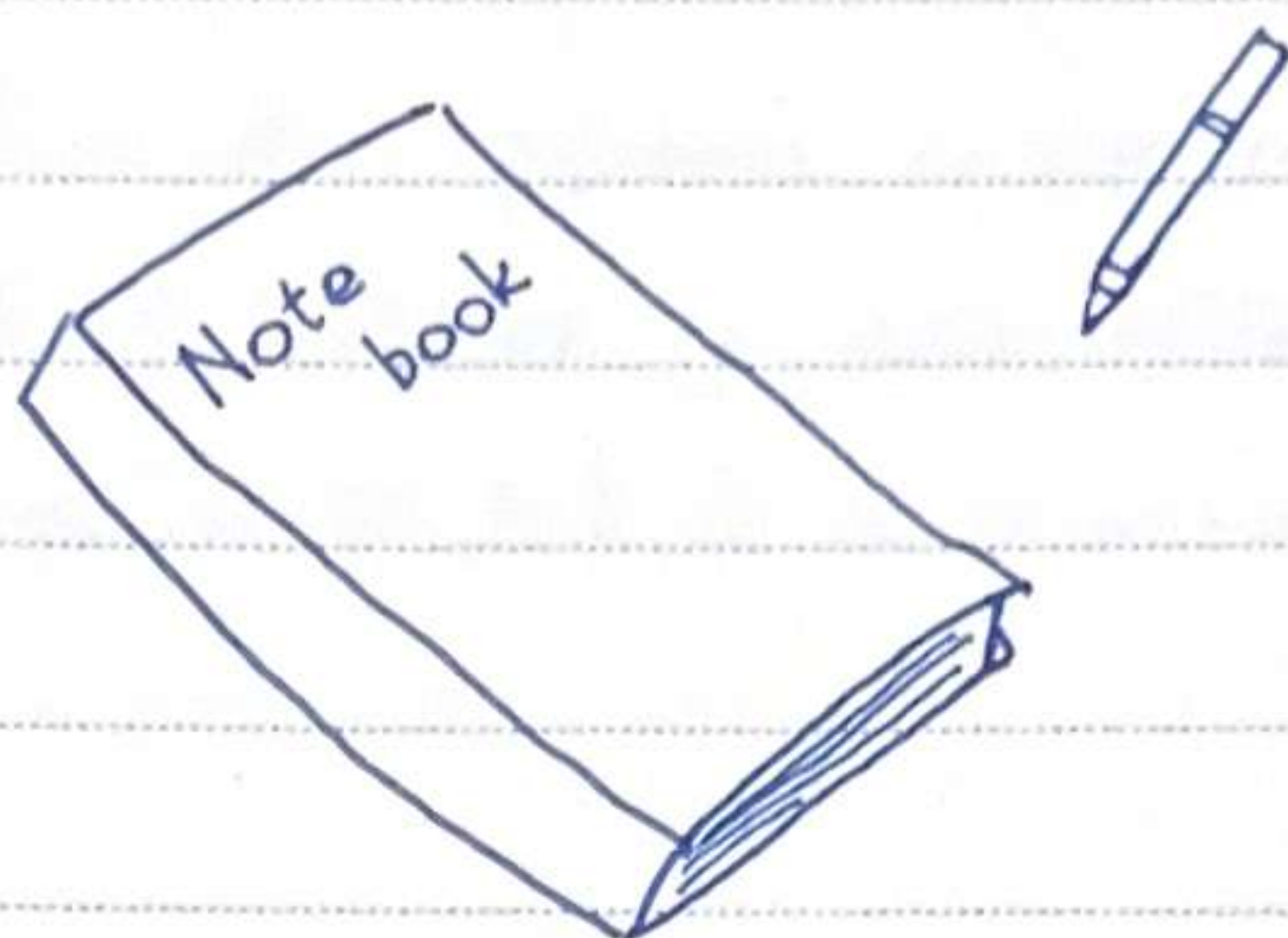
Date.

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Yr 11 Notebook

Specialist

Mathematics



Wingyee

Arithmetic Sequence

▣ An arithmetic sequence is a sequence where there is a common difference between any two successive terms.

▣ the form: $a, a+d, a+2d, \dots$

↓
first term

↘ common difference $= t_2 - t_1 = t_3 - t_2$

▣ Example:

$$\{ 3, 13, 23, \dots \}$$

$$t_n = a + (n-1)d$$

① $\{ 4, 9, 15, \dots \}$ is this sequence arithmetic?

$d = 9 - 4 \neq 15 - 9$, hence it isn't arithmetic.

② Find a, b as $\{ 41, a, 55, b \}$ is arithmetic

$$55 = 41 + 2d$$

$$d = 7$$

$\dots$

③ Find t_{16} and t_n if $t_4 = 15, t_8 = 37$

$$a + 3d = 15$$

$$a + 7d = 37$$

$$7d - 3d, 37 - 15$$

$$4d = 22$$

$$d = 5.5$$

④ CAS Active Find t_{20} $\{ 5, 40, 75, 110, 145 \}$

$\Rightarrow$ Main $\rightarrow$ Keyboard $\rightarrow$ Math2 - 'Define', keyboard - 'abc' $\rightarrow$ Define $t(n) = a + (n-1)d$ [exe]

$\Rightarrow$ Math3 ' $\Rightarrow$ ' (top right corner) $\rightarrow$ [exe]

$\Rightarrow 5 \Rightarrow a, 35 \Rightarrow d$ ↗

$\Rightarrow t(20) \rightarrow$ [exe]

Arithmetic Series

- A series is the sum of the terms of a sequence

$$S_1 = t_1$$

$$S_3 = t_1 + t_2 + t_3$$

$$S_n = t_1 + t_2 + \dots + t_n$$

- To calculate the sum:

$$S_n = \frac{n}{2} (a + l)$$

↓
first term

↓
last term

- If you don't know the last term:

$$S_n = \frac{n}{2} (2a + (n-1)d)$$

↓
first term

↓
common difference

- Example:

- find S_{20} of sequence $\{12, 25, 38 \dots\}$, $a = 12$, $d = 13$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{20} = \frac{20}{2} [24 + 19 \times 13]$$

$$= 10 (24 + 247)$$

$$= 2710$$

- if $t_3 = 4$, $t_8 = 11$, what is S_{30} [CAS Active]

- find a and d first, type in Cas

$$t_3 = a + 2d = 4$$

$$t_8 = a + 7d = 11 \quad \text{type in CAS } \left[\begin{array}{l} a+2d=4 \\ a+7d=11 \end{array} \right] a, d \rightarrow \boxed{\text{exe}}$$

- CAS: Define $s(n) = \frac{n}{2} (2a + (n-1)d) \rightarrow \boxed{\text{exe}}$

$$3) \text{ CAS: } 10 \Rightarrow a \rightarrow \boxed{\text{exe}}$$

$$-3 \Rightarrow d \rightarrow \boxed{\text{exe}}$$

CAS:

$$4) s(30)$$

$$= 1005$$

Advance Arithmetic Problems

$\{8, 1, -6, \dots\}$ is an arithmetic sequence. Which term is equal to -258 ?

m Define $t(n, a, d) = a + (n-1)d \rightarrow \boxed{\text{exe}}$ (should show 'done')

m 'Math3' & 'abc' INPUT: solve($t(n, 8, -7) = -258, n$)

m $\boxed{\text{exe}}$ (should show $n=39$)

Functional Definition

Functional Definition is a sequence of numbers expressed in the form: $t_n = f(n)$

Example: $t_n = 2n - 7$, solve $t\{1, 2, 3, \dots\}$

$$t_1 = 2 - 7 = -5$$

$$t_2 = 4 - 7 = -3$$

...

Sequence is $\{-5, -3, \dots\}$

Advance Arithmetic Problems

$\{8, 1, -6, \dots\}$ is an arithmetic sequence. Which term is equal to -258

iii Define $t(n, a, d) = a + (n-1)d \rightarrow \boxed{\text{exe}}$ (should show 'done')

iv 'Math3' & 'abc' INPUT: solve($t(n, 8, -7) = -258, n$)

iv $\boxed{\text{exe}}$ (should show $n=39$)

Functional Definition

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$$t_1 = 2 - 7 = -5$$

$$t_2 = 4 - 7 = -3$$

...

Sequence is $\{-5, -3, \dots\}$

Recursive Definition

- Repeated use of an instruction
- Each term is calculated using the previous term

Example $t_{n+1} = 3t_n - 2, t_1 = 6$

$$t_2 = 3 \times 6 - 2 = 16$$

$$t_3 = 3 \times 16 - 2 = 46$$

① $\{14, 7, 3.5 \dots\}$ find next three terms

Answer: ~~$\frac{3.5}{2}$~~ $\frac{7}{4}, \frac{7}{8}, \frac{7}{16}$

② $k_{n+1} = 2k_n + 1, k_1 = 0.5$ find k_2, k_3, k_5

$$k_2 = 2 \times 0.5 + 1 = 2$$

$$k_3 = 2 \times 2 + 1 = 5$$

t_4 must be found in order to find k_5

③ CAS Active $t_{n+1} = 3t_n - 2, t_1 = 6$

Menu second page 'sequence'

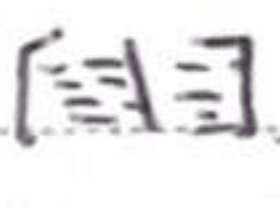
Third row 'Recursive'

Select first equation, which is $a_{n+1} =$
 $a_1 =$

INPUT Equation: $a_{n+1} = 3 \times a_n - 2$

此处的 a_n 要通过选择 ~~first~~ row
第三个选项 n, a_n 来输入

INPUT $a_1 = 6$

Press second row first icon (the one that look like excel )

To set number of terms: press ~~set~~ 2nd row second icon

(the one with X and Y)

Then enter end: 10

Logistic equation (recursive definition)

- Logistic equation is a model of population growth, an example of a recursive definition as it needs the previous term in order to generate the next term of the sequence.

$$t_{n+1} = a * t_n * (1 - t_n), \quad 0 < t_0 < 1, a \text{ is a constant}$$

- The equation can generate sequence.

The sequence can either:

- Converge on one fixed value (limiting value) $\lim_{n \rightarrow \infty} t_n = x$

A string of number which settles at a fixed value is called convergent.

A sequence t_n can converge to only one possible number, x , called the limit of the sequence.

- Diverge (grow further apart)

A sequence whose terms grow further and further apart is called diverge.

That is, a sequence is divergent if as $n \rightarrow \infty$, $t_n \rightarrow \pm \infty$

- Oscillate between two values

A sequence which fluctuates between two values is called oscillating.

* Depending on the value of a , sequence of Logistic equation could be convergent, divergent or oscillating.

Example: $t_{n+1} = a * t_n * (1 - t_n)$, $t_0 = 0.7$, $a = 2$

$$t_{n+1} = 2 * t_n * (1 - t_n)$$

$$t_1 = 2 * 0.7 * (1 - 0.7)$$

$$= 2 * 0.7 * 0.3 = 0.42$$

$$t_2 = 2 * 0.42 * 0.58 = 0.4872$$

...

CAS: Main - Sequence, 'A' is a from the question

$$\text{Input: } a_{n+1} = A * a_n * (1 - a_n)$$

$$a_1 = T_0$$

Press [=] [=]

Geometric Sequence

□ A geometric sequence is a sequence where each term is obtained by multiplying the preceding term by a certain constant factor.

□ For a geometric sequence:

$$t_n = a \times r^{n-1}$$

first term

common ratio

$$r = \frac{t_{n+1}}{t_n} = \frac{t_2}{t_1}$$

□ The middle term (y) of three consecutive terms in a geometric sequence can be calculated in terms of the outer two terms, x and z

$$\{ \dots, x, y, z, \dots \}$$

$$y^2 = xz$$

[3.5, Q6]

□ Textbook Example:

① $a=1, t_3=4$ $t_3 = 1 \times r^{(3-1)}$

$$4 = r^2$$

$$r = \pm 2 \quad [r \text{ can be } -2 \text{ or } +2]$$

Find t_{10} $t_{10} = 1 \times 2^9$ or $t_{10} = 1 \times (-2)^9$
 $= 512$ two possible answer $= -512$

② $t_2 = \frac{1}{2}$ $t_5 = \frac{27}{16}$, find r

$$\frac{1}{2} = ar$$

$$\frac{27}{16} = ar^4$$

$$\frac{ar^4}{ar} = \frac{1.6875}{0.5}$$

$$r^3 = 3.375$$

$$r = 1.5$$

③ $0.00032 = 0.2^{(k-1)}$ 求 k

CAS $\log_{0.2} (0.00032) = 5$

$$0.2^{k-1} = 0.2^5$$

$$k-1 = 5$$

$$k = 6$$

TIP: 书写格式:

$$t_n = 20 \times (1.4)^{n-1}$$

不是 $t_n = 20 \times 1.4^{(n-1)}$

Geometric Series

□ The sum of the first n terms of a geometric sequence:

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad ; \quad r \neq 1$$

$\nwarrow$ first term
 $\swarrow$ common ratio

□ The infinite sum of a geometric sequence where $|r| < 1$

$$S_{\infty} = \frac{a}{1 - r} \quad ; \quad |r| < 1$$

$\nwarrow$ first term
 $\swarrow$ common ratio whose magnitude is less than one

[3.6] Example: ① $S_4 = 30$ $S_{\infty} = 32$, find t_1, t_2, t_3

$$32 = \frac{a}{1 - r}$$

$$30 = \frac{a(r^4 - 1)}{r - 1}$$

$$\text{or } 30 = \frac{a(1 - r^4)}{1 - r}$$

$$S_n = \frac{a(r^n - 1)}{r - 1} \text{ 可变形为 } S_n = \frac{a(1 - r^n)}{1 - r}$$

$$\text{Substitute } \frac{a}{1 - r} = 32$$

$$30 = 32(1 - r^4)$$

$$r^4 = \frac{1}{16}$$

$$r = \frac{1}{2}$$

$$a = 16 \dots$$

② CAS

$\{2, 3, 4\frac{1}{2} + \dots\}$ How many terms to achieve $S_n > 100$

$$r = \frac{3}{2} \quad a = 2$$

$$100 = \frac{2(\frac{3}{2}^n - 1)}{\frac{3}{2} - 1}$$

$$25 = \frac{3}{2}^n - 1$$

$$26 = \frac{3}{2}^n$$

CAS: solve $(26 = \frac{3}{2}^x, x)$ in decimal mode, not standard

$$x = 8.035$$

9 terms needed

③ Series Sum: 155 $a = 6$ $d = 1$

$$① 155 = \frac{n}{2} (2 \times 6 + (n-1))$$

$$② 155 \times 2 = n(12 + n - 1)$$

$$③ 310 = 12n + n^2 - n$$

$$④ 310 = 11n + n^2$$

$$⑤ \text{CAS: solve}(310 = 11n + n^2, n)$$

$$⑥ n = 12.945 \dots$$

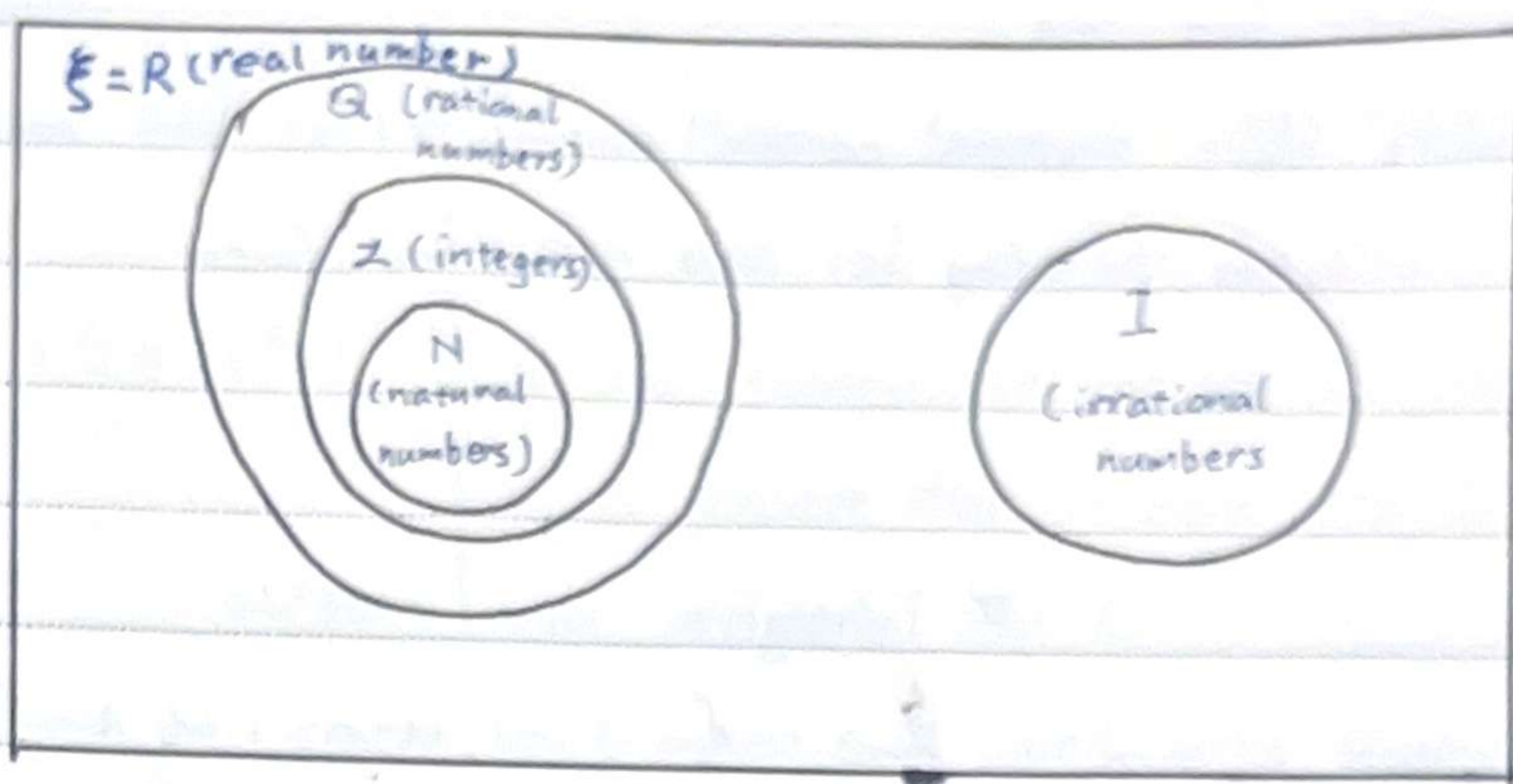
$$S_{12} = 6(12 + 11)$$

$$= 138$$

$$155 - 138 = 17$$

removes 12 full rows and 17 ~~to~~ lags from the 13th row.

Real Numbers



N = natural numbers

Z = integers $\{\dots, -3, -2, -1, 0, +1, +2, \dots\}$

Q = rational numbers

I = irrational numbers

R = real number.

Set notation 设定符号

Sets contain elements, in this topic the elements are numbers.

For example, the following are six elements: 1, 2, 3, 4, 5, 6

$\emptyset$ is the empty or null set. This set contains no elements. $\emptyset = \{\}$.

An upper case letter, such as A , represents a subset of ξ

$$A = \{1, 3, 5\} \quad B = \{1, 2, 3, 4\}$$

$\in$ is read as 'is an element of'. For example: $3 \in A$

$\notin$ is read as 'is not an element of'. For example: $2 \notin A$

$\subset$ is read as 'is a subset of'. For example: $\{1, 3\} \subset A$

$\supset$ is read as 'is a superset of'. For example: $A \supset \{1, 3\}$

A' is the complement of A

$A \cup B$ is the union of A and B

$A \cap B$ is the intersection of A and B

$C \setminus D$ is read as 'C slash D'. This set contains all the elements in C that aren't in D

质数

Prime number is a positive natural number that has exactly two factors: itself and one.

复合数

Composite numbers are positive natural numbers that have more than two factors, including at least one prime factor.

$\mathbb{Z}$ integers.

The set of integers is given by $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

They are positive and negative whole numbers, including zero.

$\mathbb{Z}^-$ is the set of negative integers: $\mathbb{Z}^- = \{\dots, -3, -2, -1\}$.

$\mathbb{Z}^+$ is the set of positive integers: $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$

Therefore, $\mathbb{Z} = \mathbb{Z}^- \cup \{0\} \cup \mathbb{Z}^+$

$\mathbb{Q}$ rational number

The set of rational numbers is given by $\mathbb{Q}$

These are numbers of the form $\frac{p}{q}$ where $p \in \mathbb{Z}$ and $q \in \mathbb{Z} \setminus \{0\}$

Whole number are also rational numbers, $\mathbb{Z} \subset \mathbb{Q}$

$\mathbb{Q}^-$ is the set of negative rational number.

$\mathbb{Q}^+$ is the set of positive rational number.

Therefore, $\mathbb{Q} = \mathbb{Q}^- \cup \{0\} \cup \mathbb{Q}^+$

Absolute value of a real number

The absolute value $|a|$ of a real number a is the distance from a to zero.

$$|-8| = 8$$

$$|11| = 11$$

$$|-\sqrt{2}| = \sqrt{2}$$

Notation

Set notation

Math language ☒

$$\{x: x \in \mathbb{Z}, 1 \leq x \leq 5\}$$

Human language ☒

the set of numbers x such that x is an element of the set of integers and x is greater than or equal to 1 and less than or equal to 5

A given set can be stated in more than one way using set notation

For Example: $\{1, 2, 3, 4, 5\}$ can be written as

$$\{x: x \in \mathbb{Z}, 0 < x < 6\}$$

$$\{x: 1 \leq x \leq 5\}$$

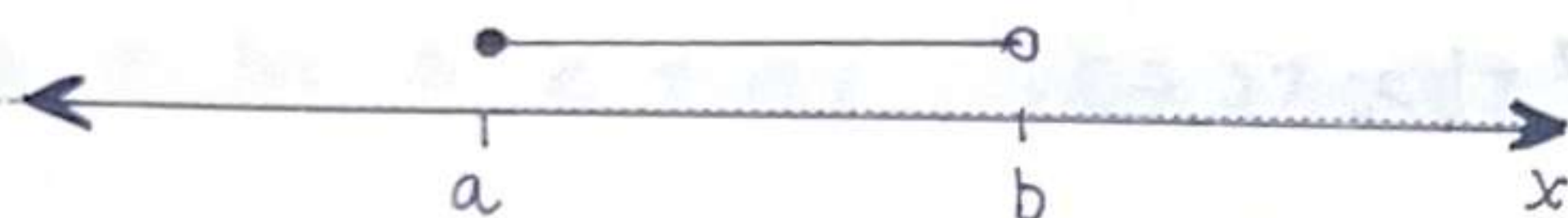
$$\{x: x \in \mathbb{Z}^+, x \leq 5\}$$

Interval notation

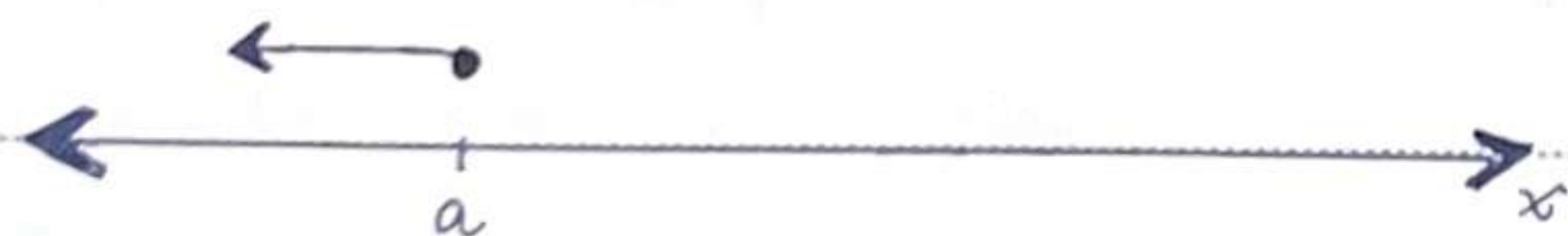
$(3, 8)$ represent all real numbers between 3 and 8, excluding both 3 and 8

$[-2, 6]$ represent all real numbers between -2 and 6, including both -2 and 6

$$[a, b) = \{x: a \leq x < b\}$$



$$(-\infty, a] = \{x: x \leq a\}$$



Surd

Factorizing

$$a\sqrt{c} + b\sqrt{c} = (a+b)\sqrt{c}$$

$$a\sqrt{c} - b\sqrt{c} = (a-b)\sqrt{c}$$

$$\sqrt{c} + \sqrt{d} = \sqrt{(cd)}$$

$$a\sqrt{c} \times b\sqrt{d} = (ab)\sqrt{cd}$$

$$\sqrt{c} \times \sqrt{c} = c$$

$$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$$

Expansions:

$$\sqrt{a}(\sqrt{b} + \sqrt{c}) = \sqrt{ab} + \sqrt{ac}$$

$$(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = (\sqrt{a})^2 - (\sqrt{b})^2 = a - b$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a-b)^2 = a^2 - 2ab + b^2$$

$$(a-b)(a+b) = a^2 - b^2$$

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\text{when } ax^2 + bx + c = 0$$

The set of complex numbers $z = a + bi$

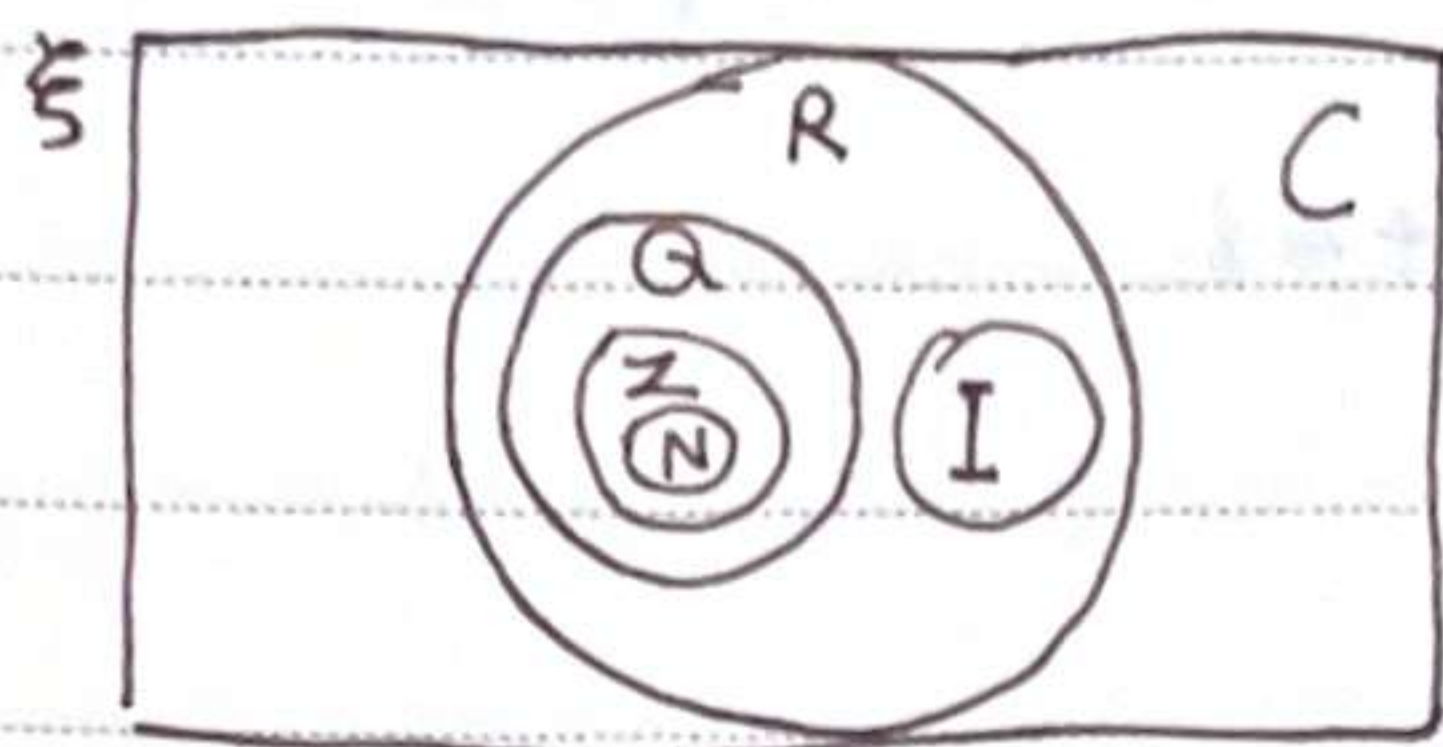
$$\star x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = 2 \pm i$$

, if the discriminant, $b^2 - 4ac$ is negative, the equation has no real solutions, but it does have two complex solutions.

A complex number is any number of the form $x + yi$, where $x, y \in \mathbb{R}$

$\mathbb{C}$ is the set of complex numbers where $\mathbb{C} = \{x + yi : x, y \in \mathbb{R}\}$.

$\mathbb{R} \subset \mathbb{C}$



Notation

✓ If $z = a + bi$, the real component of z is $\text{Re}(z) = a$

✓ and the imaginary component of z is $\text{Im}(z) = b$

For example: if $z = -2 - 2\sqrt{3}i$, $\text{Re}(z) = -2$, $\text{Im}(z) = -2\sqrt{3}$ (not $-2\sqrt{3}i$)

Equality of complex numbers

✓ If $a + bi = c + di$, then $a = c$ and $b = d$

✓ For two complex numbers z_1 and z_2 to be equal, both their real and imaginary components must be equal.

Multiplication of a complex number by a real constant

✓ If $z = a + bi$, then $kz = k(a + bi) = ka + kbi$

For example: if $z = -2 + 3i$, then $-3z = -3(-2 + 3i) = 6 - 9i$

Adding and subtracting complex numbers

if $z_1 = a + bi$ and $z_2 = c + di$,

then $z_1 + z_2 = (a+c) + (b+d)i$

and $z_1 - z_2 = (a-c) + (b-d)i$

Modulus of a complex number

The ^{大小}magnitude (or modulus or absolute value) of a complex number $z = x + yi$ is denoted by $|z|$.
(represent)

And $|z| = \sqrt{x^2 + y^2}$

Example: $z_1 = 2 - 3i$, $z_2 = -3 + 4i$

find $|z_1|$

$$|z_1| = \sqrt{x^2 + y^2}$$

$$|z_1| = \sqrt{2^2 + 3^2}$$

$$= \sqrt{13}$$

Complex number example question

① Solve to find x and y

$$(x+1) + (y-1)i = 2 + 3i$$

use equality of complex numbers

real : $x+1 = 2$

im : $y-1 = 3$

$$x = 1$$

$$y = 4$$

② Solve to find x and y

$$(2x+i) + (3-2yi) = x + 3i$$

Rearrange the equation

$$2x + 3 + i - 2yi = x + 3i$$

$$(2x+3) + (1-2y)i = x + 3i$$

use equality of complex numbers

real : $2x + 3 = x$

im : $1 - 2y = 3$

$$2x = x - 3$$

$$x = -3$$

$$y = -1$$

$$\textcircled{3} \quad z_1 = 2 - i \quad z_2 = 3 + 4i$$

find $|z_1|$

$$|z_1| = \sqrt{(2)^2 + (-1)^2} = \sqrt{5}$$

Modus of a complex number

$$\textcircled{6} \quad z = 21 - 2i + 2(10 - 5i)$$

Find $\text{im}(z)$

$$21 - 2i + 20 - 10i$$

$$41 - 12i$$

$$\text{im}(z) : -12$$

④ Express the following in terms of i

$$\sqrt{-16}$$

$$\sqrt{16} \times \sqrt{-1} \quad \because -1 = i^2$$

$$4 \times \sqrt{i^2}$$

$$4i$$

⑦ CAS ^{cplx mode} 先設置成 ^{comp} complex mode ^再 按

Main / Action / complex / re 或 im

⑤ State value of $\text{Re}(z)$ and $\text{Im}(z)$

$$(a) \quad 3 + 4i$$

$$\text{Re}(3 + 4i) = 3$$

$$\text{Im}(3 + 4i) = 4$$

$$(b) \quad \sqrt{8} - \sqrt{-40}$$

Simplify

$$z = 2\sqrt{2} - \sqrt{4 \times 10} \times \sqrt{-1} \quad \because -1 = i^2$$

$$z = 2\sqrt{2} - 2\sqrt{10}i$$

$$\text{Re}(z) = 2\sqrt{2}$$

$$\text{Im}(z) = -2\sqrt{10}$$

$$(c) \quad -6$$

$$\text{Re}(z) = -6$$

$$\text{Im}(z) = 0$$

Multiplication & Division of Complex Numbers.

The conjugate of a complex number

■ $z = a + bi$, its conjugate, $\bar{z}$, is $\bar{z} = a - bi$

■ The sum of a complex number and its conjugate :

$$z + \bar{z} = a + bi + a - bi = 2a \text{ which is a real number.}$$

■ The product of a complex number and its conjugate:

$$z \bar{z} = (a + bi)(a - bi)$$

$$= a^2 - (bi)^2$$

$$= a^2 + b^2, \text{ which is a real number.}$$

Division of complex number

$$\frac{z_1}{z_2} = \frac{a + bi}{c + di} \times \frac{c - di}{c - di}$$

Multiplication & Division example question

① Expand and simplify

$$-2i(3-4i)$$

$$-6i + 8i^2 \quad \therefore i^2 = -1$$

$$-6i - 8$$

$$-8 - 6i \quad a+bi \text{ 形式}$$

② Express in the form $a+bi$

$$\frac{3-4i}{-5i} \times \frac{5i}{5i}$$

$$\frac{5i(3-4i)}{-5i(5i)} = \frac{15i-20}{-25i^2}$$

$$\frac{3i-4}{-5} = \frac{4}{5} + \frac{3}{5}i$$

(a)

③ Prove $\overline{z_1 z_2} = \overline{z_1} \overline{z_2}$

$$z_1 = a+bi, z_2 = c+di$$

$$(a-bi)(c-di) = \overline{(a+bi)(c+di)}$$

$$ac - adi - bci + bdi = \overline{ac + adi + bci + bdi^2}$$

$$(ac - bd) - i(ad + bc) = \overline{(ac - bd) + (ad + bc)i}$$

$$= (ac - bd) - (ad + bc)i$$

(b) Prove $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$

$$(a-bi) + (c-di) = \overline{(a+bi) + (c+di)}$$

$$(a+c) - i(b+d) = \overline{(a+c) + i(b+d)}$$

$$= (a+c) - i(b+d)$$

④ $(2x+yi) \overline{(3-2i)} = 4+5i$

CAS cexpand

$$(6x+2y) - (4x-3y)i = 4+5i$$

$$\begin{cases} 6x+2y=4 \\ -(4x-3y)=5 \end{cases} \quad x, y$$

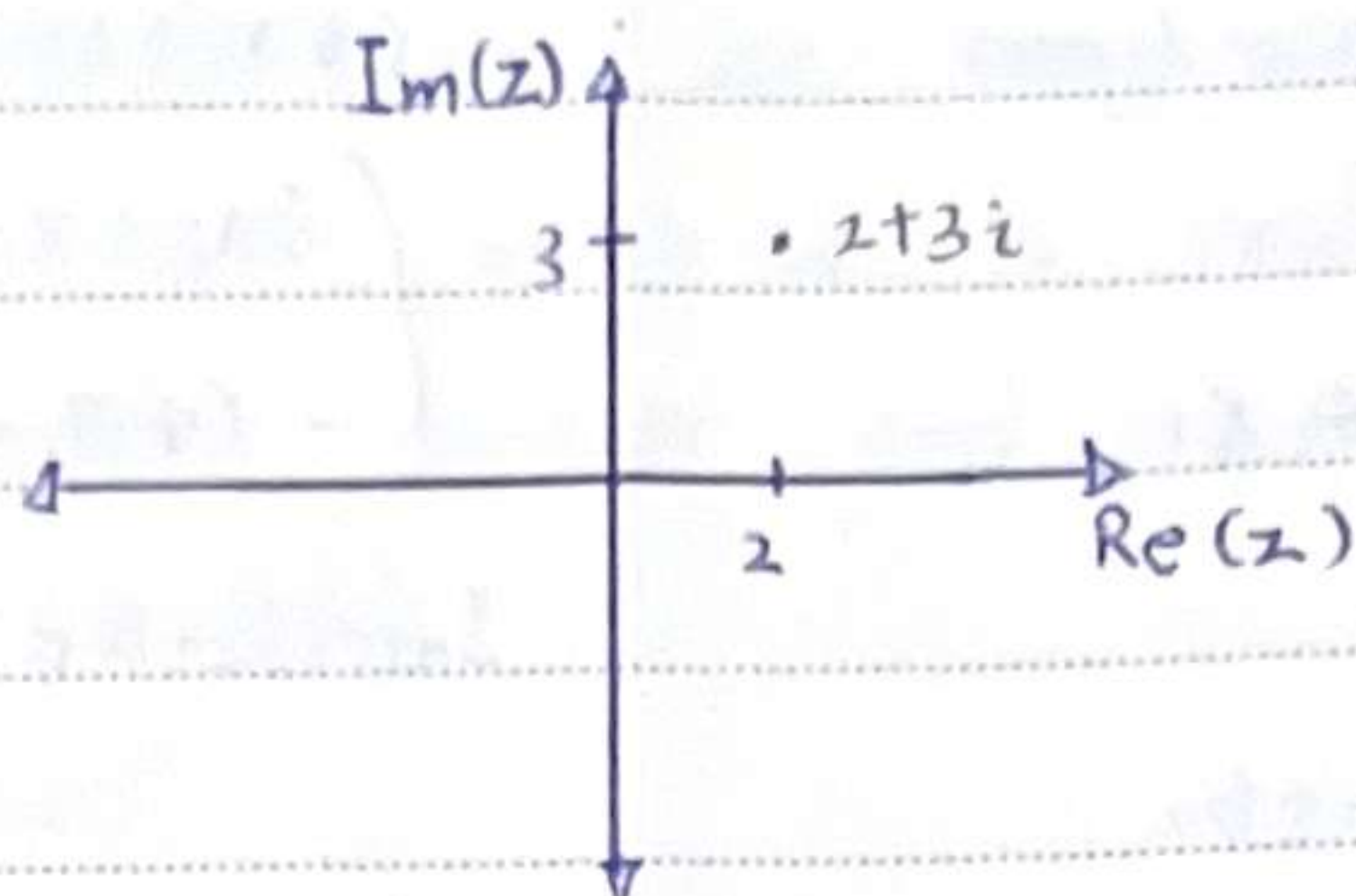
Interactive / Complex / cExpand

↓
可以把算式变成 $a+bi$ form

Interactive / Complex / conjg → 把算式变成 conjugate

Representing complex numbers on an Argand diagram

• Argand diagram



$$i^0 = 1$$

$$i^1 = i$$

$$i^2 = -1$$

$$i^3 = -i \quad \therefore -1 \times i$$

$$i^4 = 1$$

$$i^5 = i^4 \times i = i$$

← Example Question ←

① Give the following in their simplest form

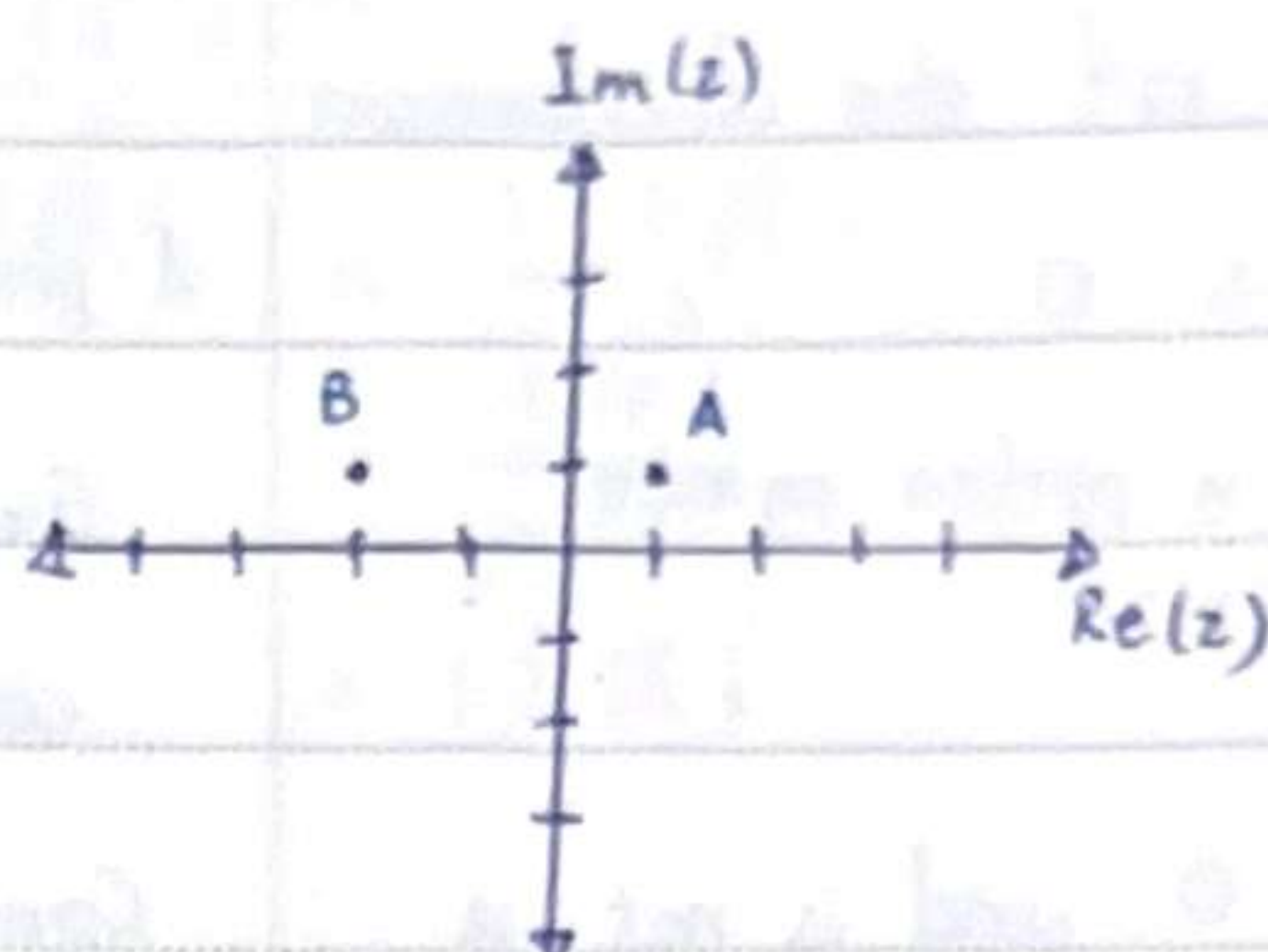
$$\begin{aligned}
 (a) \quad i^7 &= i^4 \times i^3 \\
 &= 1 \times -i \\
 &= -i
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad i^{31} &= (i^4)^7 \times i \\
 &= i
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad i^{-4} &= \frac{1}{i^4} \\
 &= \frac{1}{1} \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad i^{-15} &= \frac{1}{(i^4)^3 \times i^3} \\
 &= \frac{1}{1 \times -i} \\
 &= \frac{1}{-i} \times \frac{i}{i} \\
 &= \frac{i}{1} \quad \text{因为 } -i^2 = 1, \text{ as } i^2 = -1 \\
 &= i
 \end{aligned}$$

③ Write down the complex number



$$A: (1, 1) = 1 + i$$

$$B: (-2, 1) = -2 + i$$

④ Describe the graph, if $z = x + yi$

$$(a) \quad |z - 1| = 1$$

$$|x + yi - 1| = 1$$

$$|(x - 1) + (yi)| = 1$$

$$\sqrt{(x - 1)^2 + y^2} = 1 \rightarrow \text{radius}$$

$$(x - 1)^2 + y^2 = 1$$

$\begin{matrix} x-1=0 \\ y=0 \end{matrix}$
 This is a circle with centre at $1 + 0i$ and a radius of 1.

② Plot the following on an Argand plane.

$$(a) \quad 2 - 3i$$

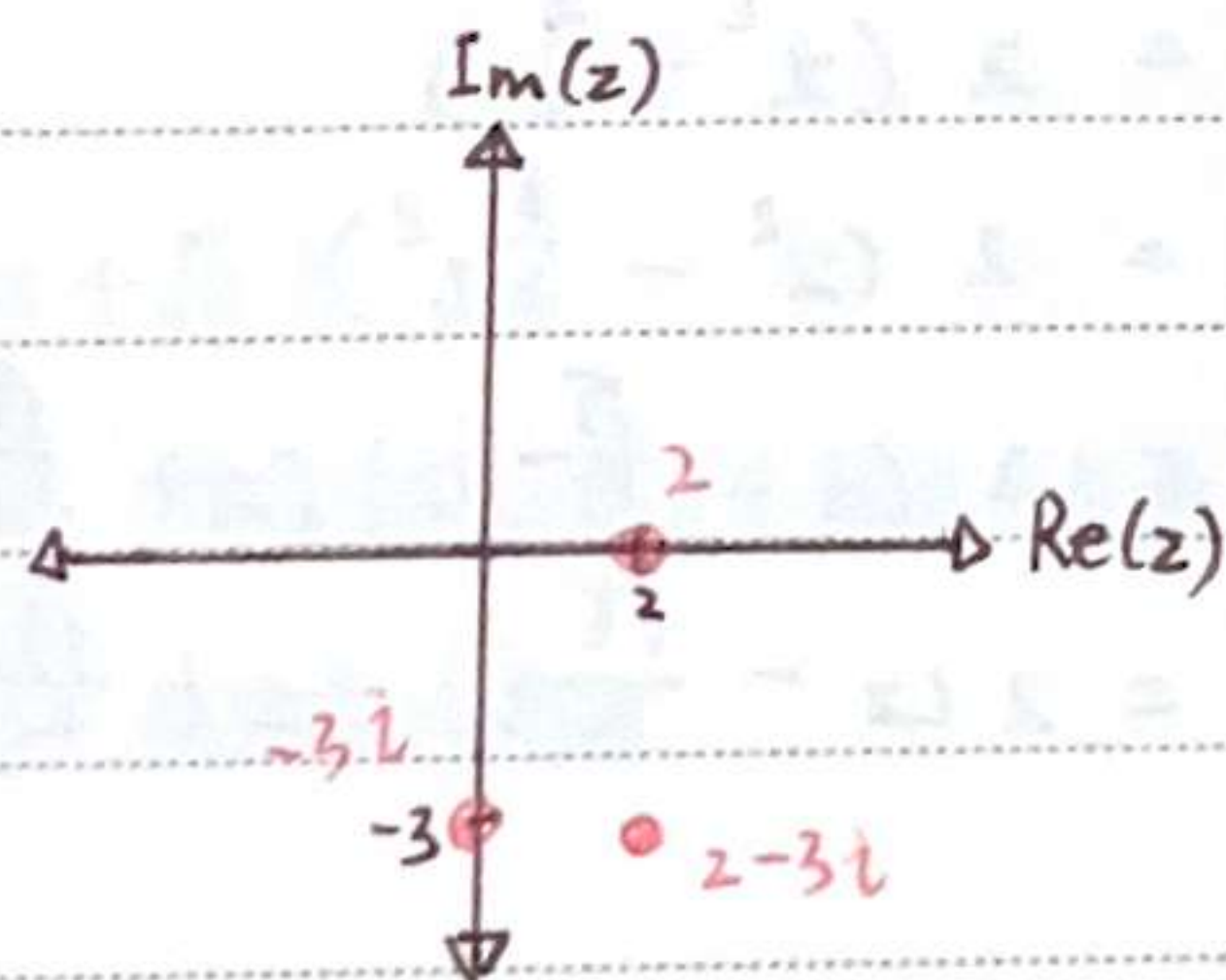
$$(2, -3)$$

$$(b) \quad -3i$$

$$(0, -3)$$

$$(c) \quad 2$$

$$(2, 0)$$



$$(b) \quad |z - 2 + 3i| = 3$$

$$|x + yi - 2 + 3i| = 3$$

$$|(x - 2) + i(y + 3)| = 3$$

$$\sqrt{(x - 2)^2 + (y + 3)^2} = 3$$

$$(x - 2)^2 + (y + 3)^2 = 9$$

This is a circle with centre at $2 - 3i$ and a radius of 3

Factorise & Solve quadratic equa with Complex number ^{over C}

Values of the discriminant	Factorising methods	Nature of solution(s)
$\Delta = 0$	A perfect ² . State the answer	Two equal rational solus
Δ is a perfect square	Factorise over Q or complete the square	Two rational solus
$\Delta > 0$ and is not a perfect square.	Complete the square	Two irrational solutions
$\Delta < 0$	Complete the square	Two complex solutions

$$ax^2 + bx + c = 0$$

Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

If $c = 0$ and $a, b \in \mathbb{R} \setminus \{0\}$, then factorise $ax^2 + bx$ by taking out the common factor $x(ax + b)$

ex: $2x^2 + 6x = 2x(x + 3)$

If $b = 0$ and $a, c \in \mathbb{R} \setminus \{0\}$, then factorise $ax^2 + c$ using completing the squares to factorise

ex: $2x^2 - 6$

$$= 2(x^2 - 3)$$

$$= 2(x - \sqrt{3})(x + \sqrt{3})$$

ex: $2x^2 + 3$

$$= 2(x^2 + \frac{3}{2})$$

$$= 2(x^2 - \frac{3}{2}i^2)$$

$$= 2(x - \frac{\sqrt{3}}{\sqrt{2}}i)(x + \frac{\sqrt{3}}{\sqrt{2}}i)$$

$$= 2(x - \frac{\sqrt{6}}{2}i)(x + \frac{\sqrt{6}}{2}i)$$

Example Question

① Factorise over $\mathbb{C}$

(a) $2z^2 - 6$ $ax^2 + c$, complete $\square^2 a^2 - b^2$

$$2(z^2 - 3)$$

$$2(z - \sqrt{3})(z + \sqrt{3})$$

(b) $2z^2 - 3$ $ax^2 + c$, $a^2 - b^2 = (a+b)(a-b)$

$$(\sqrt{2}z)^2 - (\sqrt{3})^2$$

$$(\sqrt{2}z - \sqrt{3})(\sqrt{2}z + \sqrt{3})$$

(c) $3x^2 + 6$ $ax^2 + c$, 变为 $a^2 - b^2 = (a+b)(a-b)$

$$3(x^2 + 2)$$

$$3(x^2 - (\sqrt{2}i)^2)$$

$$3(x - \sqrt{2}i)(x + \sqrt{2}i)$$

(d) $z^2 + z - 3$

$$\left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$z^2 + z + \frac{1}{4} - \frac{1}{4} - 3$$

$$\left(z + \frac{1}{2}\right)^2 - \left(\frac{\sqrt{13}}{4}\right)^2$$

$$\star \left(z + \frac{1}{2}\right)^2 - \left(\frac{\sqrt{13}}{4}\right)^2$$

$$\left(z + \frac{1}{2} - \frac{\sqrt{13}}{4}\right)\left(z + \frac{1}{2} + \frac{\sqrt{13}}{4}\right)$$

③ Solve $x^2 - 2x + 3 = 0$

$$\frac{2 \pm \sqrt{4 - 12}}{2}$$

$$\frac{2 \pm \sqrt{-8}i}{2} = \frac{2 \pm 2\sqrt{2}i}{2}$$

$$= \frac{1 \pm \sqrt{2}i}{1}$$

$$= 1 \pm \sqrt{2}i$$

④ CAS Active

(a) solve $-2z^2 + 4z - 6 = 0$

Complex (cplx) mode

solve $(-2z^2 + 4z - 6, z)$

$$z = 1 - \sqrt{2}i, z = 1 + \sqrt{2}i$$

(b) Factorise

$$z^2 + 6z + 11$$

cplx mode

INPUT: $z^2 + 6z + 11$

Highlight (select the equa)

Interactive / Transformation / factor / rFactor

$$(z + 3 + \sqrt{2}i)(z + 3 - \sqrt{2}i)$$

② Factorise and solve

(a) $3z^2 - 2 = 0$

$$(\sqrt{3}z)^2 - (\sqrt{2})^2$$

$$(\sqrt{3}z - \sqrt{2})(\sqrt{3}z + \sqrt{2})$$

① $\frac{\sqrt{2}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{6}}{3}$ 根号分母和i分母都要x乘

② $\frac{-\sqrt{2}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{-\sqrt{6}}{3}$ 根号或i消除

$$z = \pm \frac{\sqrt{6}}{3}$$

Example Question

① Factorise over $\mathbb{C}$

(a) $2x^2 - 6$ $ax^2 + c$, complete $\square^2 a^2 - b^2$

$2(x^2 - 3)$

$2(x - \sqrt{3})(x + \sqrt{3})$

(b) $2x^2 - 3$ $ax^2 + c$, $a^2 - b^2 = (a+b)(a-b)$

$(\sqrt{2}x)^2 - (\sqrt{3})^2$

$(\sqrt{2}x - \sqrt{3})(\sqrt{2}x + \sqrt{3})$

(c) $3x^2 + 6$ $ax^2 + c$, 变为 $a^2 - b^2 = (a+b)(a-b)$

$3(x^2 + 2)$

$3(x^2 - (\sqrt{2}i)^2)$

$3(x - \sqrt{2}i)(x + \sqrt{2}i)$

(d) $x^2 + x - 3$

$(\frac{1}{2})^2 = \frac{1}{4}$

$x^2 + x + \frac{1}{4} - \frac{1}{4} - 3$

$(x + \frac{1}{2})^2 - (\frac{\sqrt{13}}{2})^2$

★ $(x + \frac{1}{2})^2 - (\frac{\sqrt{13}}{2})^2$

$(x + \frac{1}{2} - \frac{\sqrt{13}}{2})(x + \frac{1}{2} + \frac{\sqrt{13}}{2})$

③ Solve $x^2 - 2x + 3 = 0$

$\frac{2 \pm \sqrt{4 - 12}}{2}$

$\frac{2 \pm \sqrt{-8}i}{2} = \frac{2 \pm 2\sqrt{2}i}{2}$

$= \frac{1 \pm \sqrt{2}i}{1}$

$= 1 \pm \sqrt{2}i$

④ CAS Active

(a) solve $-2x^2 + 4x - 6 = 0$

Complex (cplx) mode

solve $(-2x^2 + 4x - 6; x)$

$x = 1 - \sqrt{2}i, x = 1 + \sqrt{2}i$

(b) Factorise

$x^2 + 6x + 11$

cplx mode

INPUT: $x^2 + 6x + 11$

Highlight (select the equa)

Interactive / Transformation / factor / rFactor

$(x + 3 + \sqrt{2}i)(x + 3 - \sqrt{2}i)$

② Factorise and solve

(a) $3x^2 - 2 = 0$

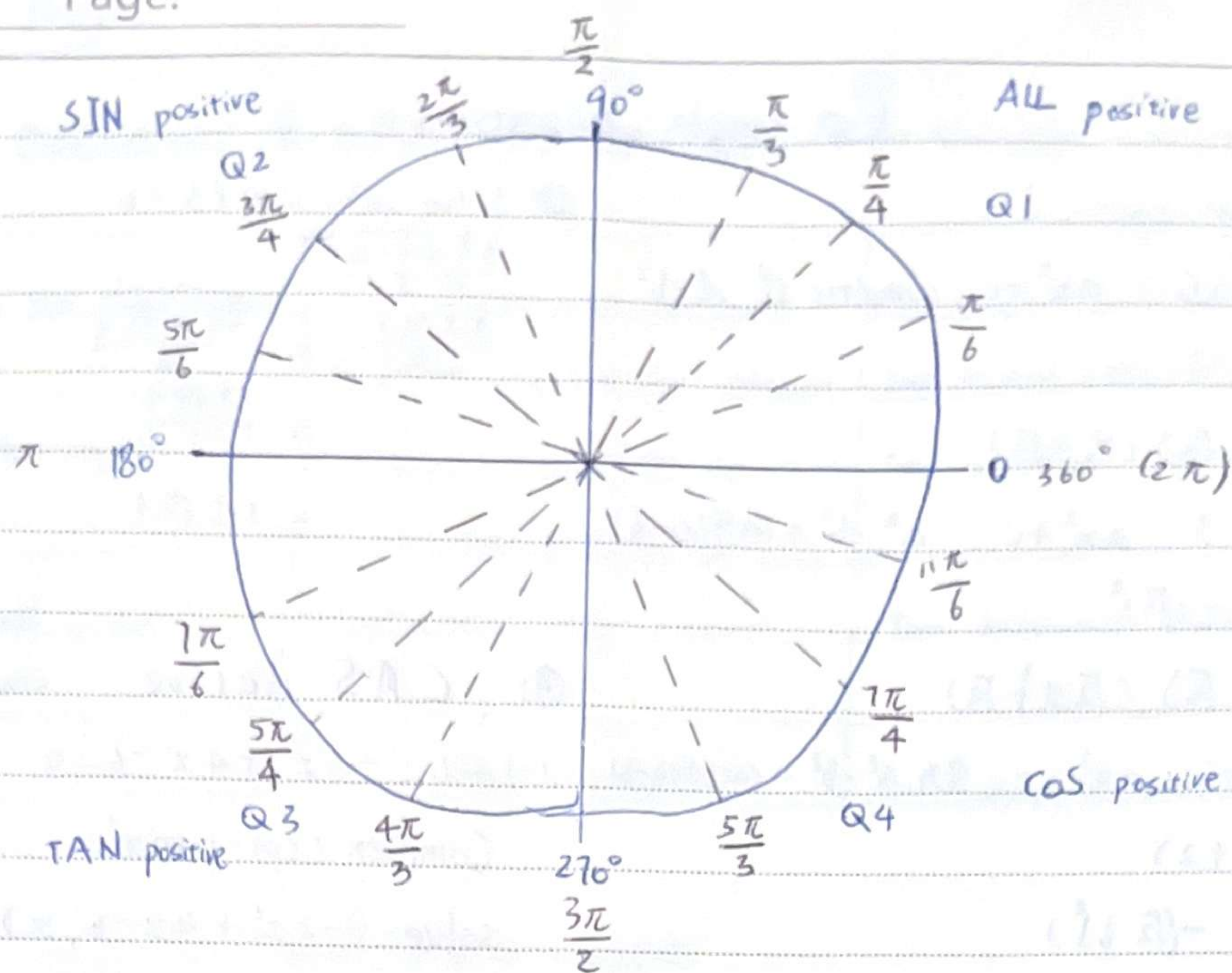
$(\sqrt{3}x)^2 - (\sqrt{2})^2$

$(\sqrt{3}x - \sqrt{2})(\sqrt{3}x + \sqrt{2})$

① $\frac{\sqrt{2}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{6}}{3}$ 根号分母和i分母都要x乘

② $\frac{-\sqrt{2}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{-\sqrt{6}}{3}$ 根号或i消除

$x = \pm \frac{\sqrt{6}}{3}$



Exact value table

	0°	30°	45°	60°	90°	180°
θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
Sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0
Cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
tan	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	∞	0

undefined

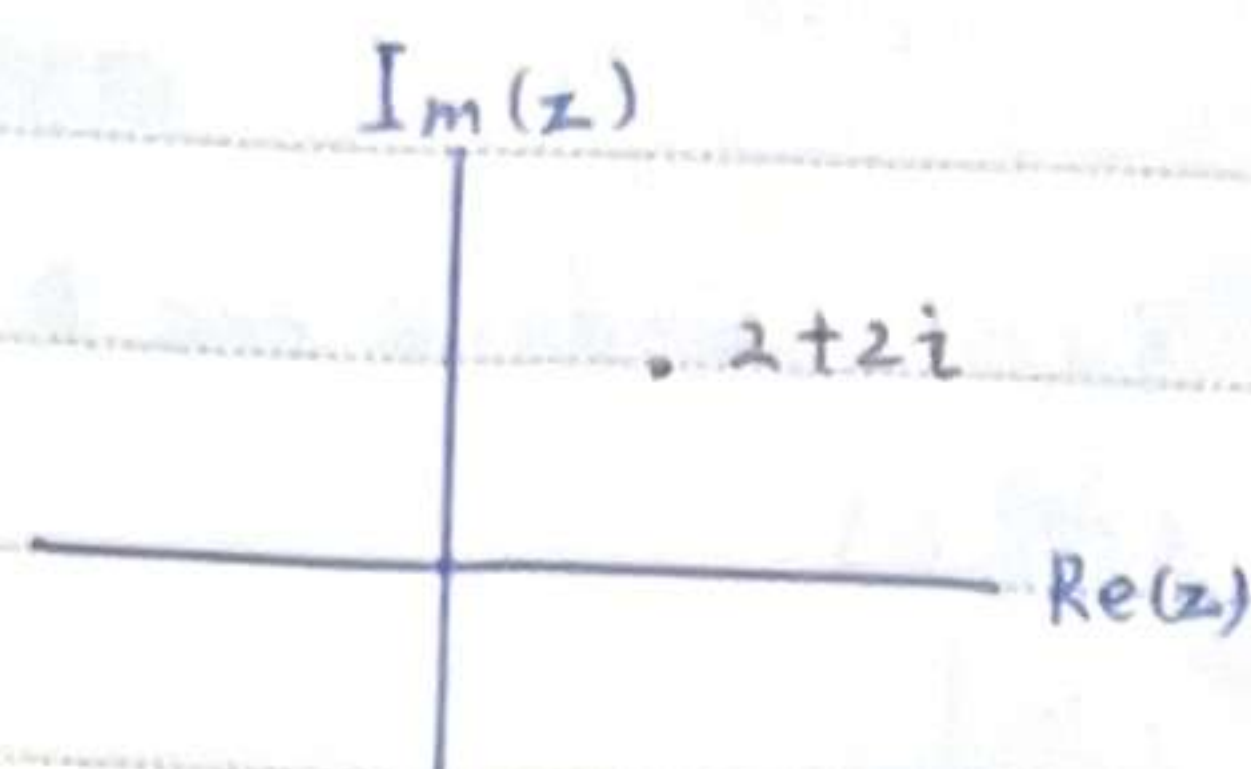
[Find Exact value] Convert $\sin\left(\frac{7\pi}{4}\right)$ to a number.

- ① what quadrant it is in - 先找在哪个 quadrant. 例 $\frac{7\pi}{4}$ $\pi = 180^\circ$ $180^\circ \times 7 \div 4 = 315^\circ$ $\bigoplus$ 从 0° 数, 315 在 Q4.
- ② Positive or negative - Q4 代表 cos, 不在它自己的象限 所以数字前加减号 " - "
- ③ match exact value - $\sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$

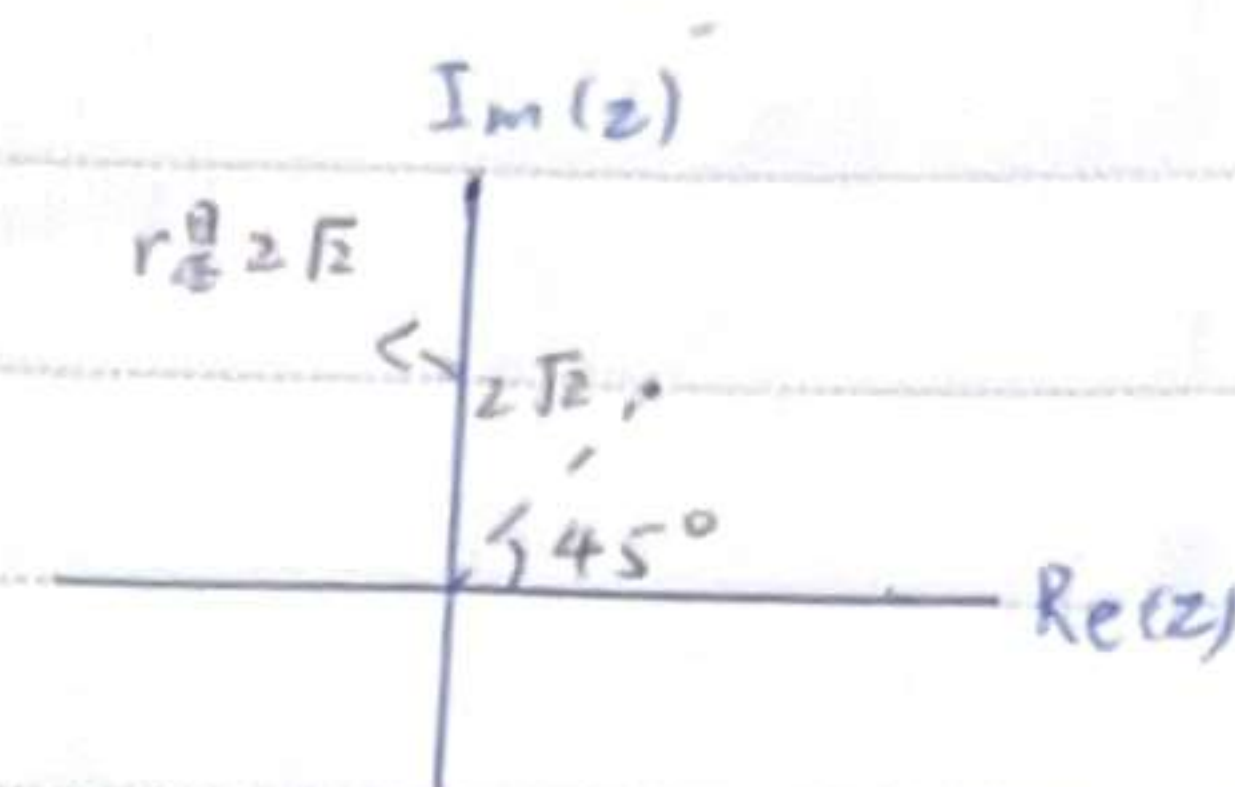
- 加负号 $-\frac{\sqrt{2}}{2}$ - 答: $-\frac{\sqrt{2}}{2}$ ★ Example: ① $\tan\left(\frac{\pi}{2}\right)$ $\frac{\pi}{2} = 90^\circ$ ∞ ② $\times \times \times \frac{9\pi}{4}$ 顺时针 zhen 继续数

Convert complex numbers in polar form.

• Cartesian form



• Polar form

□ z 可以 $= x + yi$

$$= 2 + 2i$$

$$\text{还可以} = \underbrace{2\sqrt{2}}_{\text{距离}} \text{cis} \underbrace{\frac{\pi}{4}}_{\text{角度}}$$

$$\square |z| = \sqrt{x^2 + y^2} = r$$

↓
modulus of z

★ also is the distance from the origin to the centre

□ 找 angle: 找 Argument of z

$$\arg(z) = \theta$$

In addition

If question ask you answer

written as $[-\pi < \theta \leq \pi]$ Then answer $-\frac{\pi}{4}$ 不能是 $\frac{7\pi}{4}$

$$\arg(z) = \frac{7\pi}{4}, -\frac{\pi}{4}$$

形式都行, 意思一样

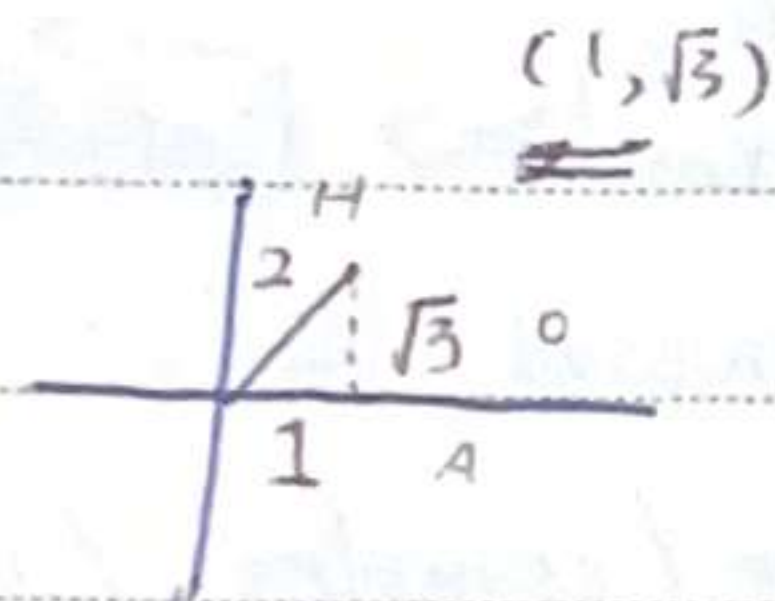
$$\text{Ang}(z) = -\frac{\pi}{4}$$

意思一样 形式有要求, 心. 需

Example Question

① $z = 1 + \sqrt{3}i$

画图:



$$\text{找 } |z|: |z| = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

$$\text{找 angle: TOA: } \frac{\sqrt{3}}{1} = \sqrt{3}, \tan(\sqrt{3})$$

$$\text{对表格: } \tan \sqrt{3} \text{ 对应 } \frac{\pi}{3}$$

$$\text{合体: } 2 \text{cis} \left(\frac{\pi}{3}\right)$$

★★★★★

$$1 + \sqrt{3}i = 2 \text{cis} \left(\frac{\pi}{3}\right)$$

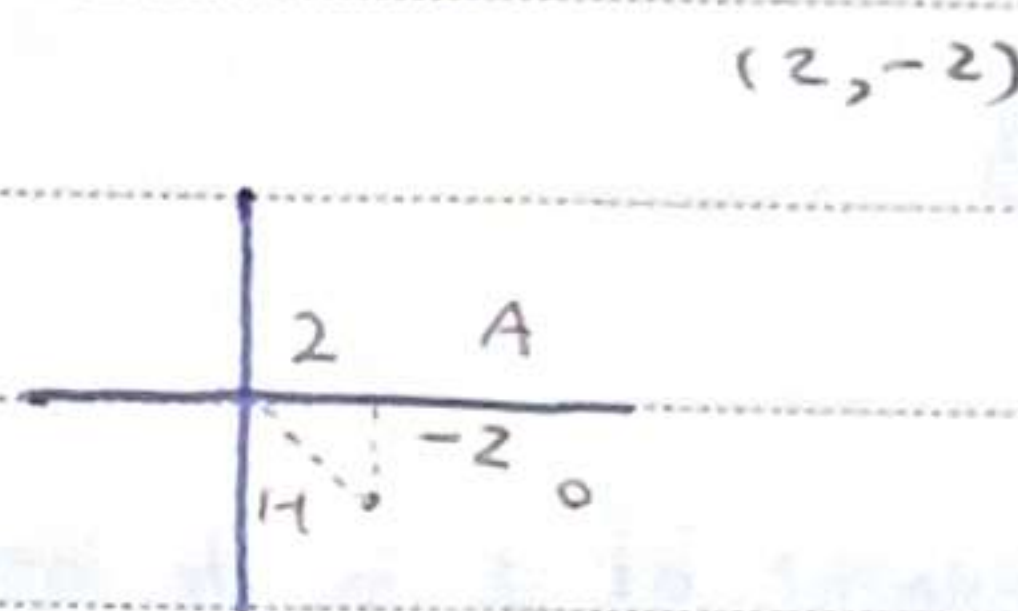
$$= 2 \cos \left(\frac{\pi}{3}\right) + i 2 \sin \left(\frac{\pi}{3}\right)$$

prim 找 exact value of $\sin \left(\frac{\pi}{3}\right)$

$$= \sqrt{3}$$

② $z = 2 - 2i$

图:



$$|z| = \sqrt{2^2 + (-2)^2} = 2\sqrt{2}$$

$$\text{Angle: TOA} = \frac{-2}{2} = -1, \tan(-1)$$

$$\text{对表格: } \tan 1 = \frac{\pi}{4}$$

$$\text{合体: } 2\sqrt{2} \text{cis} \left(-\frac{\pi}{4}\right)$$

∴ 不在 tan 表 找
or 负数 angle.

③ $z = -1$ $(-1, 0)$

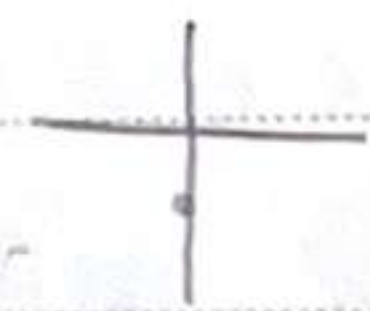


$$|z| = 1$$

$$\text{Angle: } \pi$$

$$\text{cis } \pi$$

④ $z = -i$ $(0, -1)$

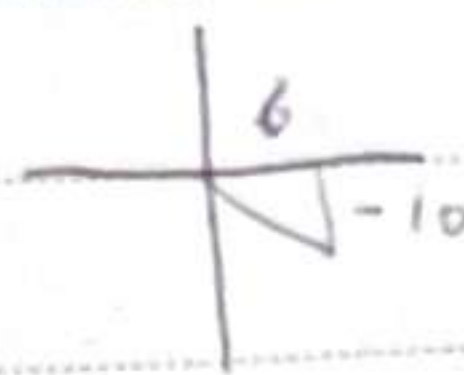


$$|z| = 1$$

$$\text{Angle: } -\frac{\pi}{2} \text{ or } \frac{3\pi}{2}$$

$$\text{cis } -\frac{\pi}{2}$$

⑤ (b) $z = 6 - 10i$ $(6, -10)$

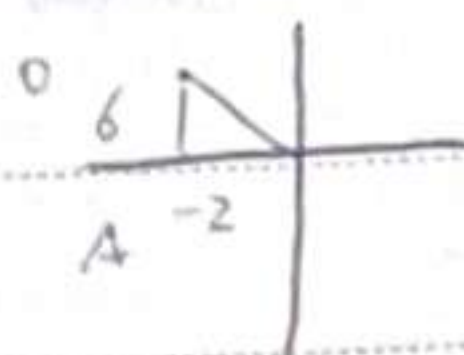


$$\tan^{-1}\left(-\frac{10}{6}\right) + 2\pi = 5.253$$

因为从 0° 开始算

(c) $6i - 2$, modulus and θ

$$(-2, 6)$$



$$\sqrt{36+4} = 2\sqrt{10}$$

$$\tan^{-1}\left(-\frac{6}{2}\right) = -3$$

$$\tan^{-1}(-3) + \pi = 1.893$$

$$2\sqrt{10} \text{ cis }$$

① $a+bi \leftarrow \text{Cartesian form} \Rightarrow \text{Polar form}$

Alg standard Cplx Rad

Interactive / complex / comToPol

$$\text{comToPol}(2+2i)$$

$$2\sqrt{2} \cdot e^{\frac{\pi i}{4}} \text{ 不是 } e^i$$

$$\text{得到这个} = 2\sqrt{2} \text{ cis } \left(\frac{\pi}{4}\right)$$

$r \text{ cis } \theta \leftarrow \text{Polar form} \Rightarrow \text{Cartesian form}$

Alg standard Cplx Rad

Interactive / complex / comToTrig $(-1-i)$

$$\sqrt{2} \left(\cos\left(-\frac{3\pi}{4}\right) + \sin\left(-\frac{3\pi}{4}\right)i \right)$$

Interactive / complex / comToRect

$$\sqrt{2} \left(\cos\left(-\frac{3\pi}{4}\right) + \sin\left(-\frac{3\pi}{4}\right)i \right)$$

$$\sqrt{2} \text{ 代表 } r \text{ 而 } -\frac{3\pi}{4} \text{ 代表 } \theta$$

$$-\sqrt{2} \text{ cis } \left(-\frac{3\pi}{4}\right)$$

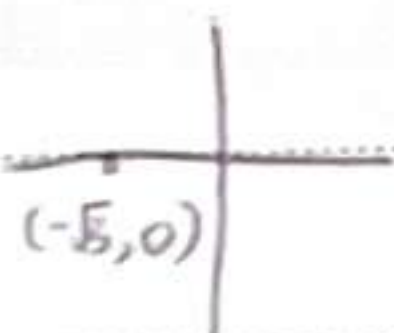
$$\text{type } \sqrt{2} \left(\cos\left(-\frac{3\pi}{4}\right) + \sin\left(-\frac{3\pi}{4}\right)i \right)$$

- select all, complex / comToRect

$$- \text{得 } -1-i$$

⑤ Express complex number in Cartesian form

$$8\sqrt{3} \text{ cis } (\pi)$$

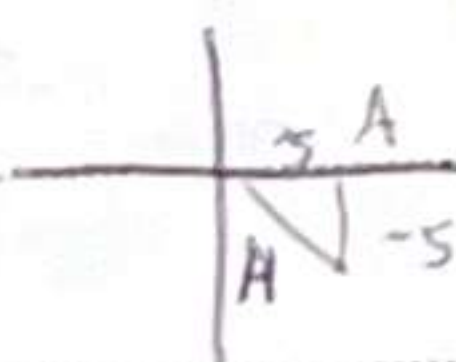


$$\text{(form: } -\sqrt{3})$$

⑥ Find argument of z in the interval $[0, 2\pi]$

give exact answer where possible

(a) $z = 5 - 5i$ $(5, -5)$



$$\tan^{-1}\left(-\frac{5}{5}\right) = -\frac{\pi}{4}$$

因为要在 $0-2\pi$ 里所以从 0° 开始算

$$-\frac{\pi}{4} + 2\pi = \frac{7\pi}{4}$$

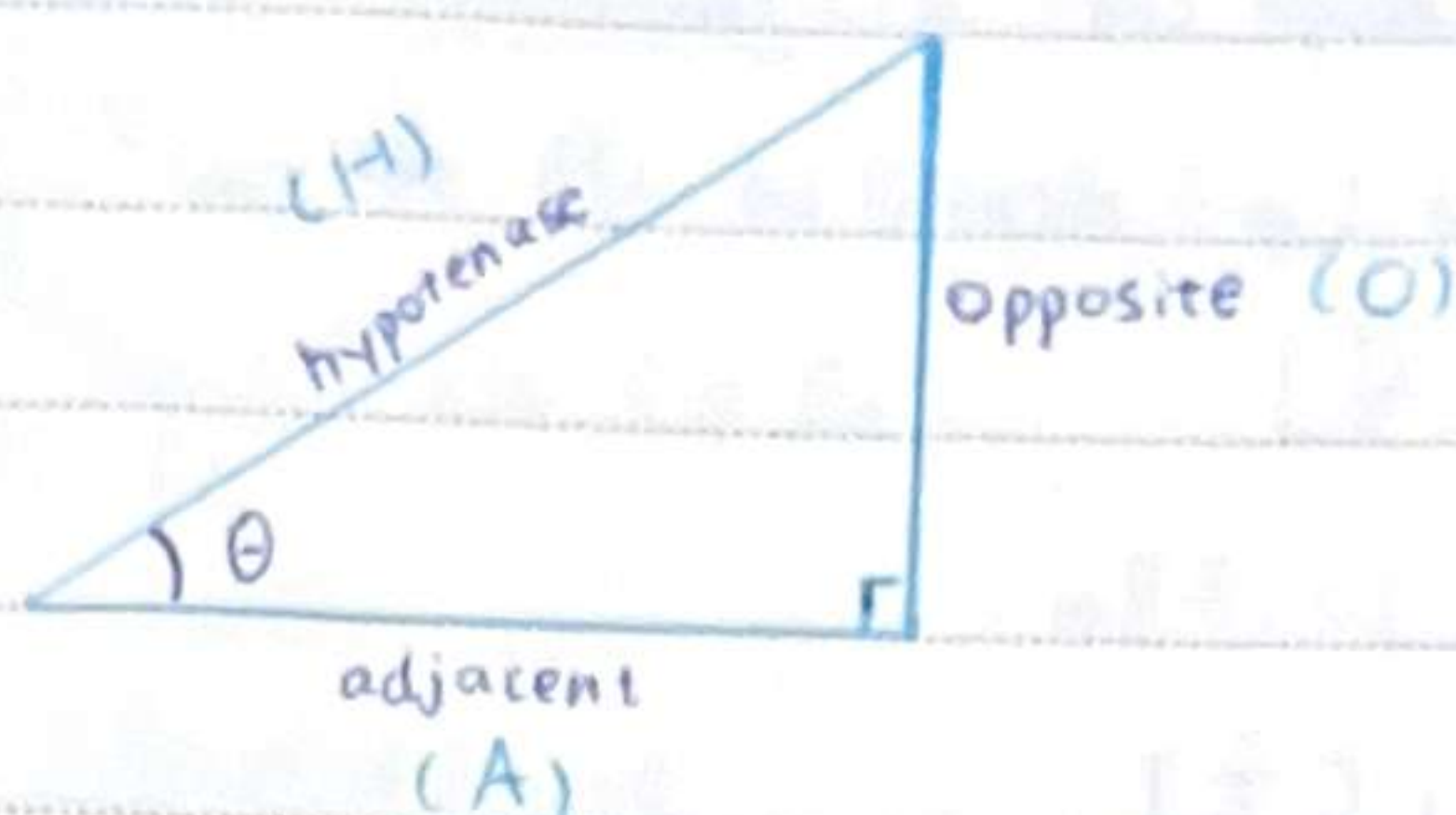
Date.

Page.

s.2 Trigonometry

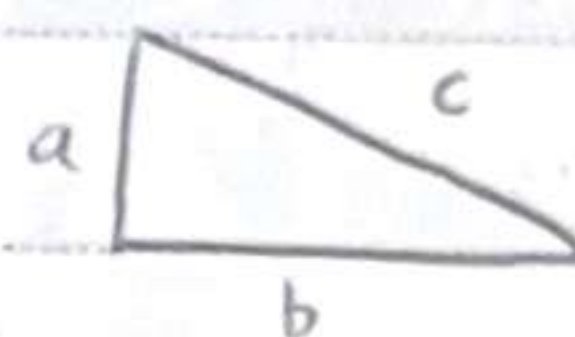
三角函数
* SOH, CAS, TOA 只适用于

直角三角形



Pythagoras' theorem states:

In any right-angled triangle, $c^2 = a^2 + b^2$.
勾股定理只用于直角三角形



CAS TIPSEY

求长度: $1 \div \cos(24^\circ 25' 0'')$

to enter degrees, minutes and seconds go to keyboard / math1
□□□ 第1第4第5

求角度: $\text{toDMS}(\cos^{-1}(\frac{A}{H}))$

go to
interactive/transformation/DMS/toDMS
or Math 1: toDMS.

$58^\circ 18' 35.75''$

if the number is 30 or greater
the minutes need to be rounded up.

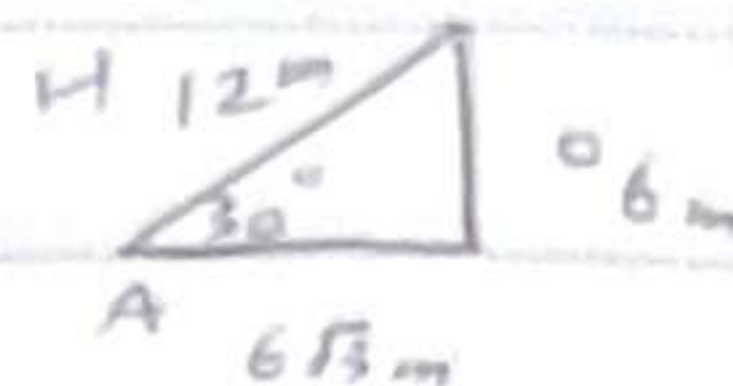
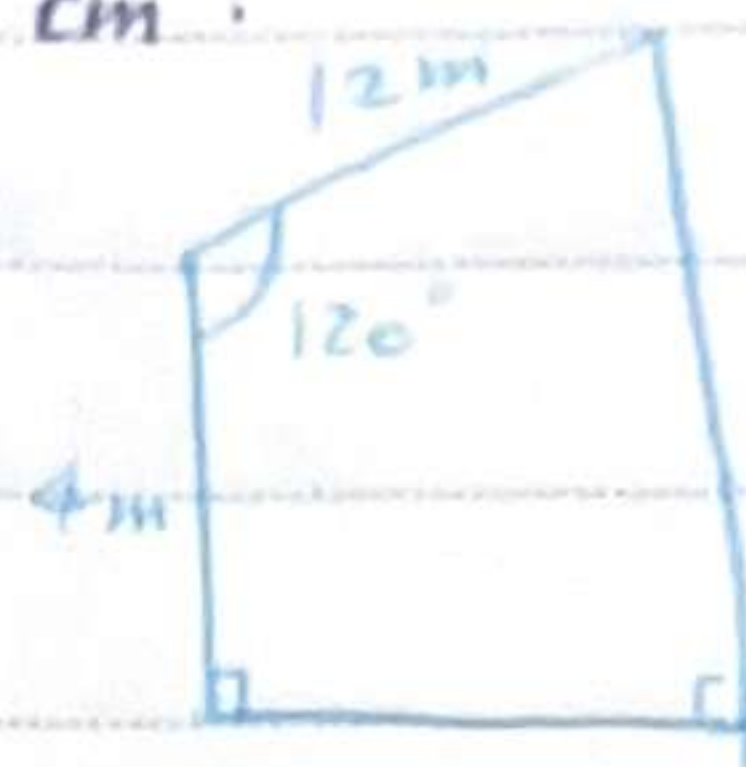
Mode: Alg Decimal Real Deg

Trigonometry Example Question.

- ① A garden bed in the shape of a trapezium. What volume of garden mulch is needed to cover it to a depth of 15 cm?

$$y: \text{CAH: } A = 12 \times \cos\left(\frac{\pi}{6}\right) \left[\frac{\sqrt{3}}{2}\right] \\ = 6\sqrt{3} \text{ m} \approx 10.39 \text{ m}$$

$$x: \text{SOH: } O = 12 \times \sin\left(\frac{\pi}{6}\right) \left[\frac{1}{2}\right] \\ = 6 \text{ m}$$



Area of trapezium: $A = \frac{1}{2}(a+b)h$

$$= \frac{1}{2}(4+x+4) \times y$$

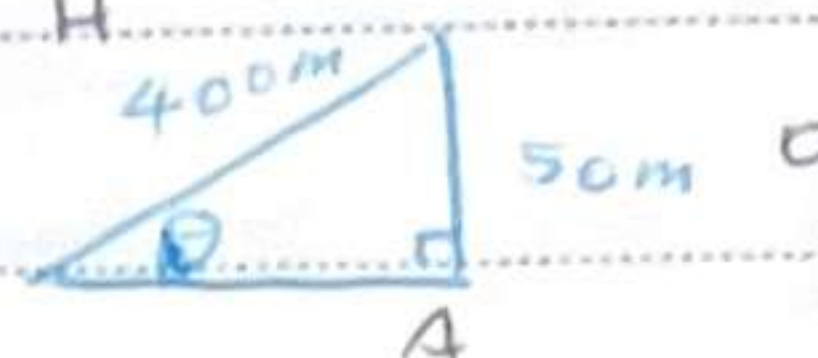
$$= \frac{1}{2}(4+6+4) \times 10.39$$

$$= 72.73 \text{ m}^2$$

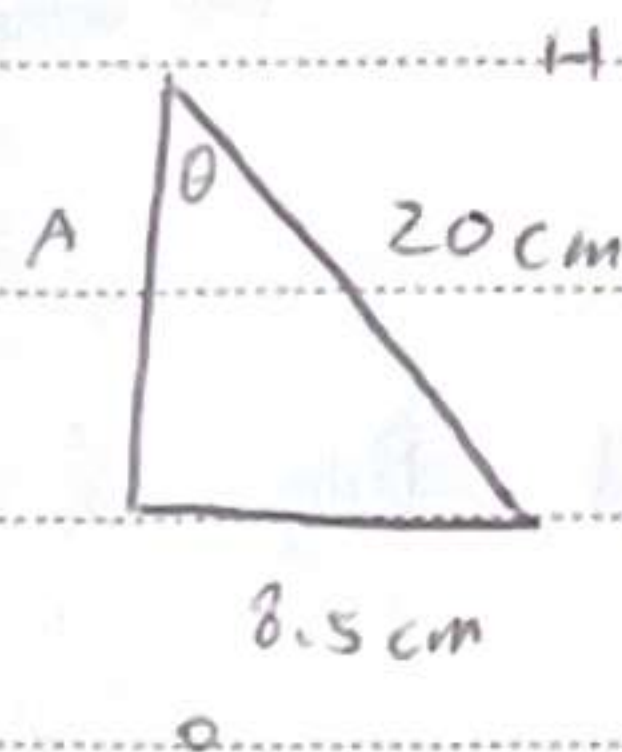
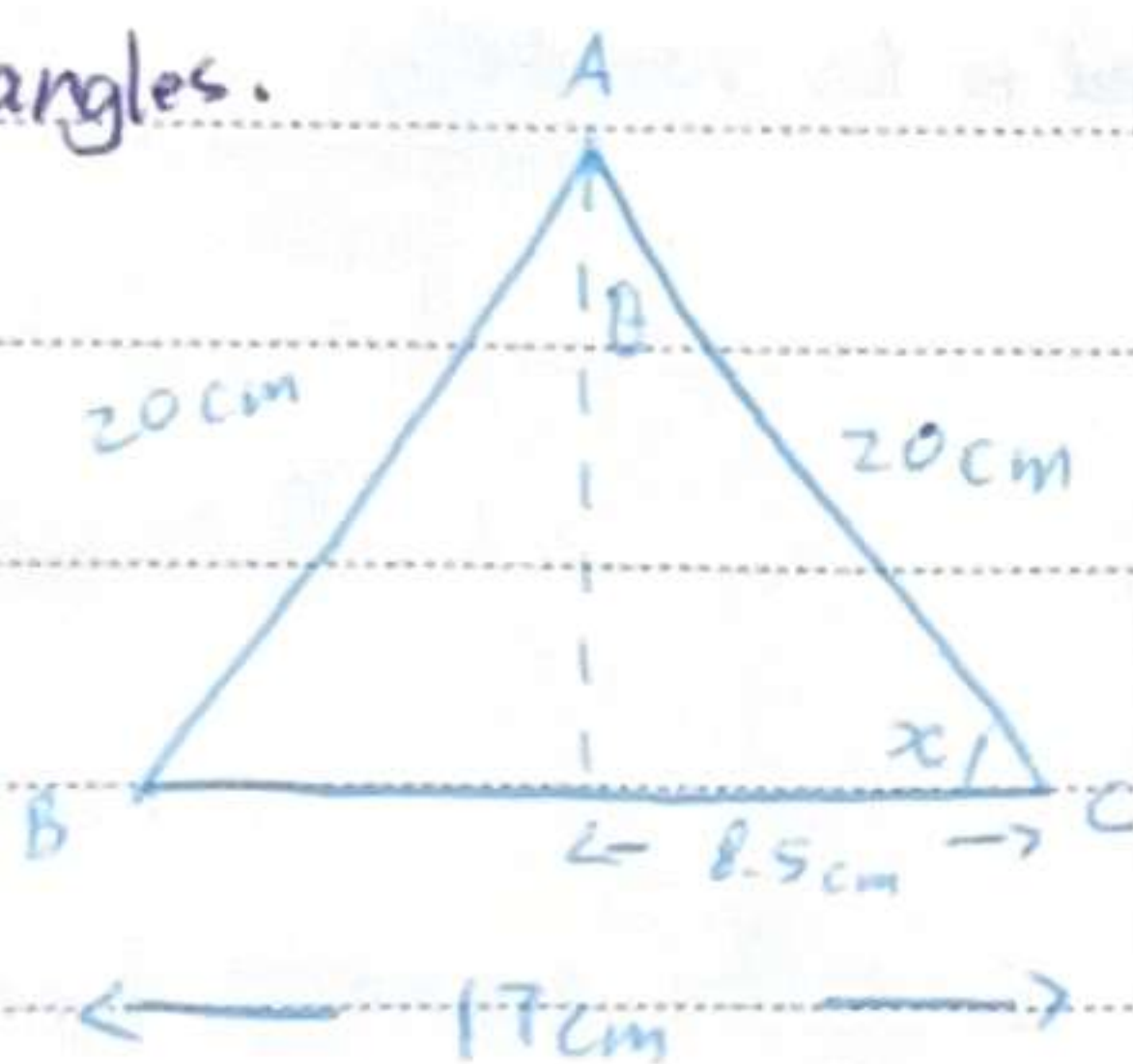
Volume: $72.73 \times 0.15 \text{ m} = 10.91 \text{ m}^3$

- ② A road 400m long goes straight up a slope. If the road rises 50m vertically, what is the angle that the road makes with the horizontal?

$$\theta = \sin^{-1}\left(\frac{50}{400}\right) = 7.11^\circ$$



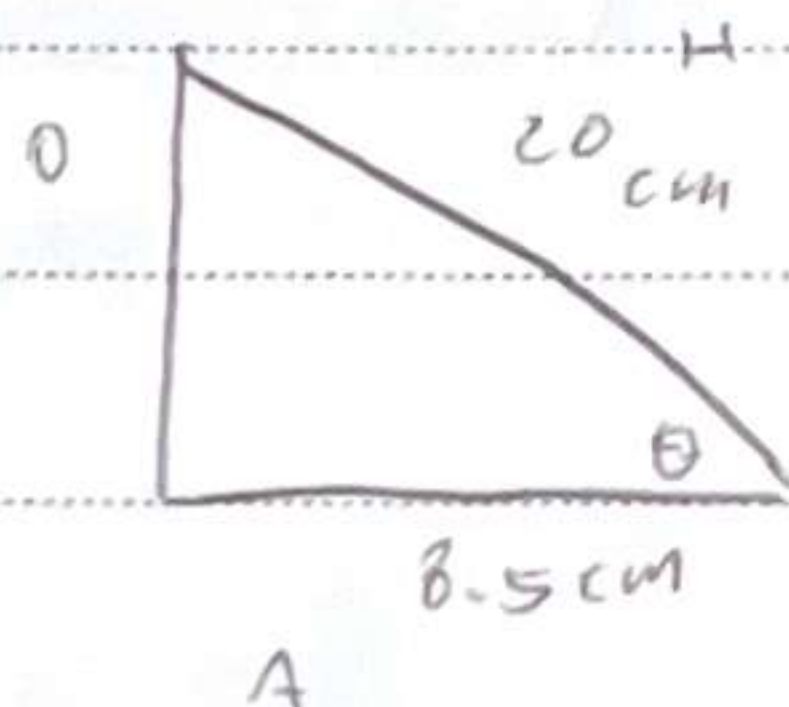
- ③ An isosceles triangle has sides of 17cm, 20cm and 20cm. Find the magnitude of the angles.



顶角: $\text{toDMS}(\sin^{-1}(8.5/20)) \times 2$

$$50^\circ 18' 4.77 \dots$$

$$\approx 50^\circ 18'$$



$x: \text{toDMS}(\cos^{-1}(8.5/20))$

$$64^\circ 50' 57.61 \dots$$

$$\approx 64^\circ 51'$$

Ans: Angles are $64^\circ 51'$, $64^\circ 51'$ and $50^\circ 18'$

3E Complex in polar form.

乘: $z_1 z_2 = r_1 r_2 \text{cis} (\theta_1 + \theta_2)$

除: $\frac{z_1}{z_2} = \frac{r_1}{r_2} \text{cis} (\theta_1 - \theta_2)$

次方: $z^n = r^n \text{cis} (n\theta_1)$

$$\bar{z} = r \text{cis} (-\theta)$$

Questions.

① $\sqrt{7} \text{cis} (-\frac{7\pi}{12}) \times 2 \text{cis} (\frac{5\pi}{12})$

express in the form $r \text{cis} \theta$, $\theta \in (-\pi, \pi]$

$$2\sqrt{7} \text{cis} (-\frac{\pi}{6})$$

④ $w = 2 \text{cis} (-\frac{\pi}{4})$, w^5 in cartesian form

$$w^5: 2^5 \text{cis} (5 \times -\frac{\pi}{4})$$

$$32 \text{cis} (-\frac{5\pi}{4}) + 2\pi$$

$$= 32 \text{cis} (\frac{3\pi}{4})$$

$$\text{SOH: } 32 \times \sin \frac{\pi}{2} = 16\sqrt{2}$$

$$\text{CAH: } 32 \times \cos \frac{\pi}{2} = 16\sqrt{2}$$

A在左边的x所以是负数

$$-16\sqrt{2} + 16\sqrt{2}i$$

② $12 \text{cis} (\frac{5\pi}{6}) \div 4 \text{cis} (\frac{\pi}{3})$ 意見是可能

express in the form $r \text{cis} \theta$, $\theta \in (-\pi, \pi]$

$$3 \text{cis} (\frac{5\pi}{6} - \frac{\pi}{6})$$

$$= 3 \text{cis} (\frac{\pi}{2})$$

③ $z = \sqrt{3} \text{cis} (\frac{3\pi}{4})$, z^3 等于? $\theta \in (-\pi, \pi]$ ⑤ $3\sqrt{5} \text{cis} (-\frac{7\pi}{12}) \div 2\sqrt{10} \text{cis} (\frac{5\pi}{6})$

(i) Polar form

$$z^3 = \sqrt{3}^3 \text{cis} (3 \times \frac{3\pi}{4})$$

$$= 3\sqrt{3} \text{cis} (\frac{9\pi}{4}) \text{ 大于 } 2\pi$$

$$\frac{9\pi}{4} - 2\pi = \frac{\pi}{4}$$

$$= 3\sqrt{3} \text{cis} (\frac{\pi}{4})$$

in the form $r \text{cis} \theta$, $\theta \in (-\pi, \pi]$

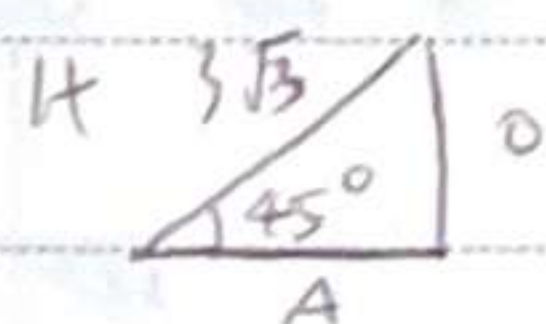
$$\text{CAS: } \frac{3\sqrt{5}}{2\sqrt{10}} = (3\sqrt{5}) \div (2\sqrt{10}) = 3 \frac{\sqrt{5}}{2\sqrt{10}} \text{ 记得约分}$$

$$\text{CAS: } (-\frac{7\pi}{12} - \frac{5\pi}{6}) = (-\frac{7\pi}{12})$$

$$= 3 \frac{\sqrt{2}}{4} \text{cis} (\frac{7\pi}{12})$$

Alg Standard Cplx Rad.

(ii) Cartesian form



$$\text{SOH: } 0 = 3\sqrt{3} \times \sin (\frac{\pi}{4}) [\frac{\sqrt{2}}{2}]$$

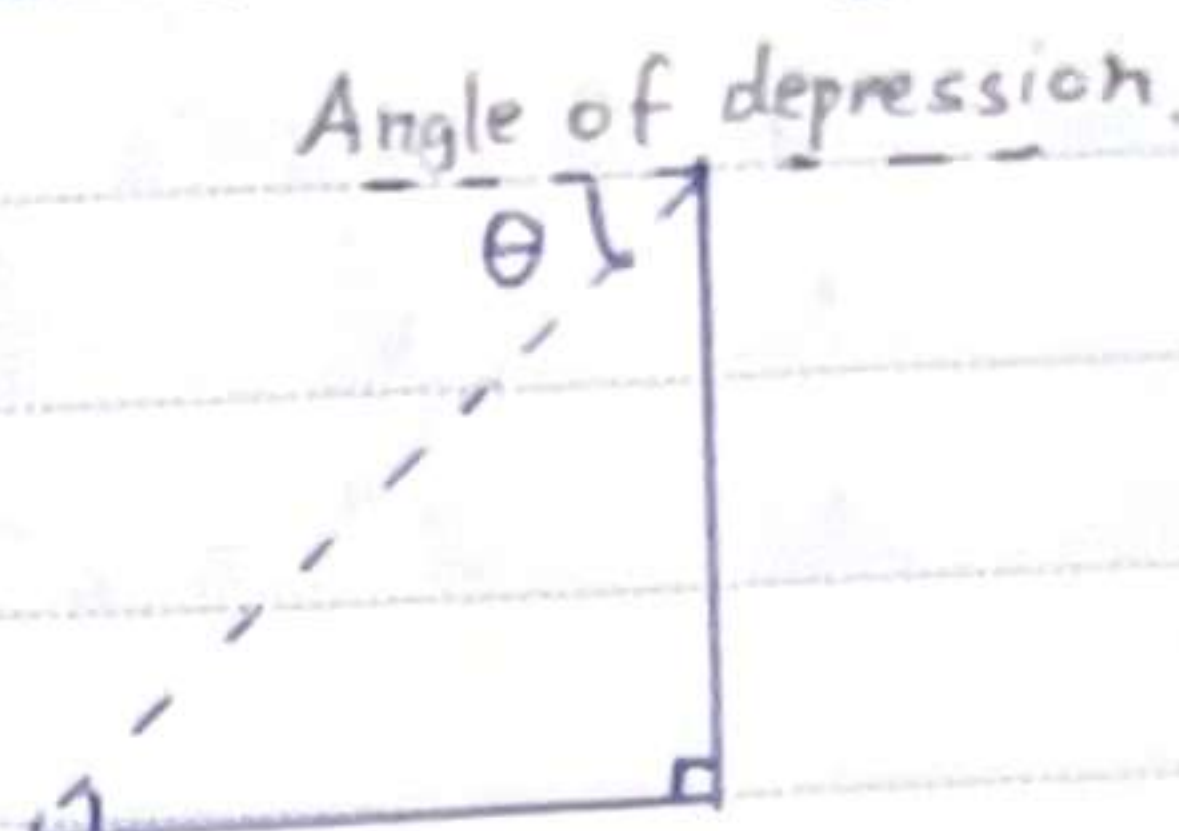
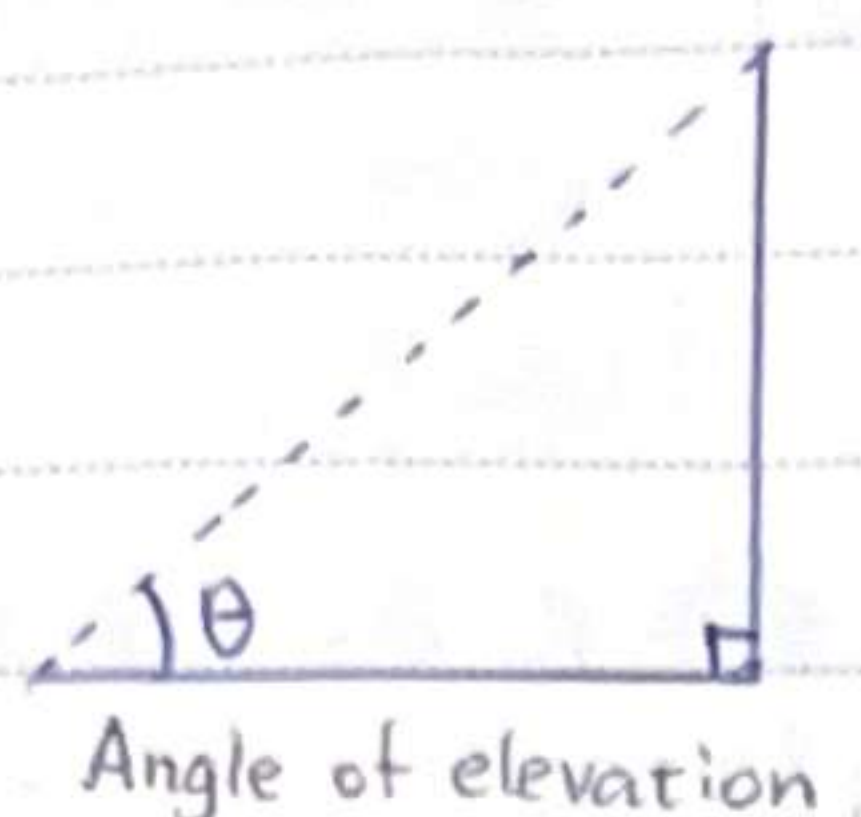
$$= \frac{3\sqrt{6}}{2}$$

$$\text{CAH: } A = 3\sqrt{3} \times \cos (\frac{\pi}{4}) [\frac{\sqrt{2}}{2}]$$

$$= \frac{3\sqrt{6}}{2}$$

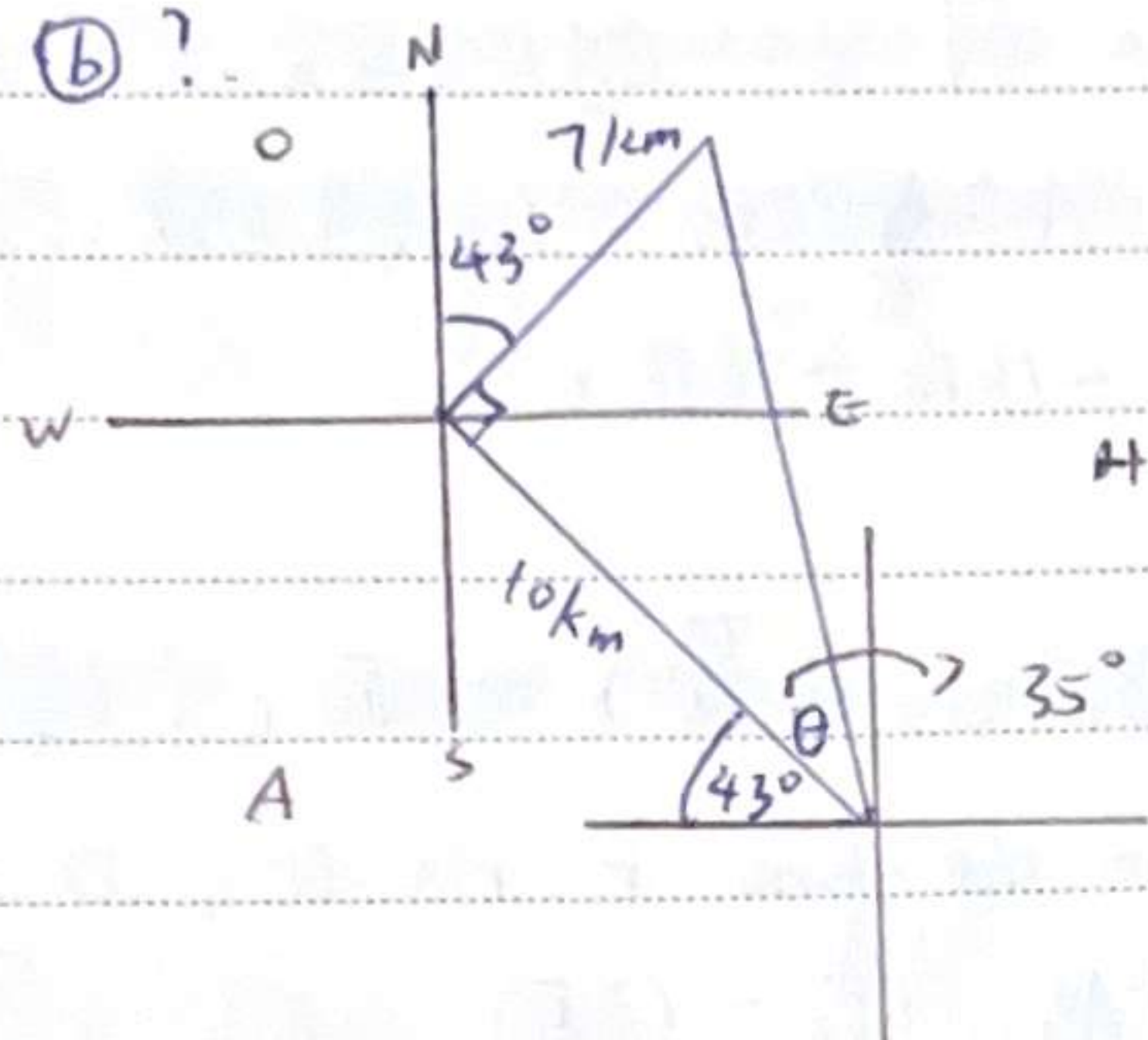
$$z = \frac{3\sqrt{2}}{2} + \frac{3\sqrt{6}}{2}i$$

5.3 Elevation, Depression and Bearings.



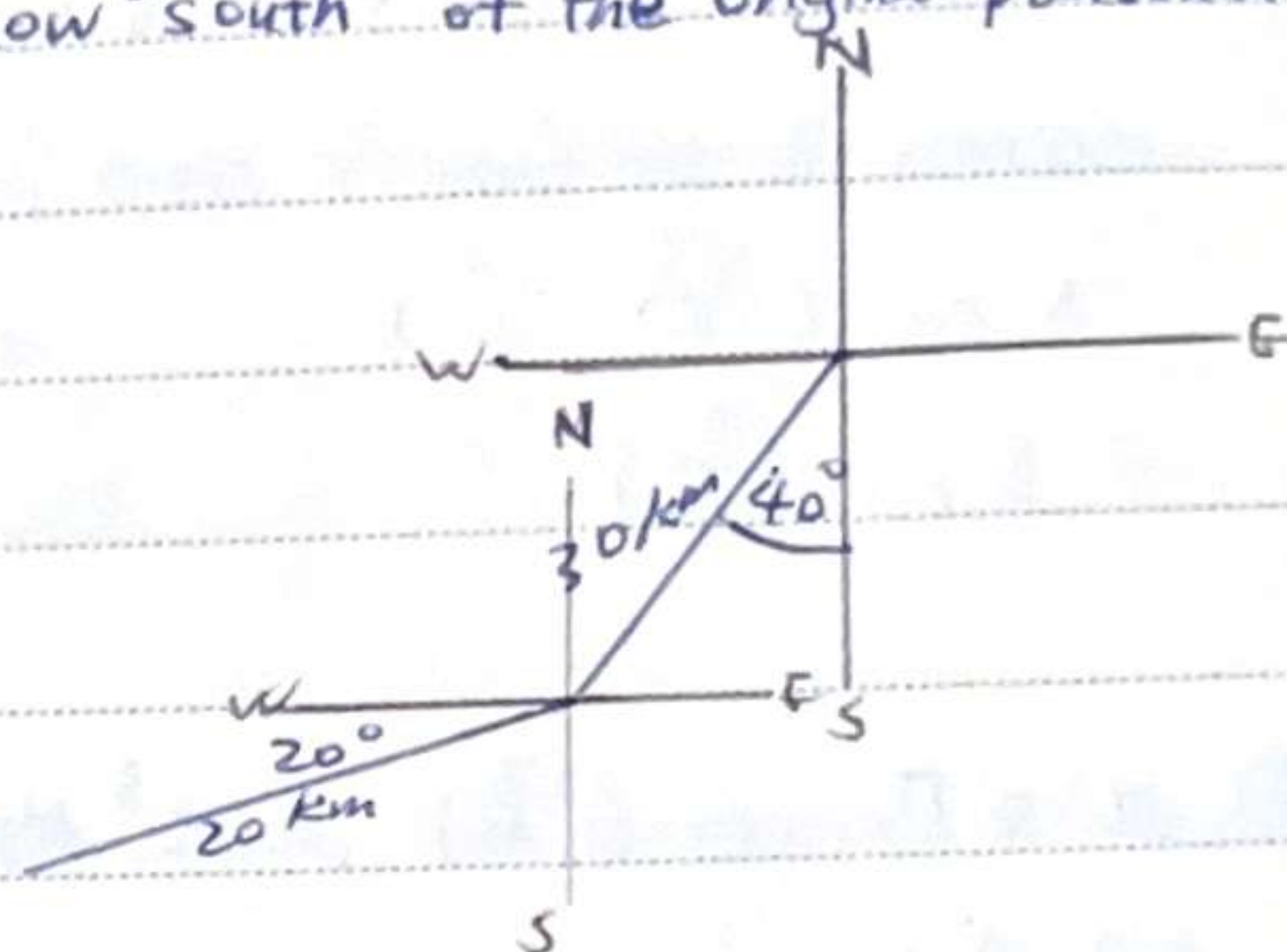
Example.

- ① A walks 7km in the direction $043^\circ T$ ② A ship sails 30km on a bearing of $220^\circ T$
 B walks 10km in the direction $133^\circ T$ then 20km on a bearing of $250^\circ T$
 What is the bearing of the A from the B? (a) Find how far south of the original position it is.



$$\tan \theta = \frac{7}{10} = \tan^{-1}\left(\frac{7}{10}\right) = 35^\circ$$

$$\text{Bearing is } 43 + 35 + 270 = 348^\circ T.$$



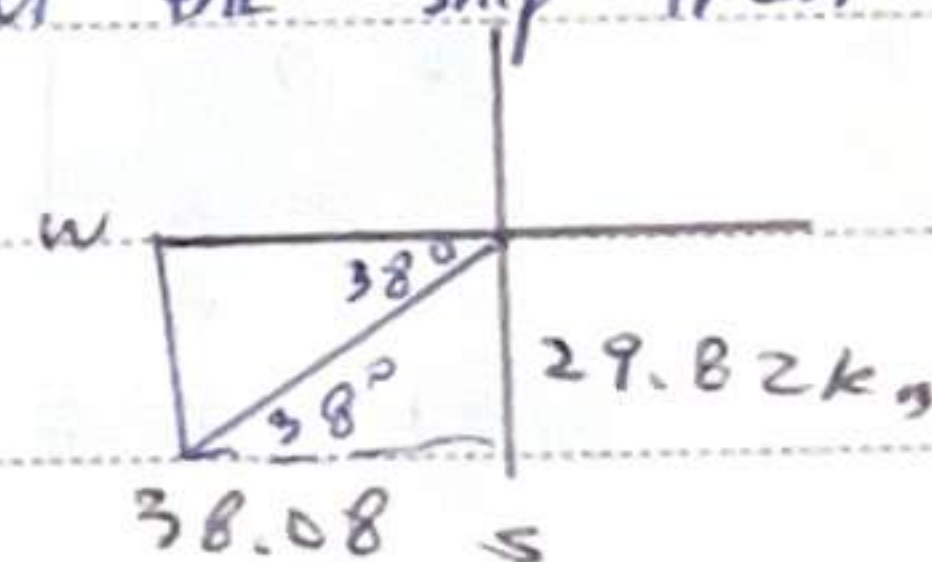
$$30 \times \cos(40) = 22.98 \text{ km}$$

$$20 \times \sin(20) = 6.84 \text{ km}$$

$$6.84 + 22.98 = 29.82 \text{ km}$$

(b) Known the ship is 38.08km west from origin

Find the true bearing of the ship from its original position.

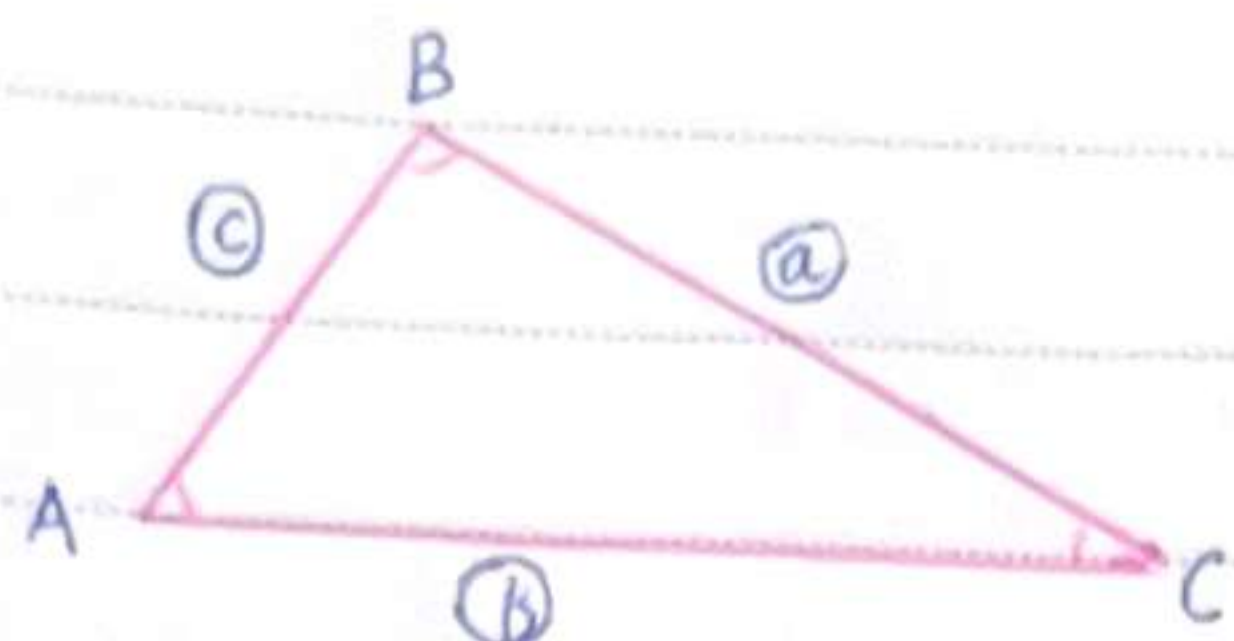


$$\tan^{-1}\left(\frac{29.82}{38.08}\right) = 38^\circ$$

$$270 - 38 = 232^\circ T.$$

5.4

The sine rule



$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$

The sine rule can be used to solve non-right-angled triangles if we are given:

1. two angles and one side length



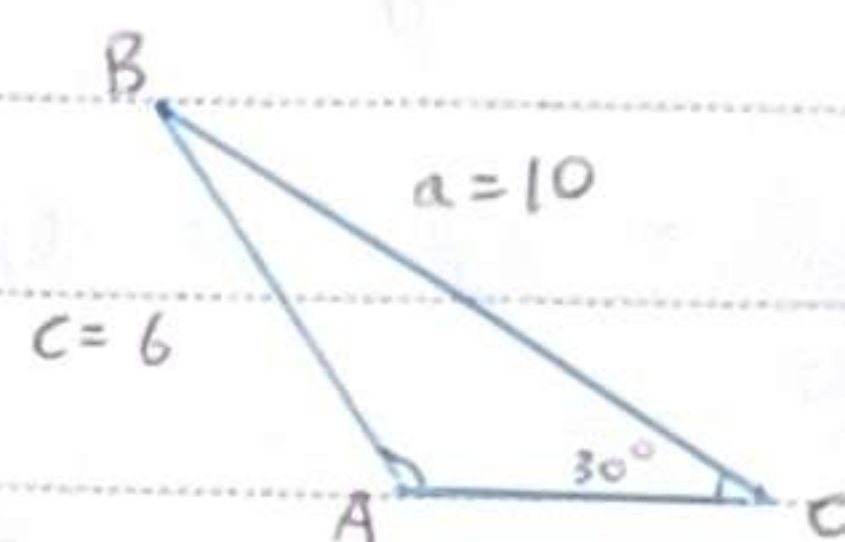
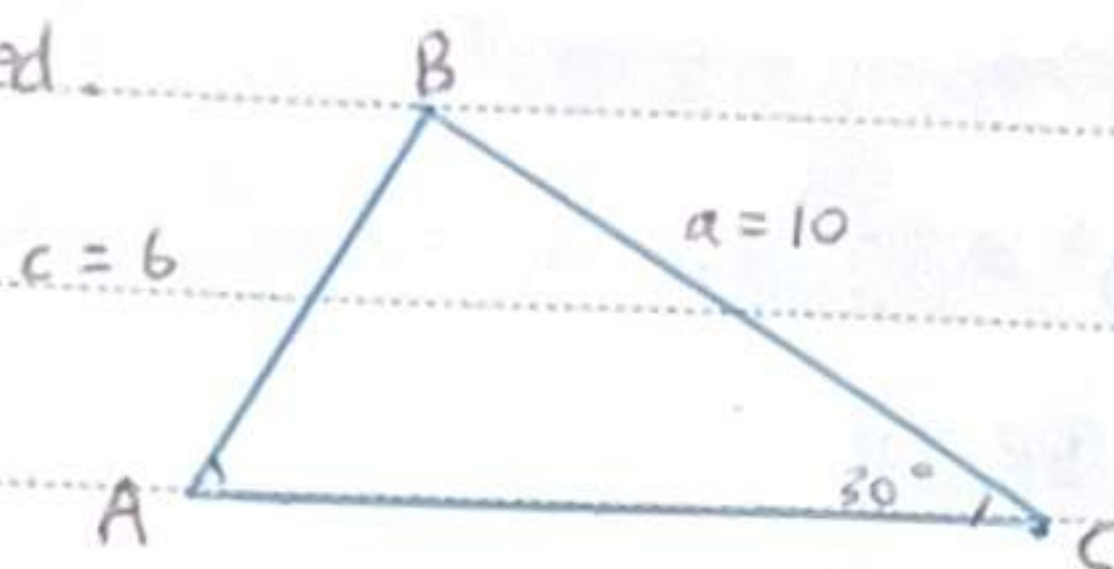
2. two side lengths and an opposite angle



The ambiguous case 模棱两可的情况

The ambiguous case exists if C is an acute angle and $a > c > a \sin(C)$, or if B is an acute angle and $a > b > a \sin(B)$, and so on.

For example 举个例子: if $a = 10$, $c = 6$, $C = 30^\circ$. Two possible triangles could be created.



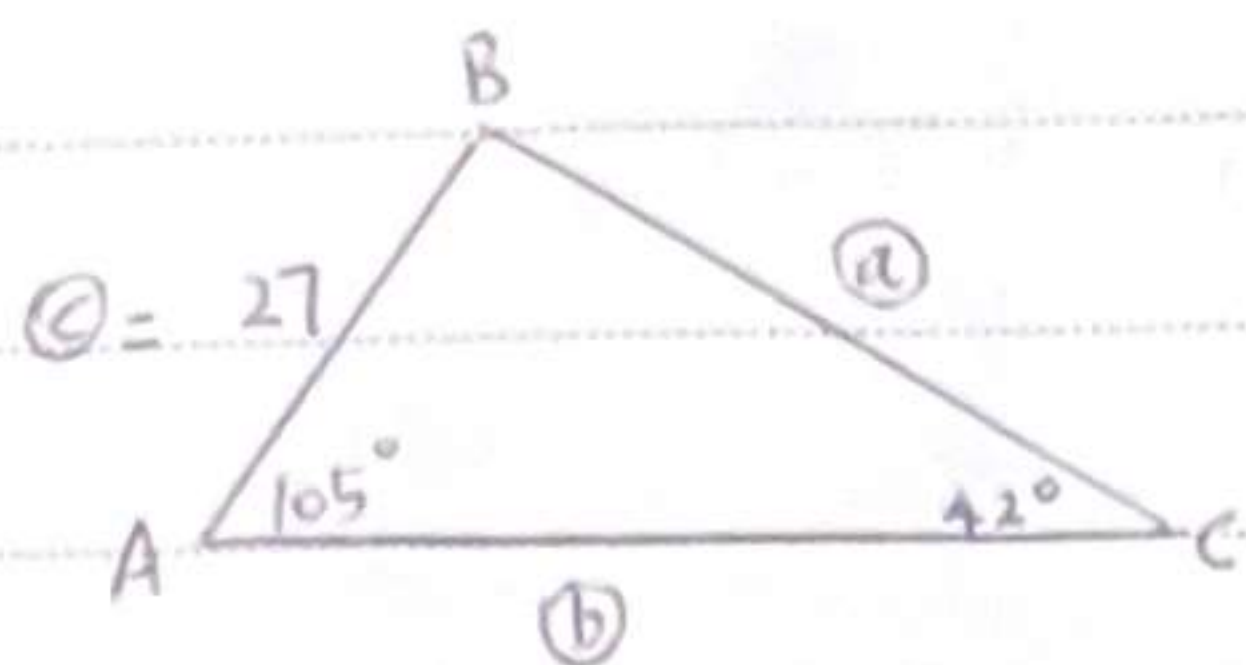
左图的 A is an acute angle, while in 右图 A is an obtuse angle. The two values for A will add to 180° .

[Always test before commencing a question]

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Example of the sine rule

- ① $a = 27$ $C = 42^\circ$ $A = 105^\circ$ Find B , b , c



$$\frac{a}{\sin(105^\circ)} = \frac{27}{\sin(42^\circ)} = \frac{b}{\sin B}$$

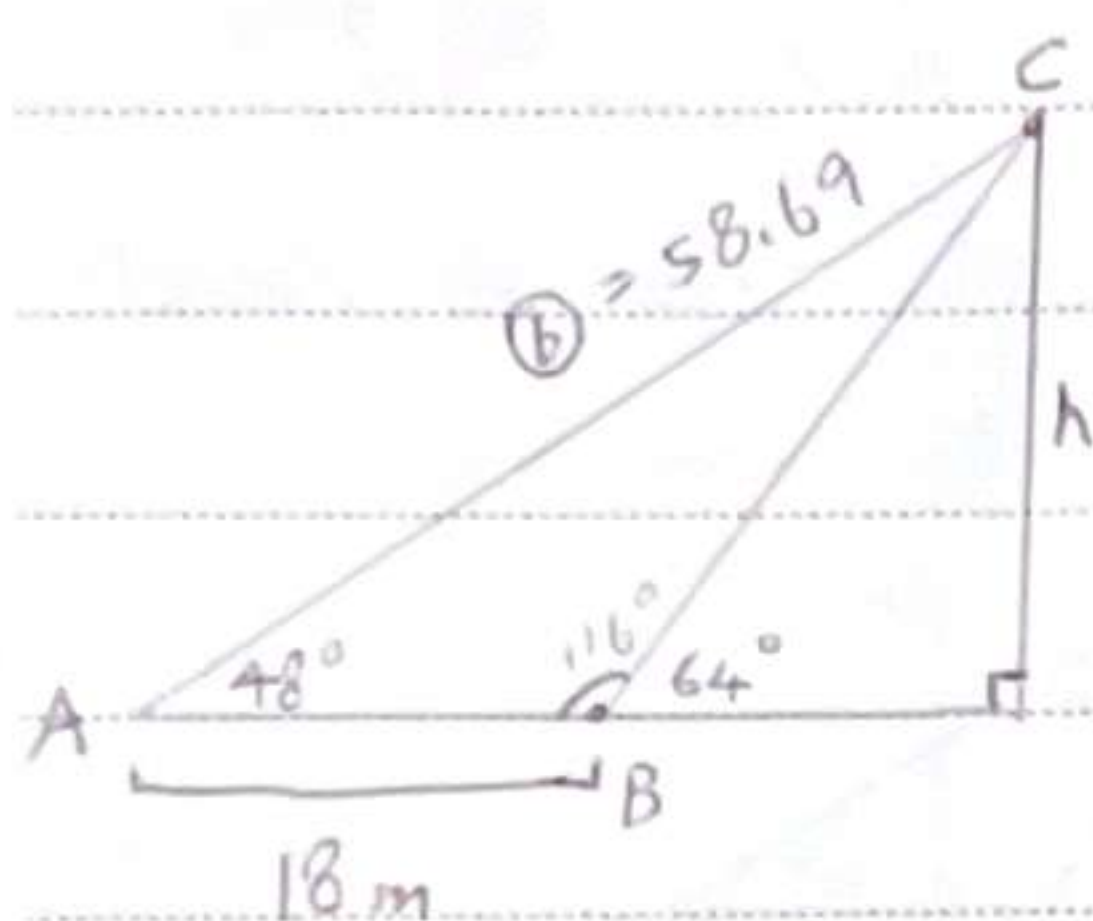
Find a : $(27 \times \sin(105^\circ)) \div \sin(42^\circ) = 38.98 \text{ cm}$

Find B : $180 - 105 - 42 = 33^\circ$

Find b : $27 \times \sin(33^\circ) \div \sin(42^\circ) = 21.98 \text{ cm}$

如果找 angle 就用 $\sin^{-1}$, 用 $\cos$ 要先调 mode

- ② Angle of elevation to the top as 48° , walks 18m closer, angle change to 64° .
How high the building?



Find B : $180 - 64 = 116^\circ$

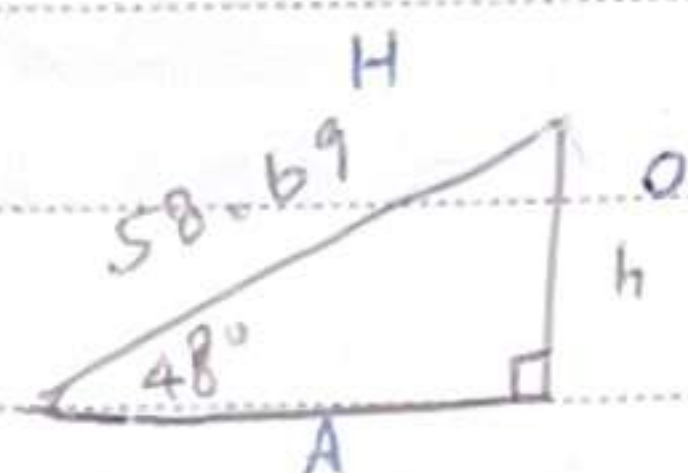
Find C : $180 - 48 - 116 = 16^\circ$

Find C' : $180 - 90 - 64 = 26^\circ$

Find b : $\frac{18}{\sin(16^\circ)} = \frac{b}{\sin(116^\circ)}$

$b = 58.69$

Find h :



$$\sin(\theta) = \frac{O}{H}$$

$$\sin(48^\circ) = \frac{O}{58.69}$$

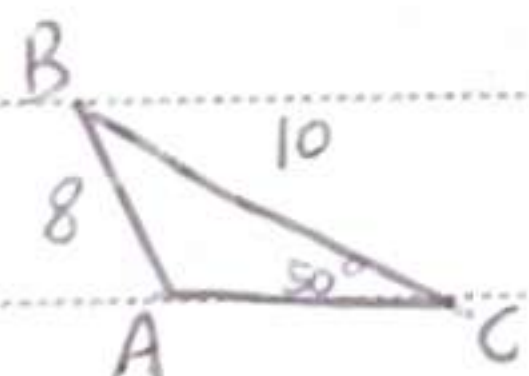
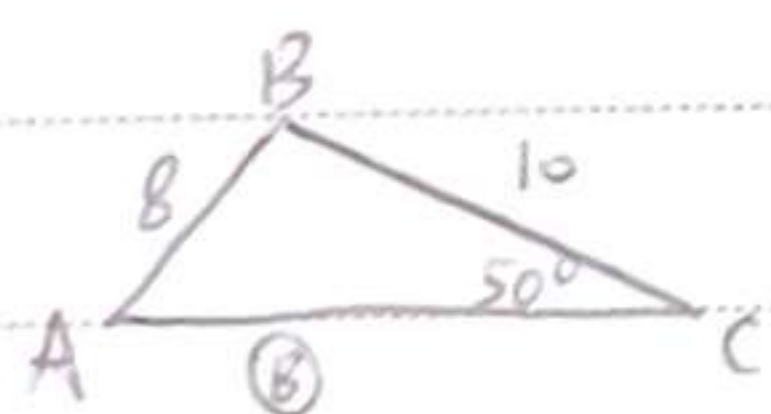
$$O = 43.62$$

Ans

The height of the building is

43.62 m

- ③ $a = 10$, $c = 8$, $C = 50^\circ$ Find two possible value of A and hence b



Find 图1 A : $\frac{8}{\sin 50} = \frac{10}{\sin A} = \sin^{-1}\left(\frac{10 \times \sin 50}{8}\right)$

$$A = 73^\circ 15'$$

Find 图2 A : $180 - 73^\circ 15' = 106^\circ 45'$

Find 图1 b : $B: 180 - 50 - 73^\circ 15' = 57^\circ 45'$

$$\frac{b}{\sin(57^\circ 45')} = \frac{8}{\sin(50)} \quad b = 8.83$$

Find 图2 b : $B: 180 - 50 - 106^\circ 45' = 23^\circ 15'$

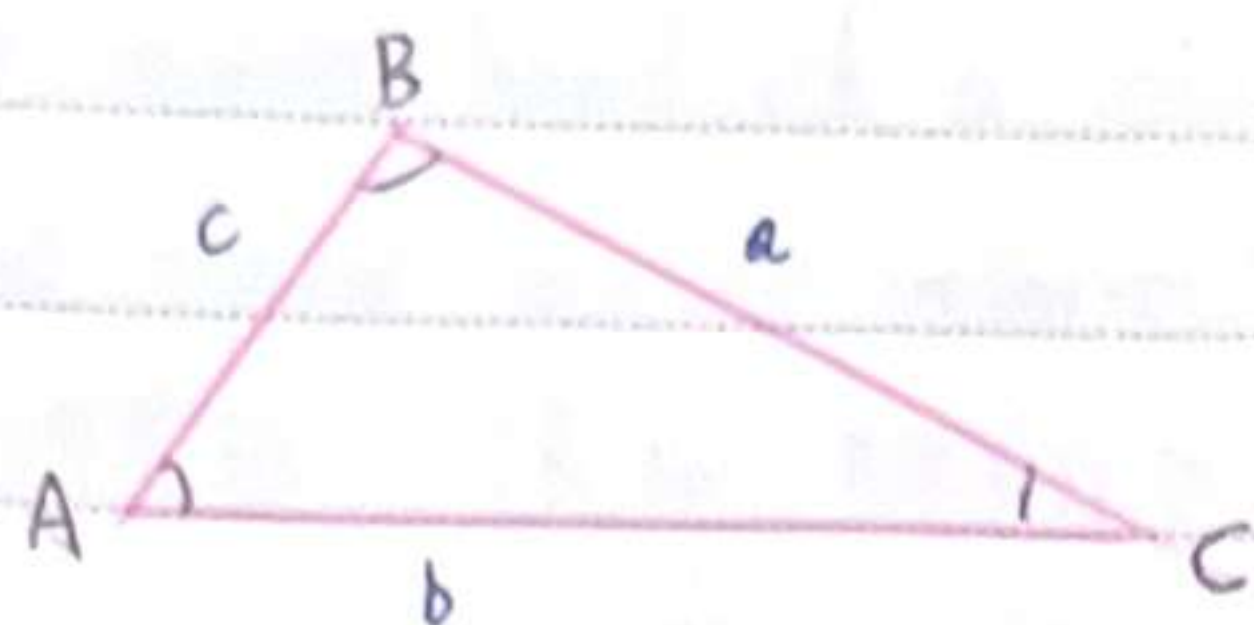
$$\frac{b}{\sin(23^\circ 15')} = \frac{8}{\sin(50)} \quad b = 4.12$$

Ans

$$A = 73^\circ 15' \text{ 和 } 106^\circ 45'$$

$$b = 8.83 \text{ 和 } 4.12$$

5.5 The Cosine Rule



求边长: $a^2 = b^2 + c^2 - 2bc \cos(A)$

$$b^2 = a^2 + c^2 - 2ac \cos(B)$$

$$c^2 = a^2 + b^2 - 2ab \cos(C)$$

已知边长求角度

$$\cos(A) = \frac{b^2 + c^2 - a^2}{2bc}$$

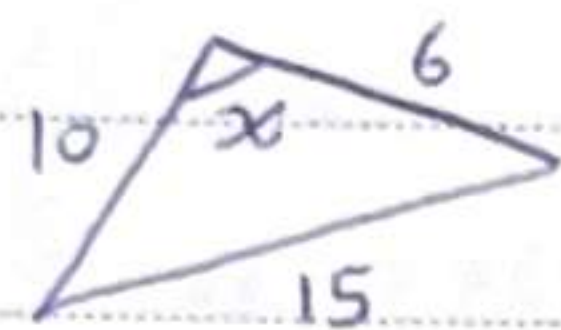
$$\cos(B) = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos(C) = \frac{a^2 + b^2 - c^2}{2ab}$$

- Largest angle is opposite to the longest side
- Smallest angle is opposite to the shortest side.

CAS

solve ($a^2 = b^2 + c^2 - 2bc \cos(A)$)



求 x

key keyboard / math (3) 第三行

CAS: solve ($15^2 = 10^2 + 6^2 - 2 \times 10 \times 6 \times \cos(x)$, x) | $0 \leq x \leq 180^\circ$

$$x = 137.8735$$

5.6 Arcs, sectors and segments

角度 $^{\circ}$ to radian measure (in terms of π):

$$135^{\circ} = 135^{\circ} \times \frac{\pi}{180^{\circ}} = \frac{3\pi}{4}$$

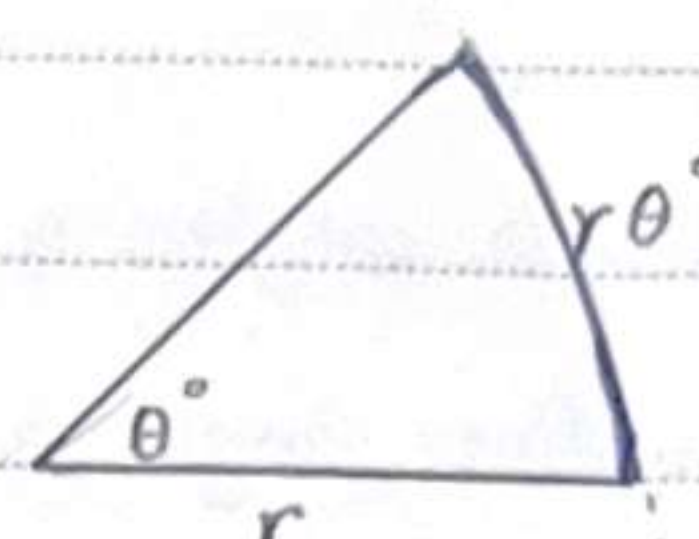
Radian measure to degrees:

$$\frac{4\pi}{5} = \frac{4\pi}{5} \times \frac{180^{\circ}}{\pi} = \frac{720^{\circ}}{5} = 144^{\circ}$$

Arcs 弧长 (圆的某一段边长)

$$L = r\theta$$

$\downarrow$ Arc length $\downarrow$ radius $\downarrow$ angle in radians

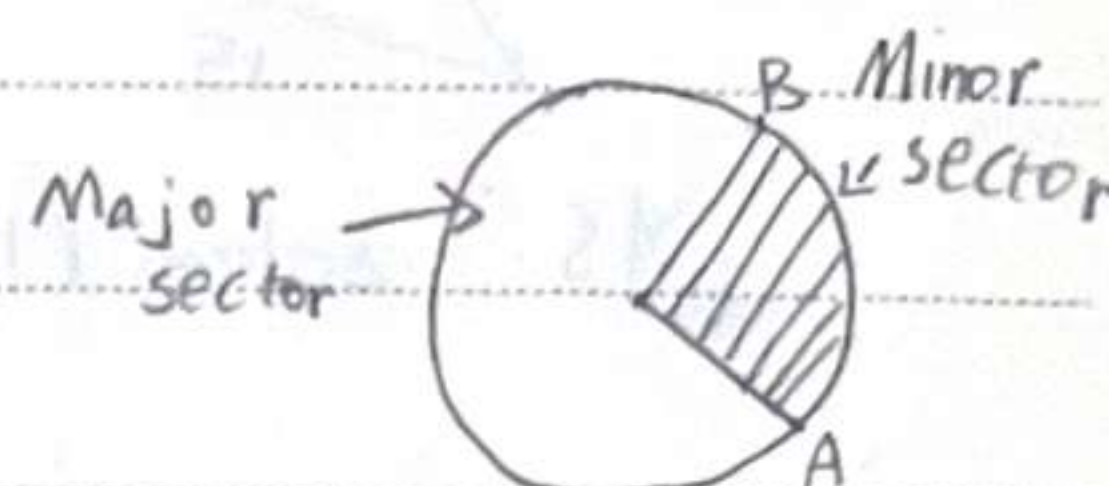
Example: Given $\theta = 75^{\circ}$, $r = 8$, find L .

① Convert θ to radian: $75^{\circ} \times \frac{\pi}{180} = 1.3090$

② 套公式: $8 \times 1.3090 = 10.47 \text{ cm}$

Sectors 扇形 (圆的某一部分) $\rightarrow$ 与弧长成正比

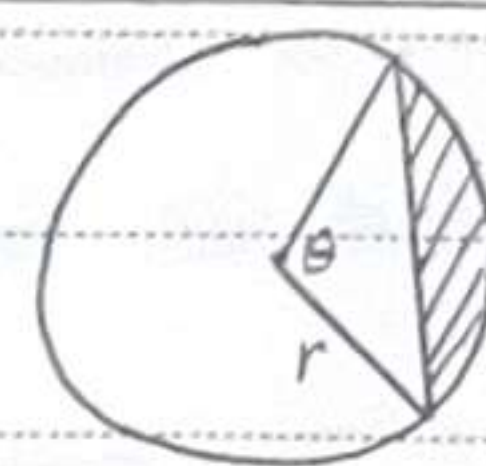
$$\text{Area of a sector} = \frac{1}{2} r^2 \theta$$



Segments 线段 (弧长和弦所界定的扇形部分)

$$\text{Area of a segment} = \frac{1}{2} r^2 (\theta - \sin(\theta^{\circ}))$$

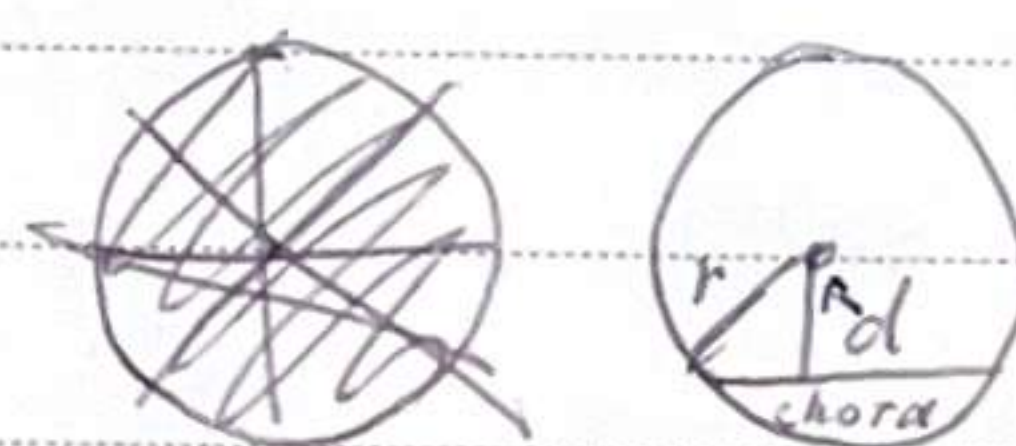
$\downarrow$ in radian $\downarrow$ in degree



* ~~Chord~~ Chord length = $2 \times \sqrt{r^2 - d^2}$

or = $2 \times r \times \sin(c/2)$

c is the angle subtended at the center by the chord



* radian mode

Question?

- ① The minute hand of a clock is 35 cm long. How far does the tip of the hand travel in 20 minutes?

整理思路: 一个表 360° 共分为12份, 每份代表5分钟, $20 \div 5 = 4$, 所以 20 minutes 代表4份。

$\frac{4}{12} = \frac{1}{3}$, 20分钟占 360° 的 $\frac{1}{3}$ which is $360 \times \frac{1}{3} = 120^\circ$. 20分钟代表 120°

Degree 转 Radian: $120^\circ \times \frac{\pi}{180} = 2.0944$

求 Arc: $L = r\theta$

$$= 35 \times 2.0944$$

$$\approx 73.3 \text{ cm}$$

- ② Find the angle subtended by a 20 cm arc in a circle of radius 75 cm:

(a) in radians

$$L = r\theta$$

$$= 20 \div 75$$

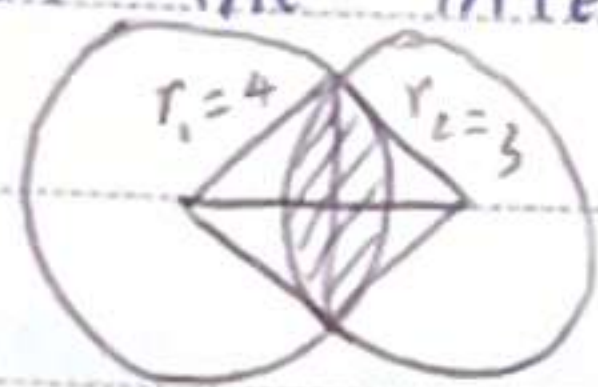
$$= \frac{4}{15} \text{ or } 0.2667^\circ$$

(b) in degrees

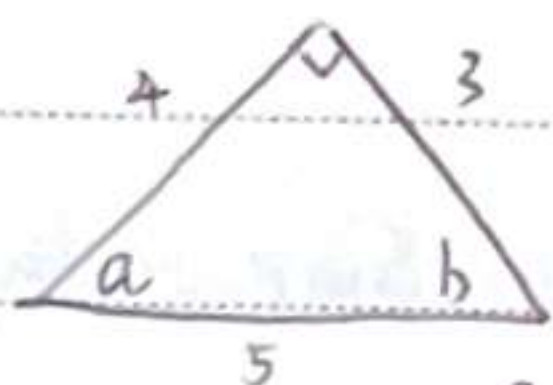
$$\frac{4}{15} \times \frac{180}{\pi} = \frac{48^\circ}{\pi}$$

$$\text{CAS: to DMS: } \frac{48^\circ}{\pi} = 15^\circ 17'$$

- ③ Two circles of radii 3 cm and 4 cm have their centres 5 cm apart. Find the area of the intersection of the two circles.



前提摘要: 3, 4 and 5 form a Pythagorean triad, they form a 直角 triangle.



$$\text{求 } a: \sin(a) = \frac{3}{5}$$

$$\sin^{-1}(0.6) = 36.87^\circ \times \frac{\pi}{180} = 0.6435^\circ$$

$$\text{得 } 2a: 2a = 73.74^\circ \text{ or } 1.287^\circ$$

$$\text{求 } b: \sin(b) = \frac{4}{5}$$

$$\sin^{-1}(0.8) = 53.13^\circ \times \frac{\pi}{180} = 0.9273^\circ$$

$$\text{得 } 2b: 2b = 106.26^\circ \text{ or } 1.8546^\circ$$

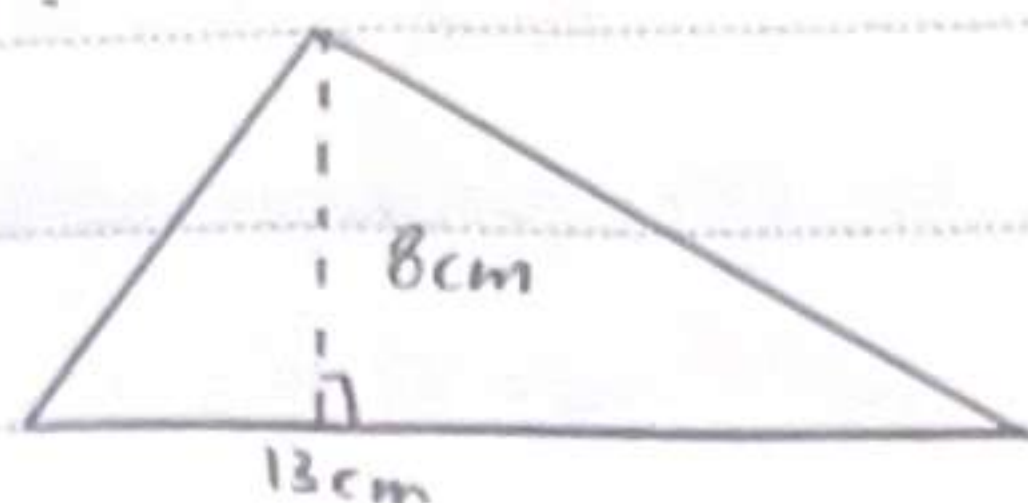
$$\frac{1}{2} r^2 (\theta - \sin(\theta))$$

$$A \text{ of two segment} = \frac{1}{2} \times 16 \times (1.287 - \sin(73.74^\circ)) + \frac{1}{2} \times 9 \times (1.8546 - \sin(106.26^\circ))$$

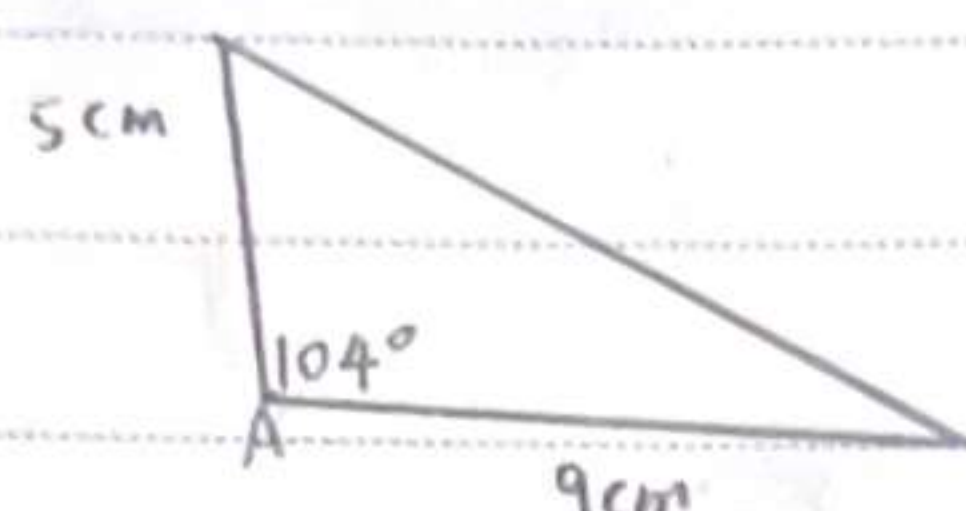
$$\approx 6.64 \text{ cm}^2$$

Area of a triangle

Given:

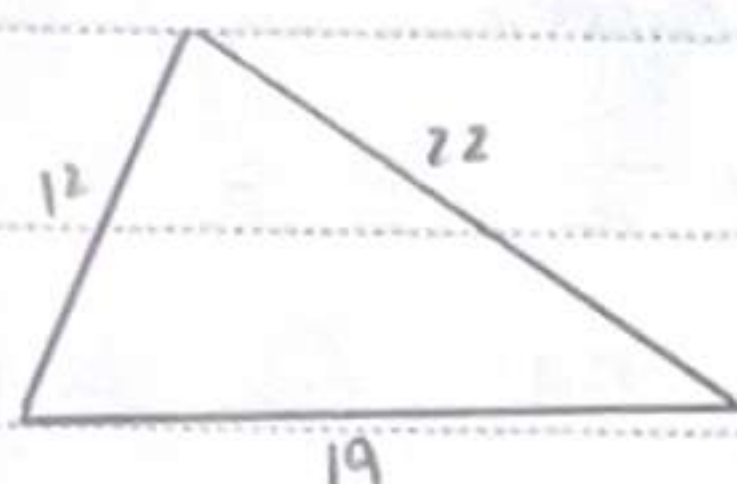


$$\text{Area} = \frac{1}{2} b h$$



$$\text{Area} = \frac{1}{2} b c \sin(A)$$

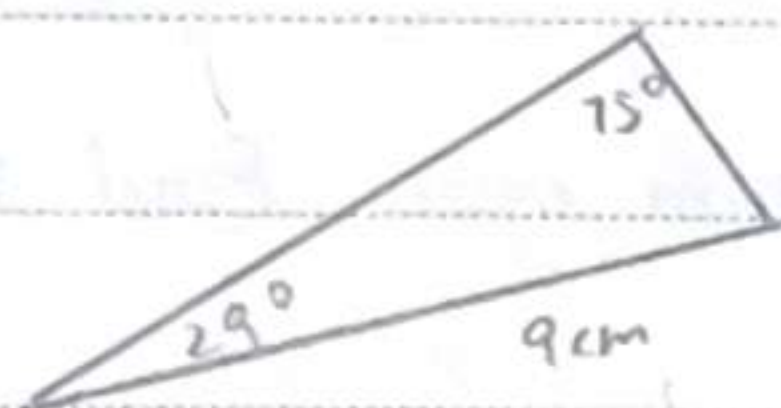
- 必须对着 bc



Heron's formula:

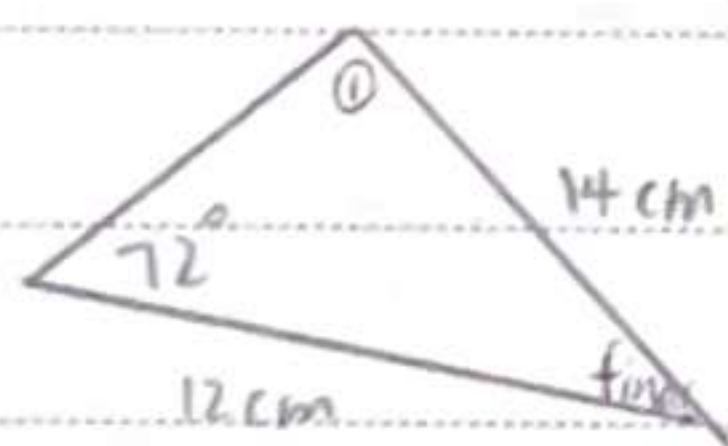
$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$s = \frac{a+b+c}{2}$$



use Sine rule to find the side length

$$\text{Area} = \frac{1}{2} b c \sin(A)$$



① use Sine rule to find the angle

② Determine the final angle

③ $\text{Area} = \frac{1}{2} b c \sin(A)$

④ Check if the ambiguous case is exist

4.2.

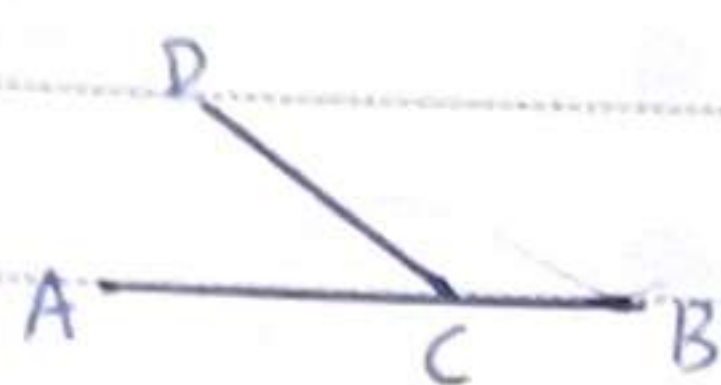
Basic Geometry



$$A + B = 90^\circ$$

Complementary

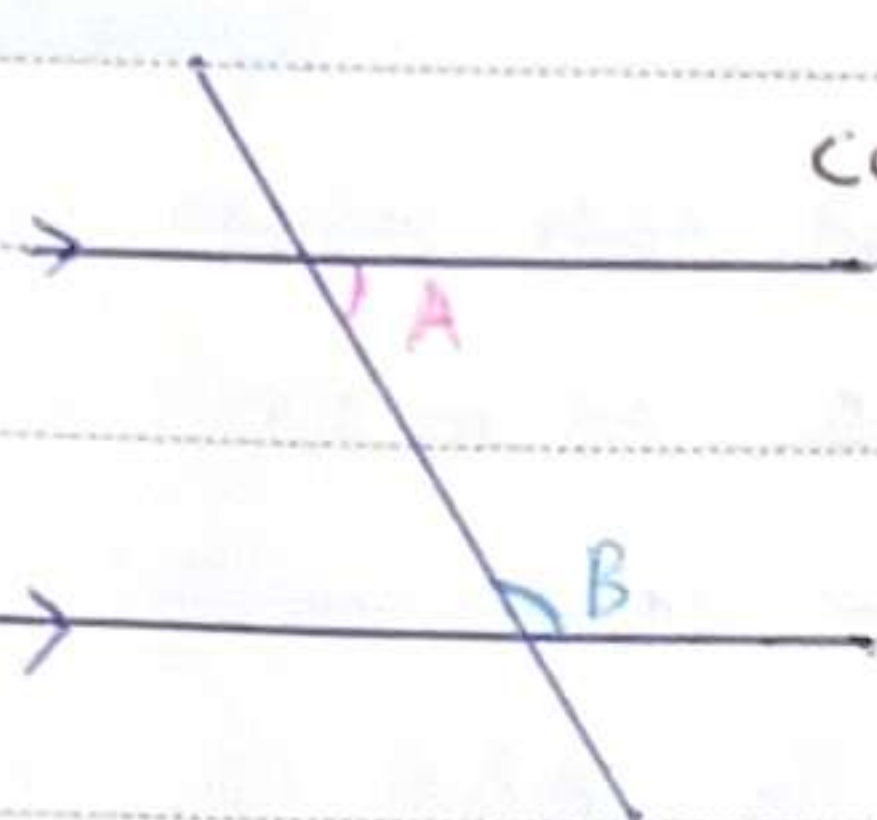
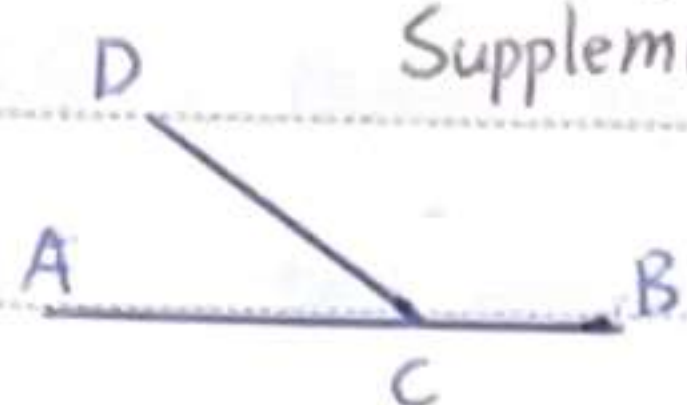
补充



$$ACD + BCD = 180^\circ$$

Supplementary

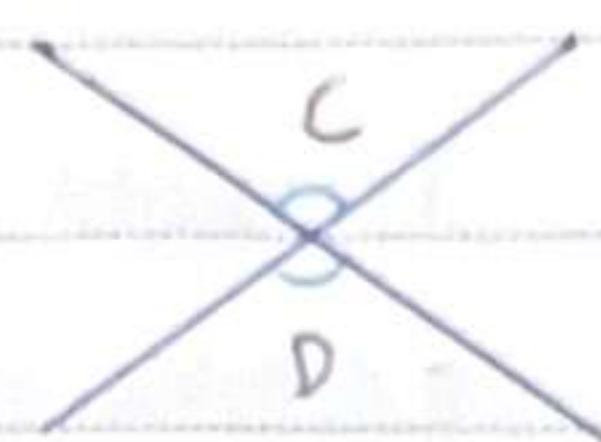
补充



$$A + B = 180^\circ$$

Co-interior angles

内角



$$C = D$$

Vertically opposite angles

垂直相反角



Corresponding angles

对应角



$$A = B$$

$$C = D$$

transversal

横线的



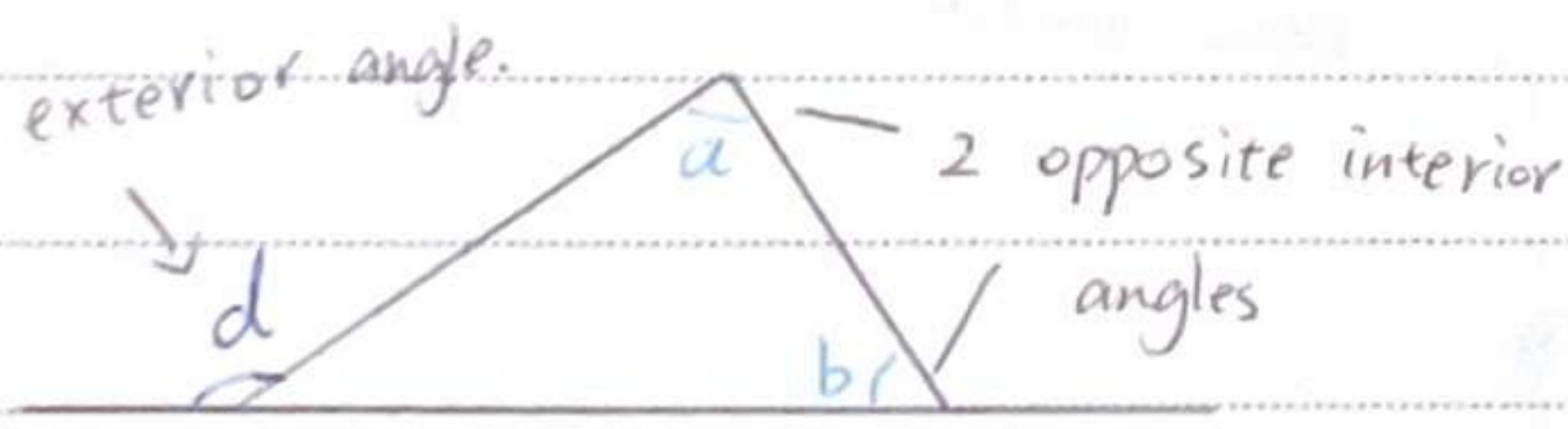
$$C = D$$

Alternate angles

交替角

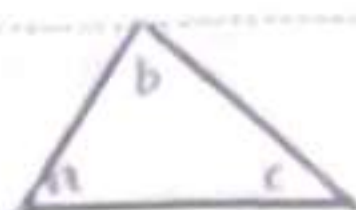
Exterior Angle Theorem

- the exterior angle of a triangle is equal to the sum of the two opposite interior angles.



$$a + b = d$$

Polygons 多边形



Triangle

$$a + b + c = 180^\circ$$

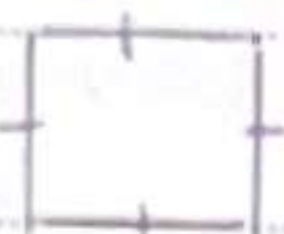
求总内角

$$\text{Sum of interior angles} = (n-2) \times 180^\circ$$

↓
number of sides



Rectangle



Square



Rhombus

菱形
(四边相等)

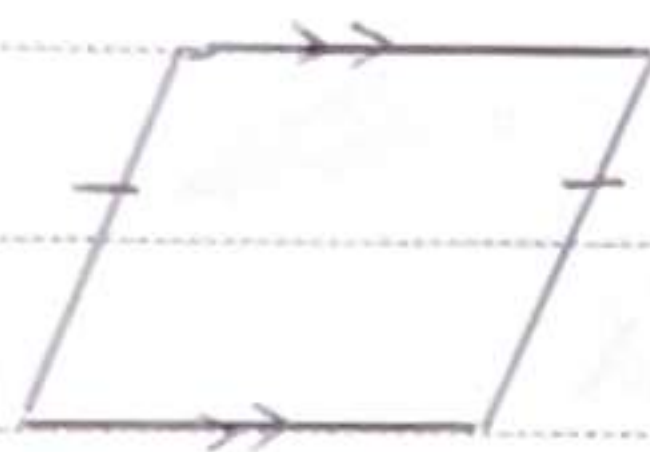
Proof a shape is Rhombus, four sides equal



Kite

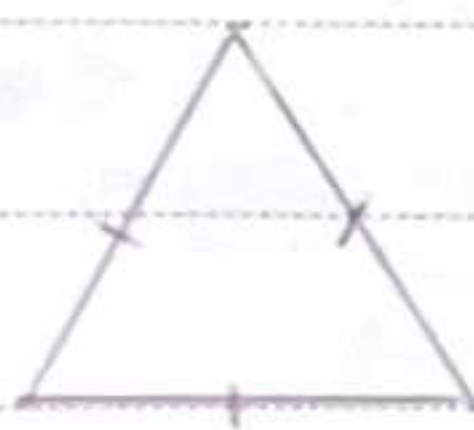
(相邻相等)

风筝

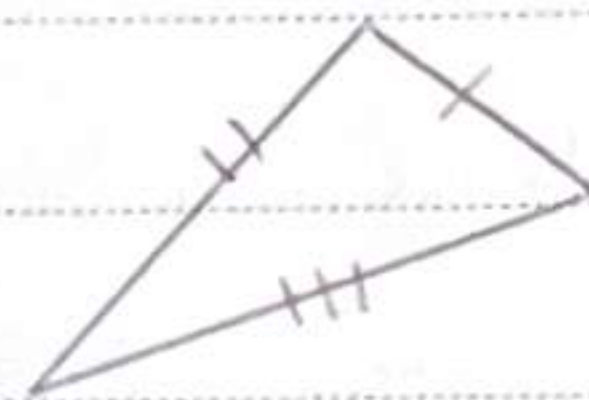


Parallelogram

平行四边形

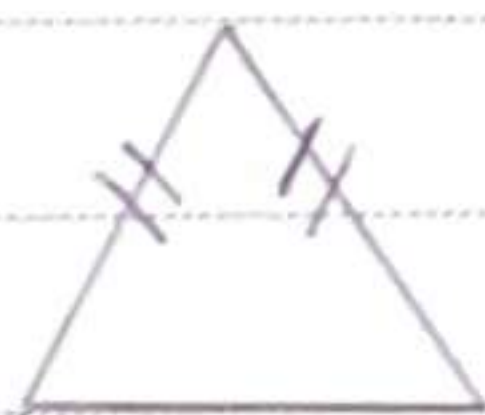


Equilateral 等边三角形

~~Equilateral~~ triangle~~等边三角形~~

Scalene triangle

不等边三角形



~~Equilateral~~ Isosceles 等腰三角形
triangle

4.3

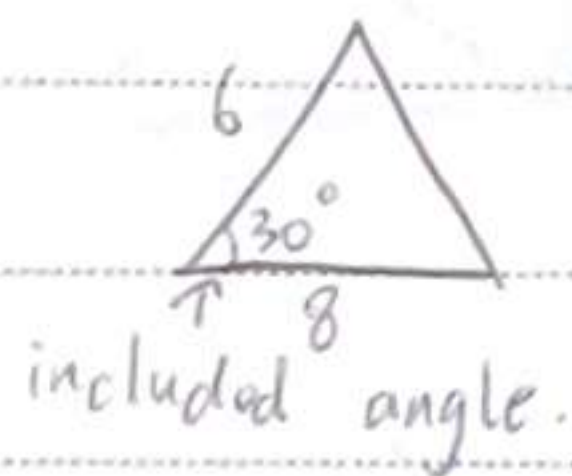
^{同余性} Congruence & ^{相似性} Similarity

o Congruence

- shapes that have the exact same shape and size
- Written as $\triangle ABC \cong \triangle DEF$
- Four tests to prove:
 - ① SSS : three equal sides
 - ② SAS : two equal sides and the included angle
 - ③ ASA : two equal angles and the included side
 - ④ RHS : a right-angle, equal hypotenuse and another equal side

o Similarity

- shapes that have the same shape but different size.
- Written as $\triangle ABC \sim \triangle DEF$
- Three tests to prove:
 - ① AAA: all the corresponding angles are equal.
 - ② SSS : the ratio between the corresponding side lengths are equal
 - ③ SAS : two of their corresponding ^{sides} ~~angles~~ are in the same ratio.
the included angle is the same in both triangles

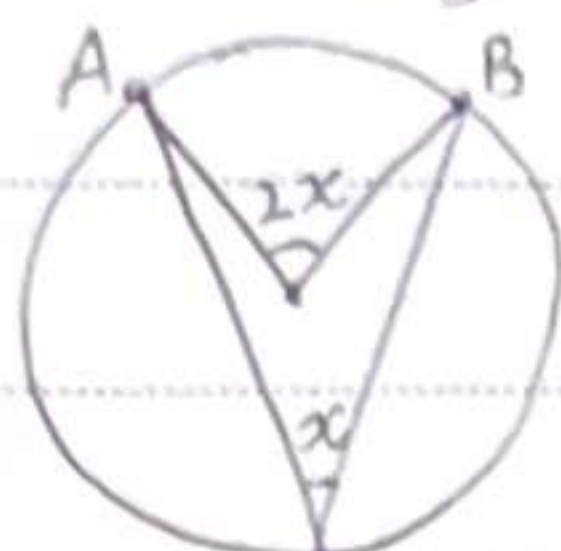


EX 4.6

Circle geometry

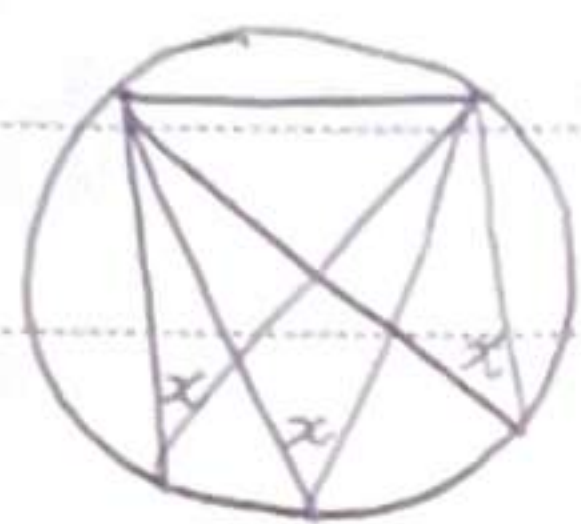
○ Circle theorem 1

- the angle subtended by the chord at the centre is twice the value of the angle subtended by the same chord at the circumference.



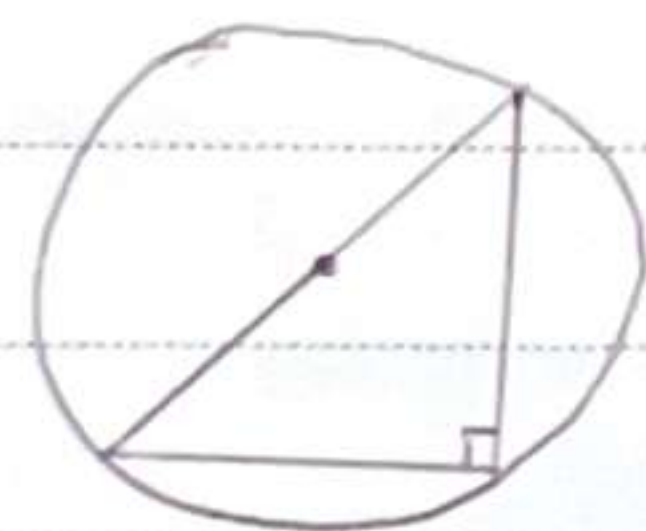
○ Circle theorem 2

- angles subtended at the circumference, in the same segment, by the same chord are equal.



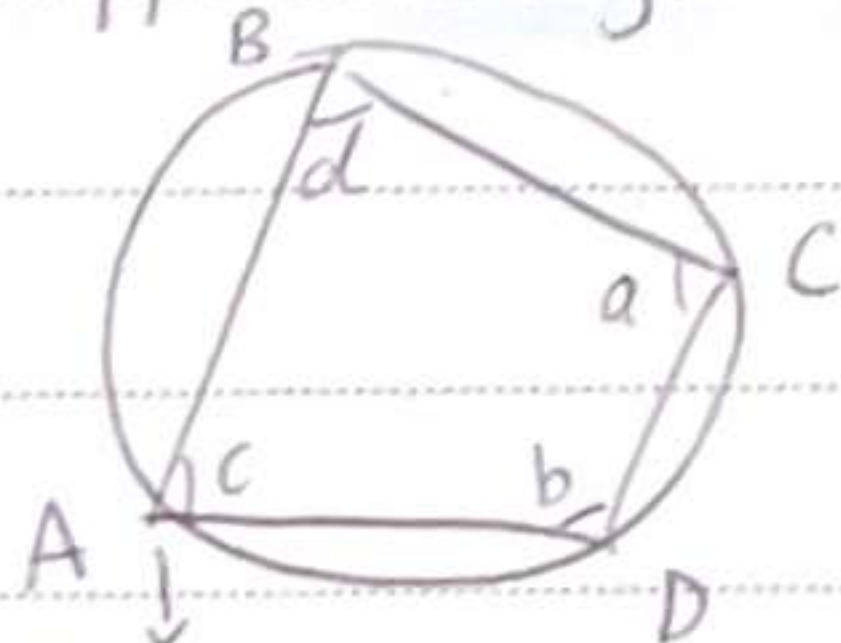
○ Circle theorem 3

- the angle subtended by a diameter is a right angle (90°)



○ Circle theorem 4

- Opposite angles in a cyclic quadrilateral add up to 180 degrees.



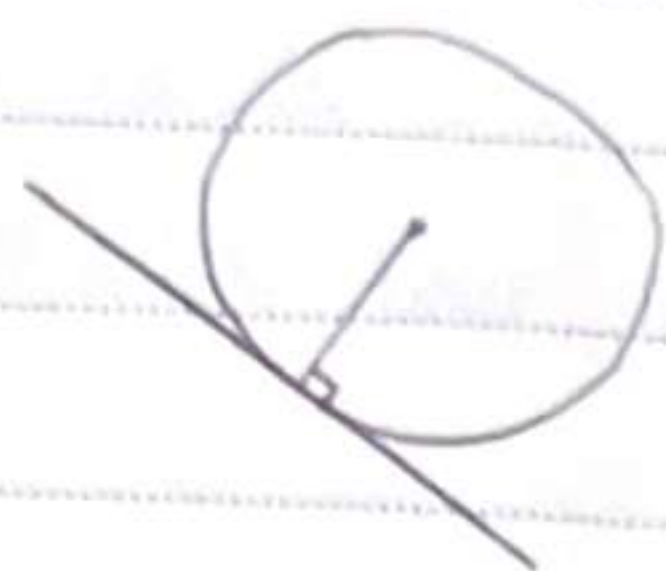
$$a + c = 180^\circ$$

$$b + d = 180^\circ$$

要碰到圓。

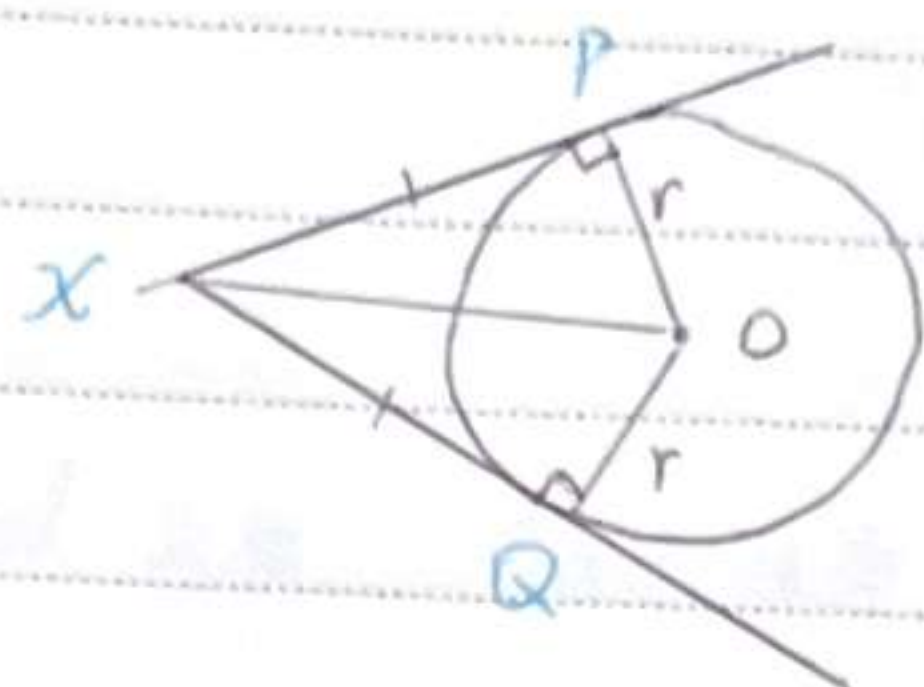
○ Circle Theorem 5 (tangent is perpendicular to radius)

- radius 和 tangent 形成的角永远是 90°



○ Circle Theorem 6 Two tangent from the same point.

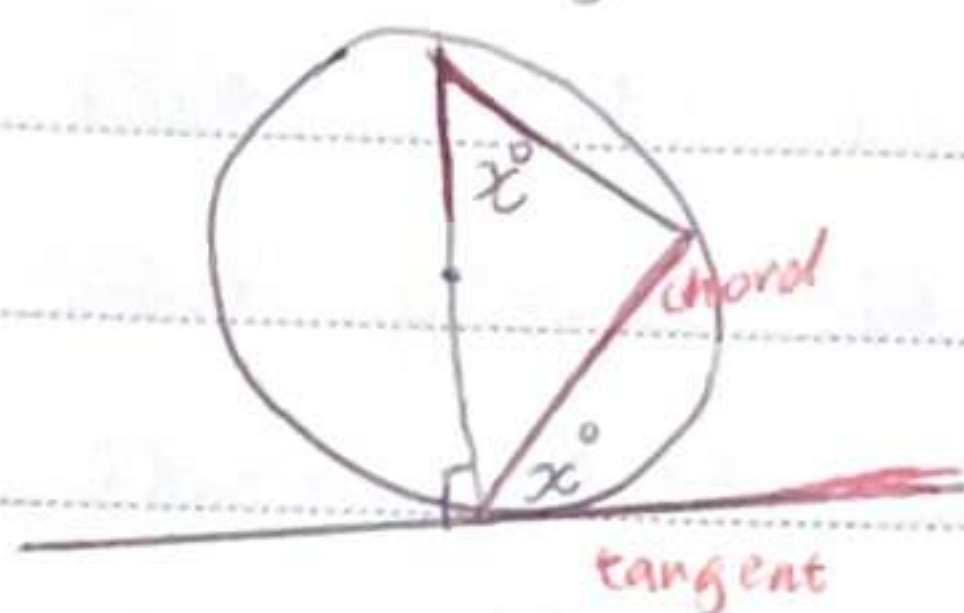
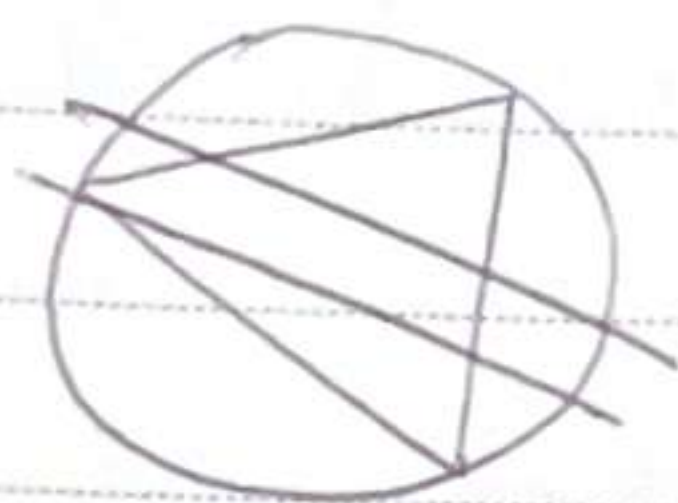
- The two tangents drawn from an external point to a circle are the same length



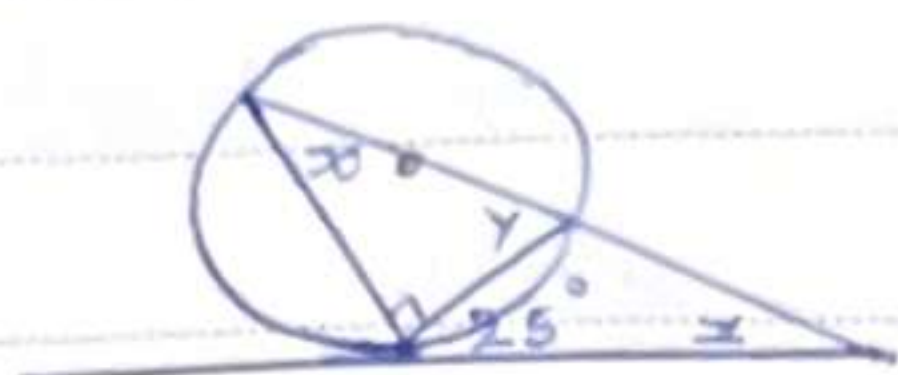
Proof: $\triangle XPO$ is congruent to $\triangle XQO$ using RHS test
therefore $XP = XQ$

○ The alternate segment theorem

- the angle between a tangent and a chord drawn from the point of contact is equal to any angle in the alternate segment



4.1 tangents chords and circles Example question

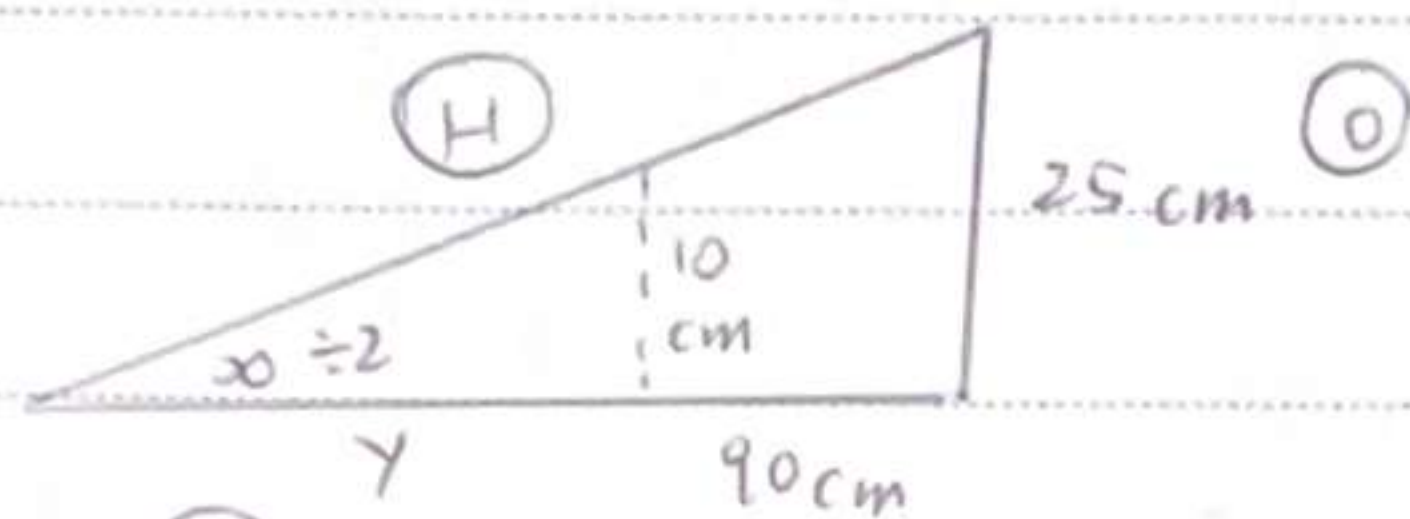
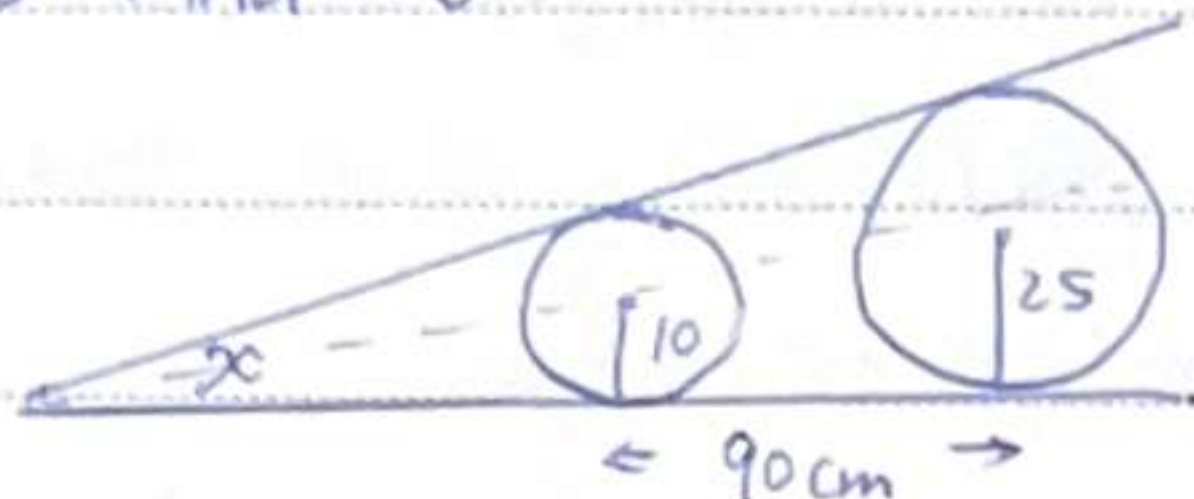
① Find x, y, z 

$$x = 25^\circ \text{ (alternate segment)}$$

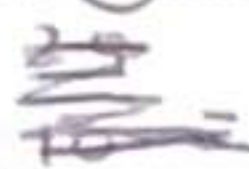
$$y = 180 - 25 - 90 = 65^\circ \text{ (circle + 3)}$$

$$z = y - 25^\circ$$

$$= 40^\circ \text{ (y is exterior angle)}$$

② Find x 

③



$$\frac{10}{y} = \frac{25}{y+90}$$

$$y = 60$$

$$\tan^{-1}\left(\frac{10}{60}\right) \times 2 = 18.92^\circ$$

求 $\frac{x}{2}$

6.2

Random experiments, events and event spaces

Playing cards

四种花色 four suits

Spade ♠ 黑桃 club ♣ 梅花

Heart ♥ 红心 Diamond ♦ 方块

每种花色有 1-13 的编数的卡

特殊卡名: Ace (1), Jack (11), Queen (12), King (13).

↳ 也称 picture, court, face cards. 花卡

Coin

Head 正面, Tail 反面

Dices

- Two dices are rolled. List the elements of the event space. and list the sample space of W , the event of both dice being ≥ 4 .

$$E = \{ (1,1) (1,2) (1,3) (1,4) (1,5) (1,6) \\ (2,1) (2,2) (2,3) (2,4) (2,5) (2,6) \\ (3,1) (3,2) (3,3) (3,4) (3,5) (3,6) \\ (4,1) (4,2) (4,3) (4,4) (4,5) (4,6) \\ (5,1) (5,2) (5,3) (5,4) (5,5) (5,6) \\ (6,1) (6,2) (6,3) (6,4) (6,5) (6,6) \}$$

$$W = \{ (4,4) (4,5) (4,6) \\ (5,4) (5,5) (5,6) \\ (6,4) (6,5) (6,6) \}$$

Discrete Random Variable distribution

 x - discrete random variable.

Probability distribution table.

条件 = 0, 1, 2, 3 时	x	0	1	2	3
Number sample proportion	$\hat{p}$	0	$\frac{1}{3}$	$\frac{2}{3}$	1
Probability of success	$Pr(\hat{p}=p)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$$\hat{p} = \frac{\text{number in the sample with the attribute}}{\text{sample size}} = \frac{x}{n}$$

$(0 \leq p \leq 1)$ 成功机率 $\int$ probability distribution
 $x=0$ 出现的

O CAS

Interactive $\rightarrow$ distribution $\rightarrow$ discrete $\rightarrow$ binomial (PDF)

x 拿球 $x: 0, 1, 2, 3$
 n 物不更的次 $n: 3$
 p 成功机率 $\hat{p} = p_{os}: 0.5$ (红球占全部球的 percentage)

O Question

① Find the probability distribution of x , obtain a head = 0.6, toss twice.
 the number of heads observed.

x	0	1	2
$Pr(X=x)$	0.16	0.48	0.36

对的是 $Pr(\hat{p}=p)$

② A swimming club has 20 members (12 males and 8 females).

(a) What is p , the proportion of males in the swimming club? $\frac{12}{20} = \frac{3}{5}$

(b) A team of five swimmers is to be selected from the club at random.

What are the possible values of the sample proportion $\hat{p}$ of male on the team?

x	0	1	2	3	4	5
$\hat{p}$	0	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	1

$\frac{0}{5}$ $\uparrow$ $\frac{1}{5}$ $\uparrow$

The sample mean as a random variable

- The mean of the distribution ($\mu_{\bar{x}}$), is always same as the population mean, μ

$$\mu_{\bar{x}} = \mu$$

- The standard deviation of the distribution of sample means can be found by:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \rightarrow \begin{array}{l} \text{standard deviation of population} \\ \downarrow \\ \text{sample size} \end{array}$$

$$sd(\hat{p}) = \sqrt{\frac{p(1-p)}{n}} \rightarrow \begin{array}{l} p \text{ is } p \text{ of success or population proportion} \\ \downarrow \\ \text{sample size} \end{array}$$

- Expect value ^{完全} (不是 mean, 而是可能得到的实际的数)

$$E(x) = np \rightarrow \begin{array}{l} \text{real balls} \\ \text{total balls} \end{array}$$

$x = 1, 2, 3$

- $n \geq 35$ is a large sample

- Example question ?

A population of 100 where $p = 0.3$

What is the probability that in a sample of size 10, $0.1 \leq \hat{p} \leq 0.3$?

用 Interactive / Distribution / Discrete / binomial CDF (Lower, Upper, Numtrial, pos)

lower and upper: $0.1 \times 10^{\text{sample size}} = 1, 0.3 \times 10 = 3$

Numtrial: Numtrial = n = sample size = 10

pos: pos = probability of success = $p = 0.3$

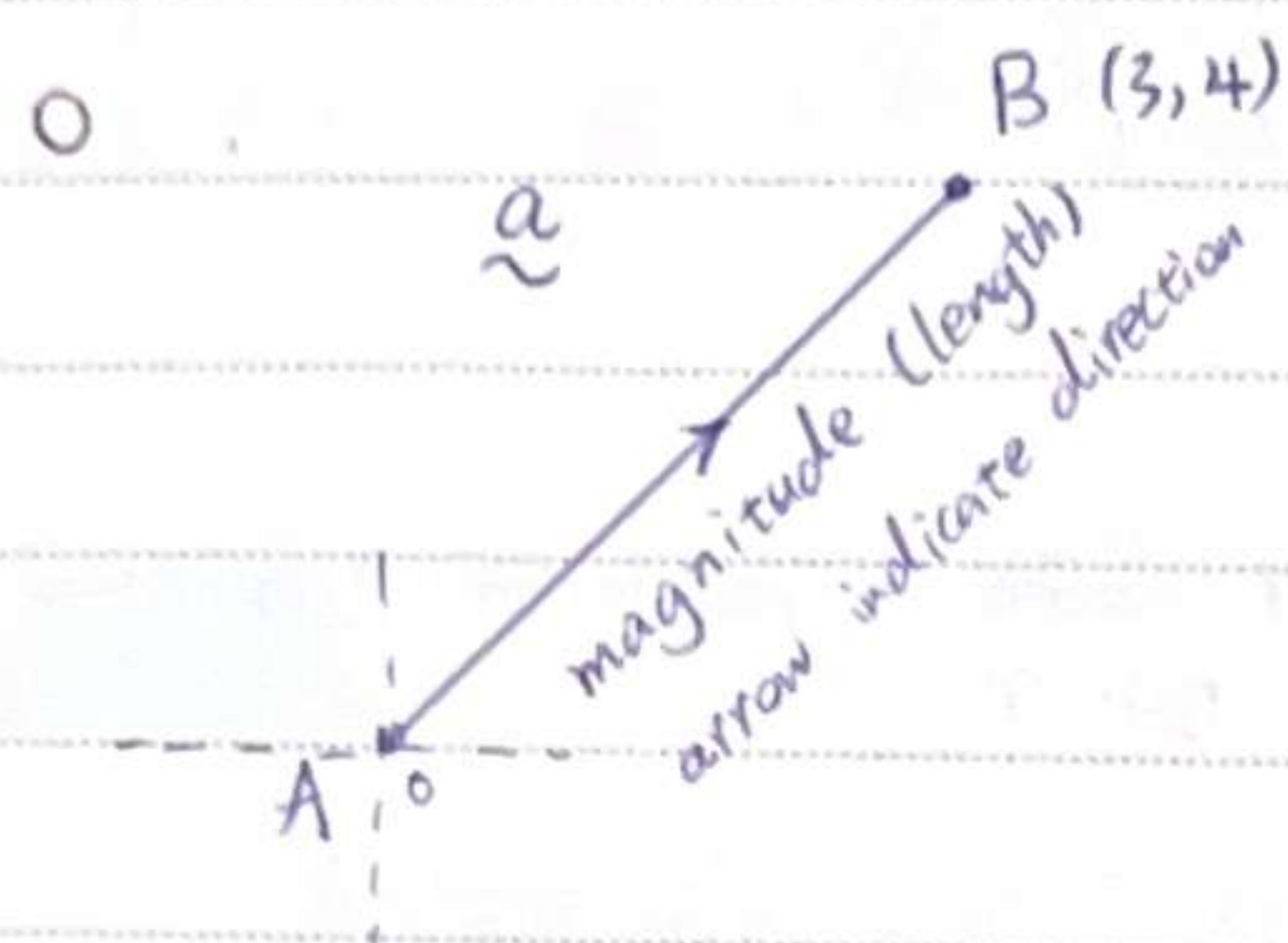
用 cas 求: binomialCDF(1, 3, 10, 0.3) ≈ 0.62

8.2

Vectors

○ Vectors: magnitude (size) + direction, E.g: gravity, displacement ^{位移}

Scalars: magnitude, E.g: time, volume, distance

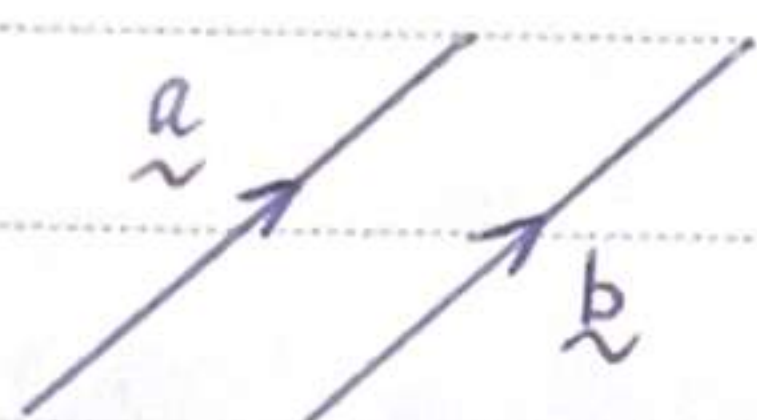


can write as $\vec{AB}$, the vector from A to B

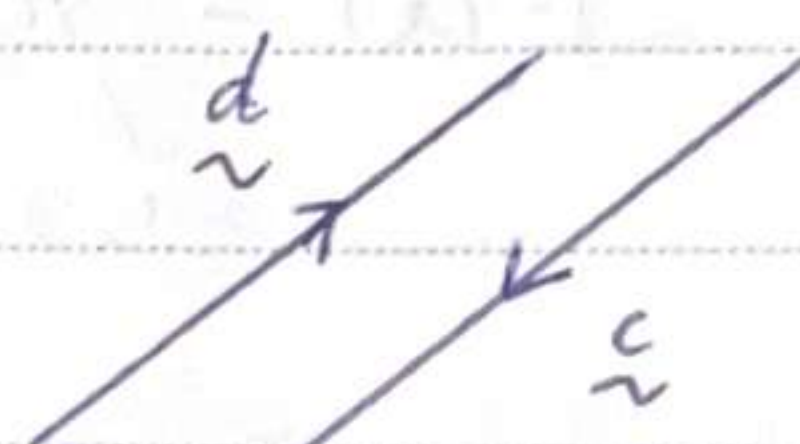
$$|\vec{AB}| = \sqrt{x^2 + y^2}$$

↑
is magnitude (length)

○ Equality of vectors



$$\vec{a} = \vec{b}$$



$\vec{c} \neq \vec{d}$ as they have different direction

$$|\vec{c}| = |\vec{d}|$$

as they have the same magnitude.

○ CAS

* magnitude: ① Action / vector / norm

② type norm([3 4])

→ 这个在 math 2

③ get 5

8.3

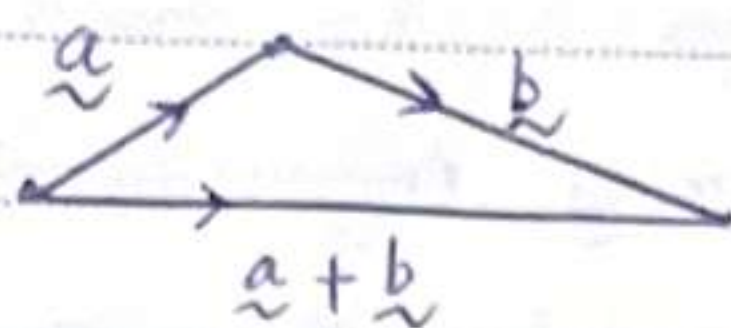
Operations on Vectors

o Addition of vectors 加法

$$\vec{AB} + \vec{BC} = \vec{AC}$$



- The method for adding two vectors is called the triangle rule for vectors.
- The addition of $\underline{a}$ and $\underline{b}$ can be shown by form a vector

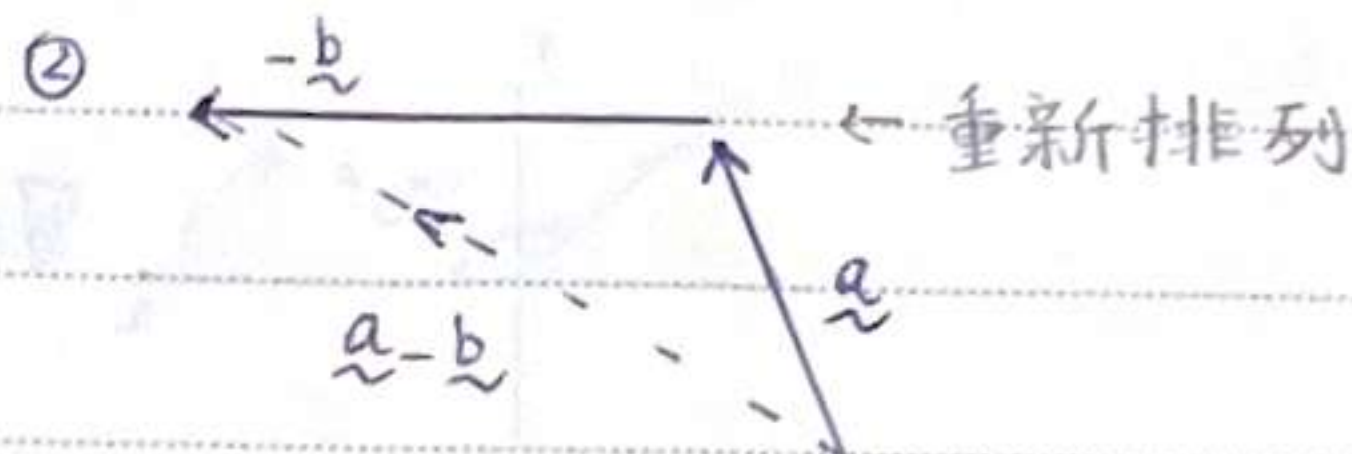
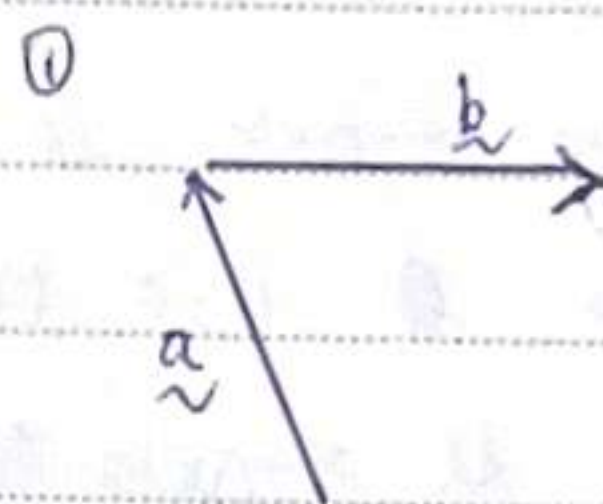


a 和 b 必须是顺流 (a and b must be in the same direction)

b 和 a+b 必须相撞 (b and a+b must collide)

o subtraction of vectors 减法

$$\underline{a} - \underline{b} = \underline{a} + (-\underline{b})$$



o other questions 其他

In simplest form, $\vec{MN} - \vec{QP} + \vec{NP} + \vec{QR}$ equals

$$= \vec{MN} + \vec{NP} + \vec{QR} - \vec{QP}$$

$$= \vec{MP} - \vec{QP} + \vec{QR}$$

$$\text{* 重点: } -\vec{QP} = +\vec{PQ}$$

$$= \vec{MP} + \vec{PQ} + \vec{QR}$$

$$= \vec{MQ} + \vec{QR}$$

$$= \vec{MR}$$

8.4

Magnitude, direction of vectors

o Magnitude / Length

- The magnitude of a vector $\underline{a}$ is denoted by $|\underline{a}|$ or a .

- ~~the~~ $\underline{a} = (x, y)$

$$|\underline{a}| = \sqrt{x^2 + y^2}$$

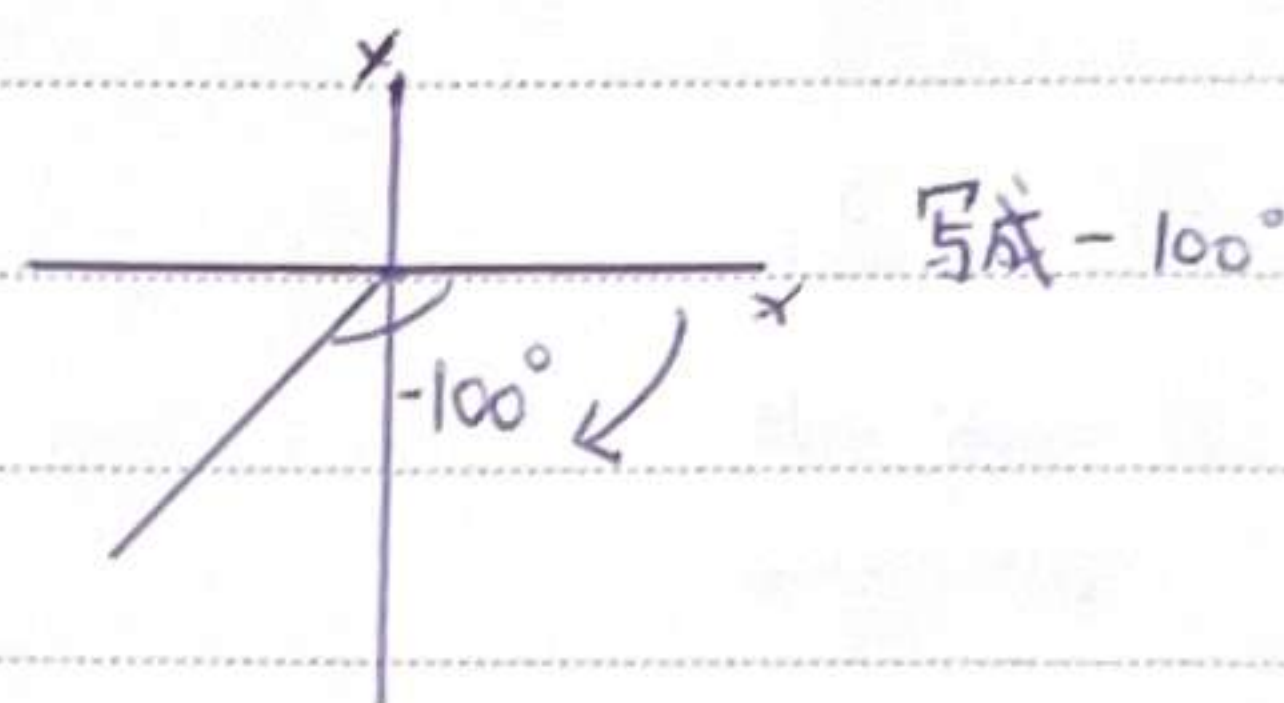
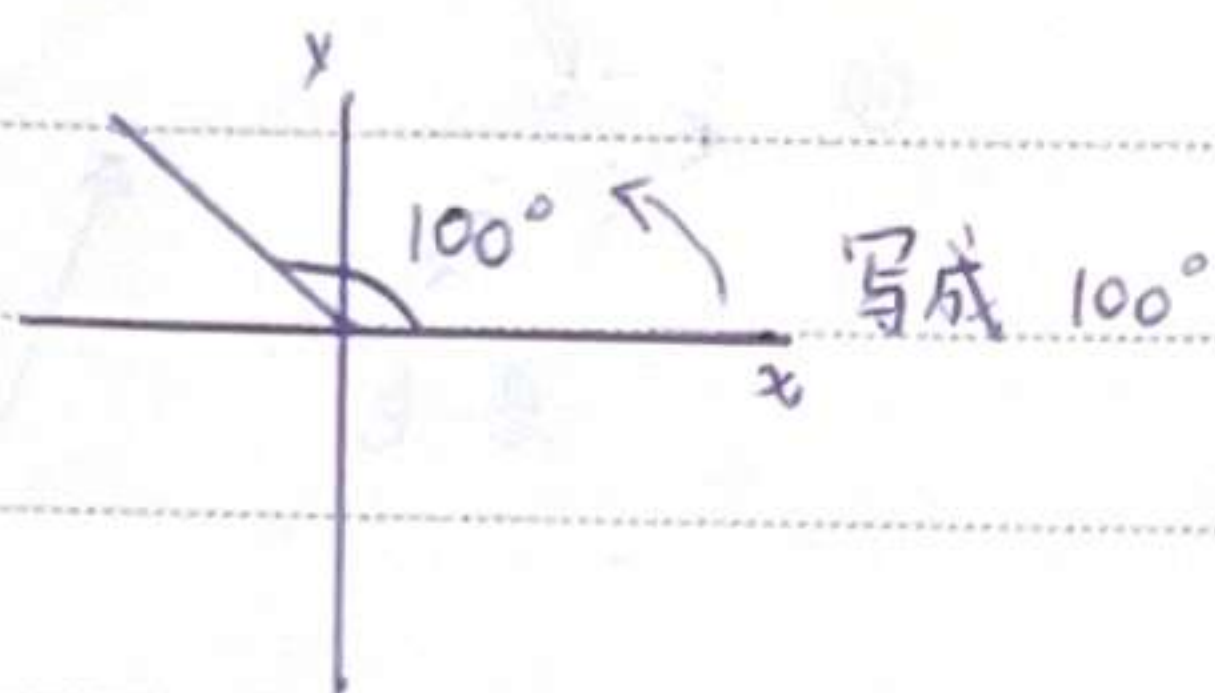
o Direction

- Direction found by using trigonometric

- angle usually will compare to a given direction, such as: north, positive x -axis

- If $\underline{a} = (x, y)$, then the direction compared to the positive x -axis is found by appropriately adjusting θ where $\theta = \tan^{-1}(\frac{y}{x})$

- Angle 从 x -axis 右边开始记如下:



8.5

 $\hat{i}$, $\hat{j}$ notation

○ Unit vectors $\underline{x} = x\hat{i} + y\hat{j} = \begin{bmatrix} x \\ y \end{bmatrix}$ 写作 $\begin{pmatrix} x \\ y \end{pmatrix}$

- A unit vector is any vector with a magnitude or length of 1 unit.
- Any vector in two dimensions can be represented as a combination of $\hat{i}$ and $\hat{j}$ vectors. $\hat{i} \rightarrow$ $\hat{j} \uparrow$
- the coefficient of $\hat{i}$ represent the size of the horizontal component.
- the coefficient of $\hat{j}$ represent the size of the vertical component.

○ Addition / Subtraction / Multiplication in $\hat{i}$, $\hat{j}$ form. 加减乘用 $\hat{i} + \hat{j}$

- For $\underline{a} = 2\hat{i} + 3\hat{j}$ and $\underline{b} = 3\hat{i} + 4\hat{j}$

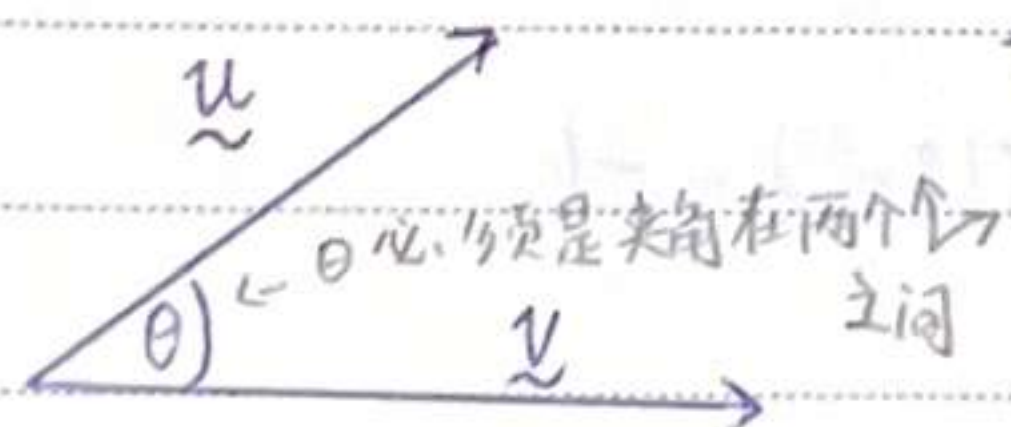
$$\begin{aligned} \underline{a} + \underline{b} &= (x_1 + x_2)\hat{i} + (y_1 + y_2)\hat{j} \\ &= (2+3)\hat{i} + (3+4)\hat{j} \\ &= 5\hat{i} + 7\hat{j} \end{aligned}$$

$$\underline{a} - \underline{b} = (x_1 - x_2)\hat{i} + (y_1 - y_2)\hat{j}$$

$$k\underline{a} = kx_1\hat{i} + ky_1\hat{j}$$

○ Multiplying two vectors

- use Method 'dot product', also know as 'scalar product' as the result of this multiplication is a scalar (magnitude only).



The dot product is given by:

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos(\theta)$$

- Another way to find dot product

For $\underline{u} = 2\hat{i} + 5\hat{j}$, $\underline{v} = 3\hat{i} + \hat{j}$ 求 $\underline{u} \cdot \underline{v}$

$$\begin{aligned} \underline{u} \cdot \underline{v} &= (2\hat{i} + 5\hat{j}) \cdot (3\hat{i} + \hat{j}) \\ &= 2 \times 3 + 5 \times 1 \\ &= 11 \end{aligned}$$

0 Find angle between two vectors

- use dot product formula to find angle

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos(\theta)$$

- Rearrange the formula, we get

$$\cos(\theta) = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

$$\theta = \cos^{-1} \left(\frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|} \right)$$

$\underline{x} = x_1 \underline{i} + y_1 \underline{j}$
or $\tan^{-1} \left(\frac{y}{x} \right)$
求和 x -axis
的角度

0 Perpendicular vectors 垂直相交

- If two vectors are 垂直相交, the angle between them is 90° .

$$\text{- so } \underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos(90^\circ)$$

$$= |\underline{u}| |\underline{v}| \times 0, \text{ since } \cos(90^\circ) = 0$$

$$= 0$$

- so if $\underline{u} \cdot \underline{v} = 0$, then $\underline{u}$ and $\underline{v}$ are 垂直相交.

0 Parallel vectors 平行

- If two vectors are 平行, the angle between them is either 0° or 180°

- For $\theta = 0^\circ$ (same direction)

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos(\theta)$$

$$= |\underline{u}| |\underline{v}| \cos(0^\circ), \text{ since } \cos(0^\circ) = 1$$

$$= |\underline{u}| |\underline{v}|$$

$\nearrow \nearrow$ or $\nwarrow \swarrow$
 $= \searrow$
 $\theta = 180^\circ$

- For $\theta = 180^\circ$ (opposite direction)

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos(\theta)$$

$$= |\underline{u}| |\underline{v}| \cos(180^\circ), \text{ since } \cos(180^\circ) = -1$$

$$= -|\underline{u}| |\underline{v}|$$

* if $\underline{a} = k \underline{b}$, where $k \in \mathbb{R} (k \neq 0)$, then $\underline{a}$ and $\underline{b}$ are parallel.

O ★ Example Question

① $\underline{u} = 2\hat{i} - 4\hat{j}$, Find a vector parallel to $\underline{u}$ such that its dot product with $\underline{u}$ is 40.

理论 $\underline{u} \cdot \underline{v} = 40$, as it is positive, the vectors are in the same direction.

公式 $\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}|$

$$\text{求 } |\underline{v}| |\underline{u}| = \sqrt{2^2 + 4^2}$$

$$= 2\sqrt{5}$$

$$\text{求 } |\underline{v}| \quad |\underline{v}| = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}|}$$

$$= \frac{40}{2\sqrt{5}}$$

$$= 4\sqrt{5}$$

结论: $\underline{v}$ is double $\underline{u}$

答案: $|\underline{v}| = 2(2\hat{i} - 4\hat{j})$

$$= 4\hat{i} - 8\hat{j}$$

② $\vec{AC} = \underline{u} + \underline{v}$, 求 $\vec{AC}$ 的 length

用 dot product: $\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos(\theta)$

$$\vec{AC} \cdot \vec{AC} = (\underline{u} + \underline{v}) (\underline{u} + \underline{v})$$

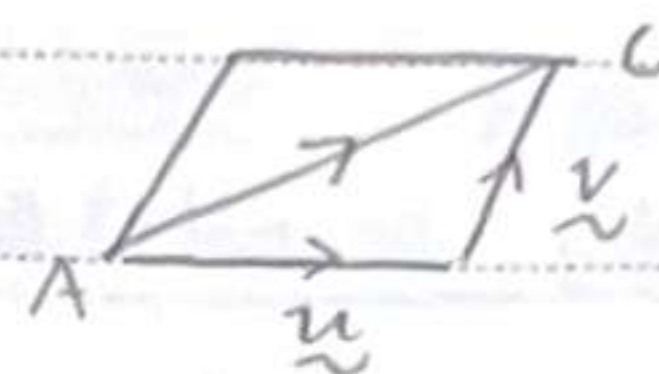
$$= |\underline{u}|^2 + \underline{u} \cdot \underline{v} + \underline{u} \cdot \underline{v} + |\underline{v}|^2$$

$$\underline{u} \cdot \underline{u} = |\underline{u}|^2 \text{ 因为 } \underline{u} \cdot \underline{u} = |\underline{u}| |\underline{u}| \cos(0)$$

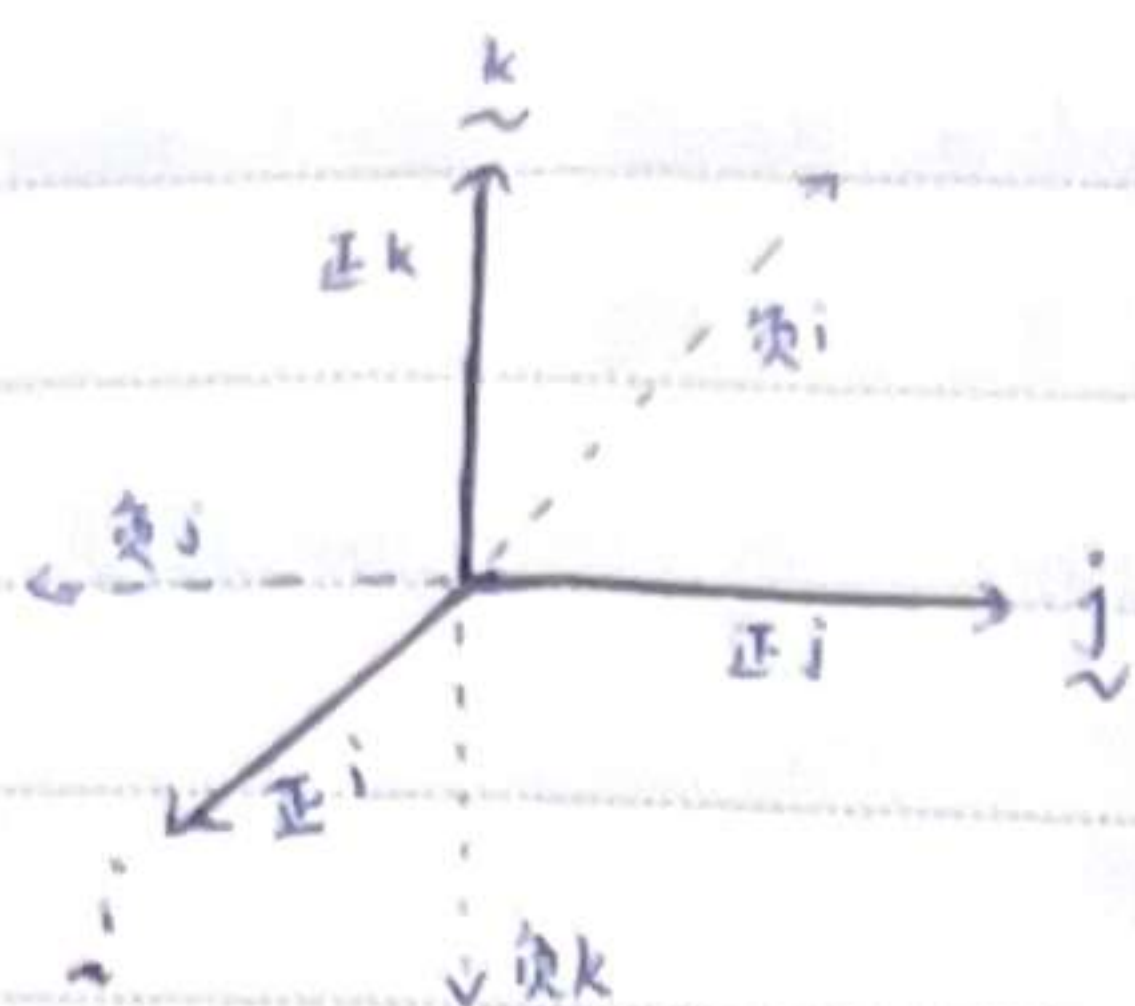
$$\text{然后 } \cos(0) = 1$$

$$= |\underline{u}|^2 + |\underline{v}|^2 + |\underline{u}| |\underline{v}| \cos(\theta) + |\underline{u}| |\underline{v}| \cos(\theta)$$

$$= |\underline{u}|^2 + |\underline{v}|^2 + 2 |\underline{u}| |\underline{v}| \cos(\theta)$$



3C & 3D



$$\vec{OA} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$|\vec{OA}| = \sqrt{x^2 + y^2 + z^2}$$

$$\text{Unit vector } \hat{OA} = \frac{\vec{OA}}{|\vec{OA}|}$$

$$= \frac{x\vec{i} + y\vec{j} + z\vec{k}}{\sqrt{x^2 + y^2 + z^2}}$$

$$\vec{OB} = p\vec{i} + q\vec{j} + r\vec{k}$$

$$\text{dot length } \vec{OA} \cdot \vec{OB} = px + qy + rz$$

$$\vec{OA} \cdot \vec{OB} = |\vec{OA}| |\vec{OB}| \cos \theta$$

CAS ^{模式} [Alg Decimal Rnd Deg]

- 求 unit vector of a vector

- 求 length Action/vector/norm

Action/vector/unit V

unitV (

[↑] $2\vec{i} + 3\vec{j} + 4\vec{k}$

Type [2 3 4] math 2 最左侧第四行 []

双击才有3位

- 求 angle between two vectors

Action/vector/angle

Type angle ([2 3 4], [-1 -2 -3])

↑ 只有这里有逗号

- 求 dot Product of two vectors 乘积

Action/Vector/dot P

Type dotP ([2 3 4], [-1 -2 -3])

↑ 中间有逗号

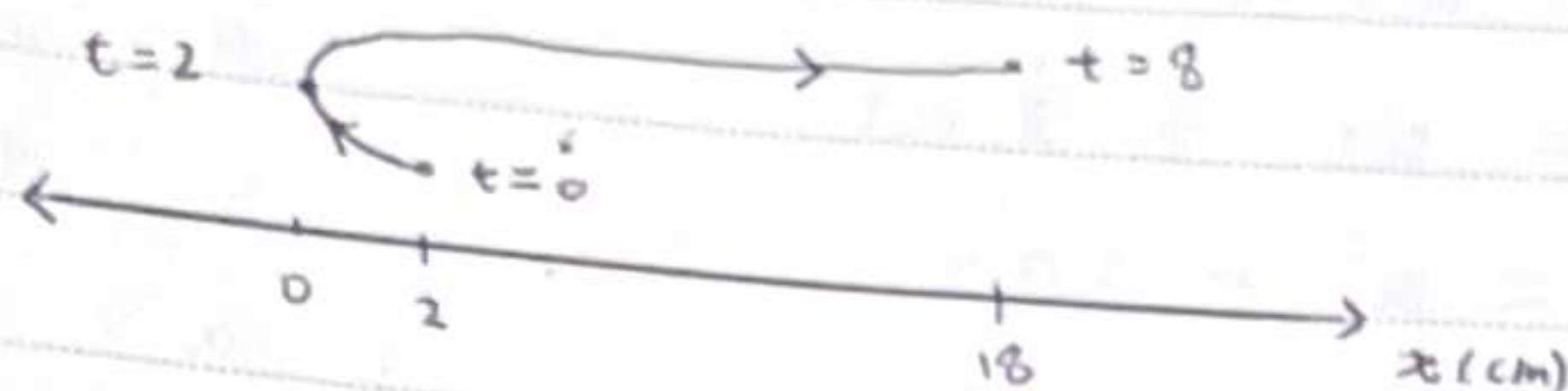
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9.2

Kinematics 运动学

Position - time line



9.3

$$\text{Average Speed} = \frac{\text{change in distance}}{\text{change in time}}$$

$$\text{Average Velocity} = \frac{\text{change in displacement}}{\text{change in time}}$$

$$\text{Average acceleration} = \frac{\text{change in velocity}}{\text{change in time}}$$

$$\text{km/h} \div 3.6 = \text{m/s}$$

$$\text{m/s} \times 3.6 = \text{km/h}$$

例

$$90 \text{ km/h} \div 3.6 = 25 \text{ m/s}$$

9.4

Constant acceleration formula

以下公式前提建立于 acceleration 是 constant 的

$$v = u + at$$

v = final velocity

$$s = ut + \frac{1}{2}at^2$$

u = initial velocity

$$v^2 = u^2 + 2as$$

s = distance travelled / displacement

$$s = \frac{1}{2}(u+v)t$$

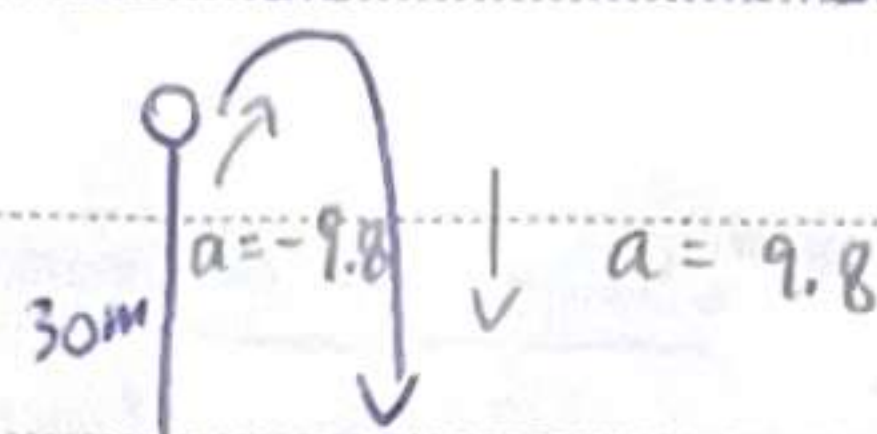
a = acceleration

$$s = vt - \frac{1}{2}at^2$$

t = time

- 例题

① A ball $\nearrow$ upward at 9.8 m/s from a tower that is 30m above ground



(a) Total time taken to reach ground

这部分
先算

$$u = 9.8 \quad v = 0 \quad a = -9.8$$

$$v = u + at$$

$$0 = 9.8 - 9.8t$$

$$t = 1s$$

这部分
再算 distance

$$s = \frac{1}{2}(u+v)t$$

$$= \frac{1}{2}(0+9.8) \times 1$$

$$= 4.9m$$

得 distance
这部分的

$$s = 4.9 + 30 = 34.9m$$

往下 $u = 0, a = 9.8, s = 34.9$ 求 t

$$s = ut + \frac{1}{2}at^2$$

$$t \approx 2.67s$$

$$\text{总 time} = 2.67 + 1 = 3.67s$$

Date.

Page.

9.5

Instantaneous rates of change

$x(t)$ position

→ Distance: $x(t) - x(0)$

↑ 如果 time 从 0 开始的话

↓

$v(t)$ velocity

↓

$a(t)$ acceleration

10.3

倒三角函数
Reciprocal Trigonometric Functions (Exact value)

Cosecant function

$$\operatorname{cosec}(x) = \frac{1}{\sin(x)}, \sin(x) \neq 0$$

$$= \frac{H}{O}$$

Secant function

$$\sec(x) = \frac{1}{\cos(x)}, \cos(x) \neq 0$$

$$= \frac{H}{A}$$

Cotangent function

$$\cot(x) = \frac{1}{\tan(x)}, \tan(x) \neq 0$$

$$= \frac{A}{O}$$

$$\star = \frac{\cos(x)}{\sin(x)} \text{ since } \frac{AH}{OA}$$

角度转 $\frac{\pi}{180}$: $45^\circ \times \frac{\pi}{180}$

$$= \frac{\pi}{4}$$

例: $\cot(30^\circ)$

$$= \frac{1}{\tan(30^\circ)}$$

$$= \frac{1}{\frac{1}{\sqrt{3}}} = \frac{3}{1} = \sqrt{3}$$

(AS [Find Exact value])

当角度为单位如 45° , 240° 等① Main (Alg Standard Real Deg) 用 Degree mode② Type 题如: $\sec(240^\circ)$

③ 'EXE'

当 π 为单位时, 如 $\frac{\pi}{6}$, $-\frac{\pi}{6}$ 等 (题目限制为 $[0, 2\pi]$ 时也可能是要 Rad mode.

① Main (Alg Standard Real Rad) 用 Radian

② Type 如: $\sin(\frac{\pi}{6})$

③ 'EXE'

10.5

Trigonometric identities

$$\sin^2(x) + \cos^2(x) = 1$$

$$1 + \tan^2(x) = \sec^2(x), \cos(x) \neq 0$$

$$1 + \cot^2(x) = \operatorname{cosec}^2(x), \sin(x) \neq 0$$

$$* \cos^2(x) \neq \cos(x^2)$$

$$\cos^2(x) = (\cos(x))^2$$

$$\star \frac{\sin(x)}{\cos(x)} = \tan(x)$$

10.6

Compound- and Double-angle formulas

Compound-angle formulas:

$$\sin(A+B) = \sin(A)\cos(B) + \cos(A)\sin(B)$$

$$\cos(A+B) = \cos(A)\cos(B) - \sin(A)\sin(B)$$

$$\tan(A+B) = \frac{\tan(A) + \tan(B)}{1 - \tan(A)\tan(B)}$$

$$\sin(A-B) = \sin(A)\cos(B) - \cos(A)\sin(B)$$

$$\cos(A-B) = \cos(A)\cos(B) + \sin(A)\sin(B)$$

$$\tan(A-B) = \frac{\tan(A) - \tan(B)}{1 + \tan(A)\tan(B)}$$

Double-angled formulas:

$$\sin(2A) = 2\sin(A)\cos(A)$$

$$\cos(2A) = \cos^2(A) - \sin^2(A)$$

$$= 1 - 2\sin^2(A)$$

$$= 2\cos^2(A) - 1$$

$$\tan(2A) = \frac{2\tan(A)}{1 - \tan^2(A)}$$

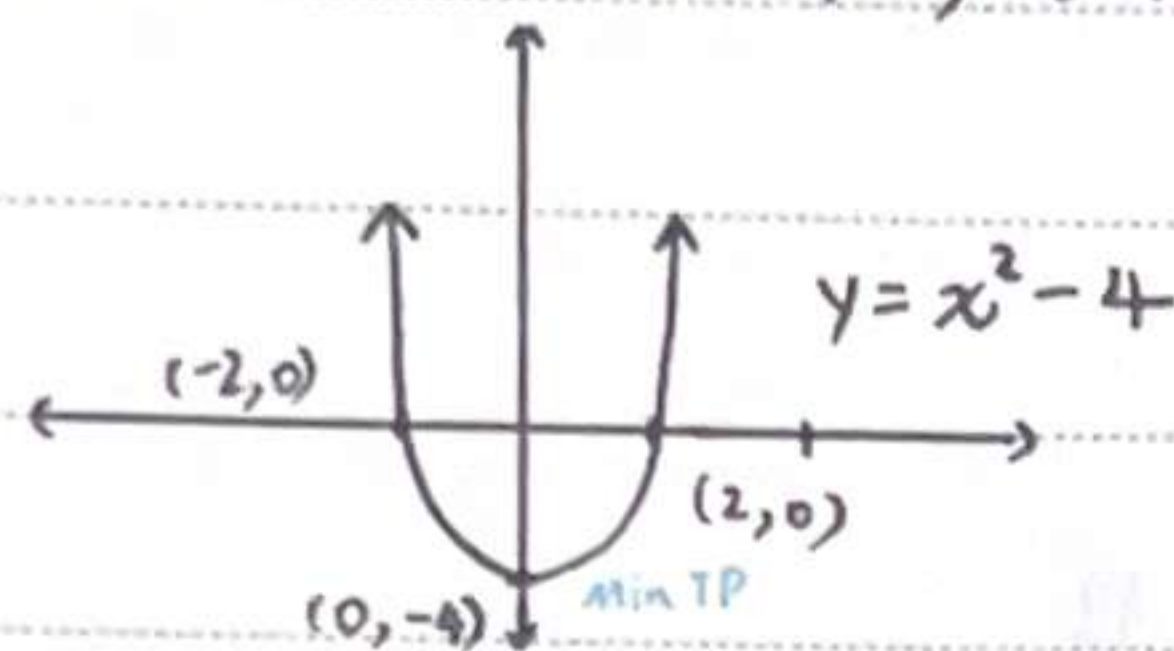
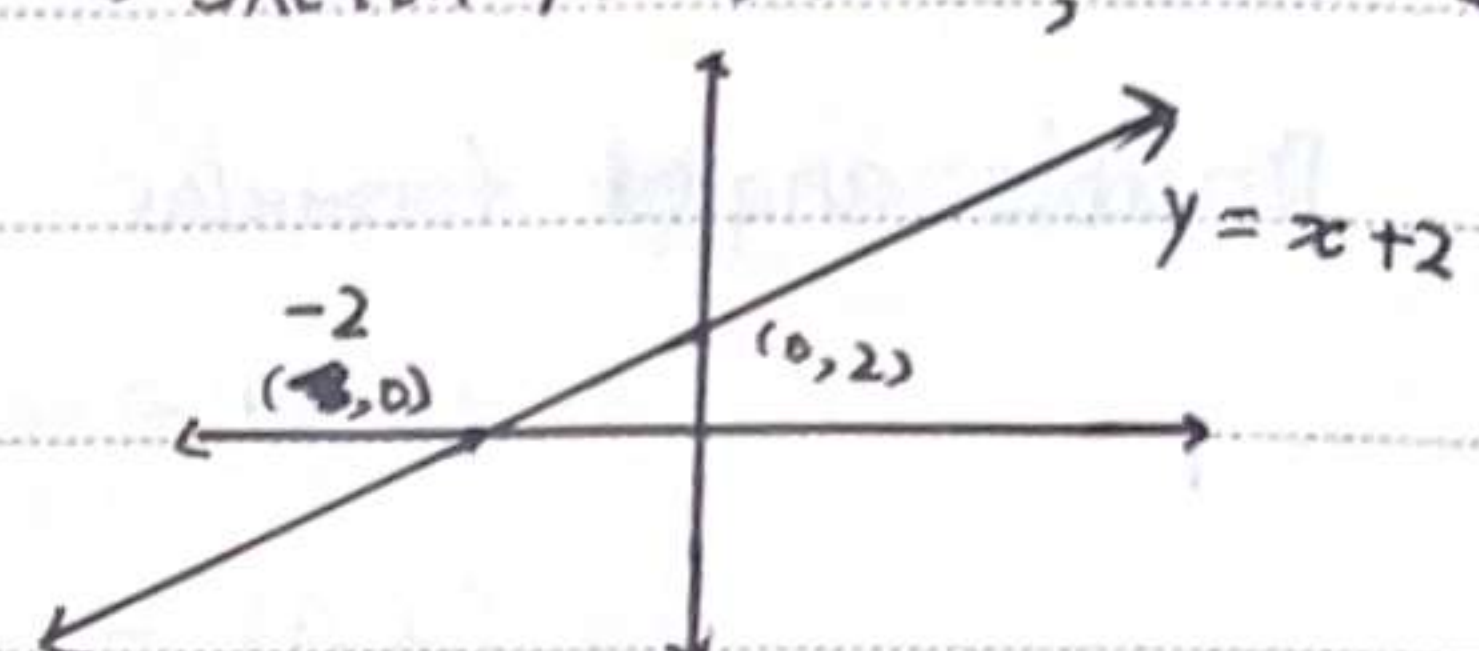
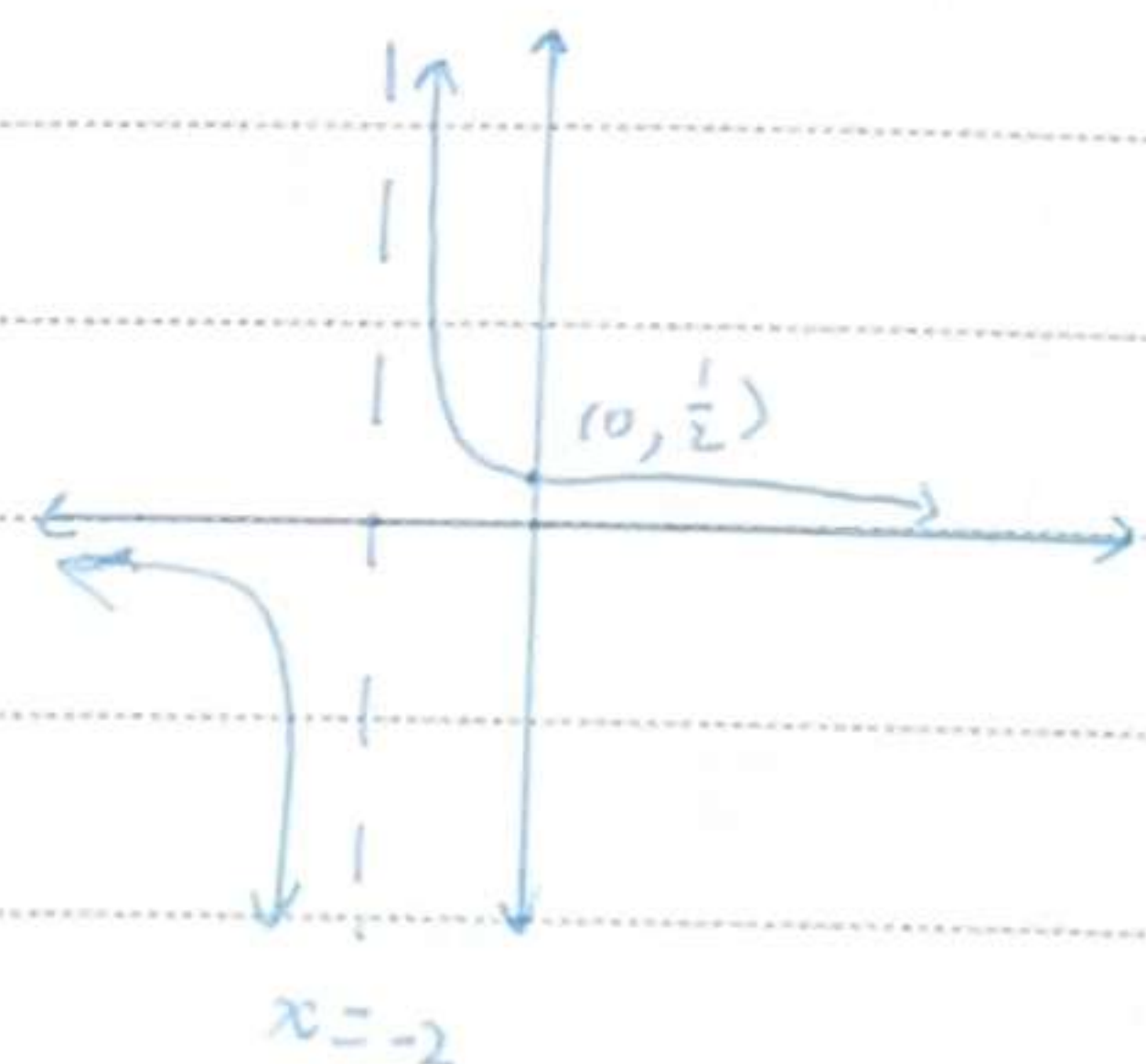
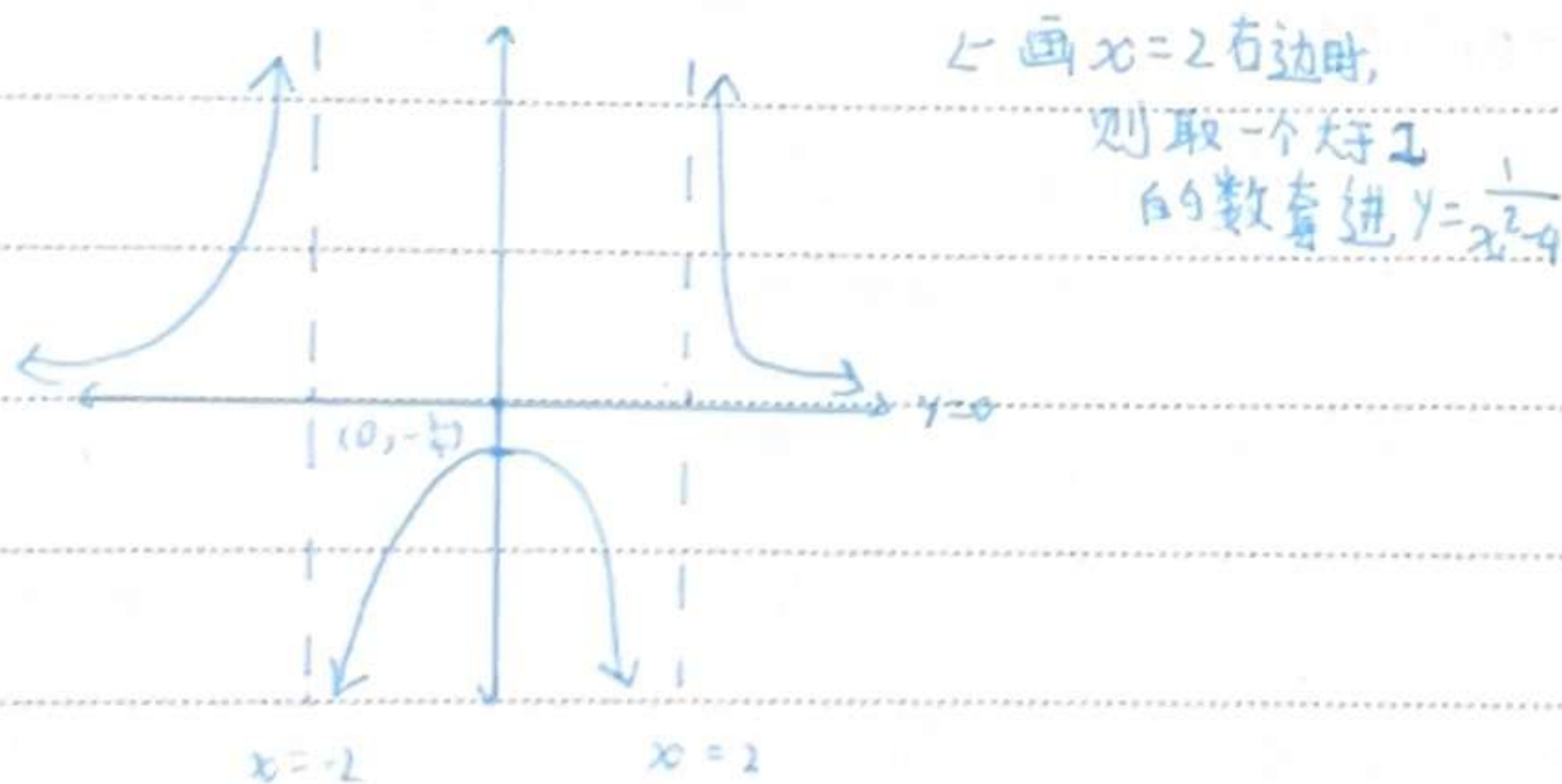
11.2

Reciprocal Graphs 倒数图

Aim: 根据给的图或公式画倒数图

Regular graph $y = f(x)$	Reciprocal $y = \frac{1}{f(x)}$ 倒数图
x -intercept $= (a, 0)$	Vertical asymptote at $x = a$
$y = \pm 1$	$y = \pm 1$
y -intercept $= (0, b)$	y -intercept $= (0, \frac{1}{b})$
maximum TP at (x, y)	minimum TP at $(x, \frac{1}{y})$
minimum TP at (x, y)	maximum TP at $(x, \frac{1}{y})$

例题: sketch from given graph

(a) Sketch $y = \frac{1}{x^2 - 4}$, $x \neq \pm 2$ (b) Sketch $y = \frac{1}{x+2}$, $x \neq -2$ ① Vertical asymptote at $x = \pm 2$ ② y -intercept $= (0, -\frac{1}{4})$ ③ Max TP at $(0, -\frac{1}{4})$ ① Vertical asymptote at $x = -2$ ② y -intercept $= (0, \frac{1}{2})$ ③ Horizontal asymptote: $y = 0$ 

④ Horizontal asymptote = 0!! (y=0) always

⑤ 画一个画 $x = -2$ 左边时, 把小于 -2 的数套进 $y = \frac{1}{x+2}$

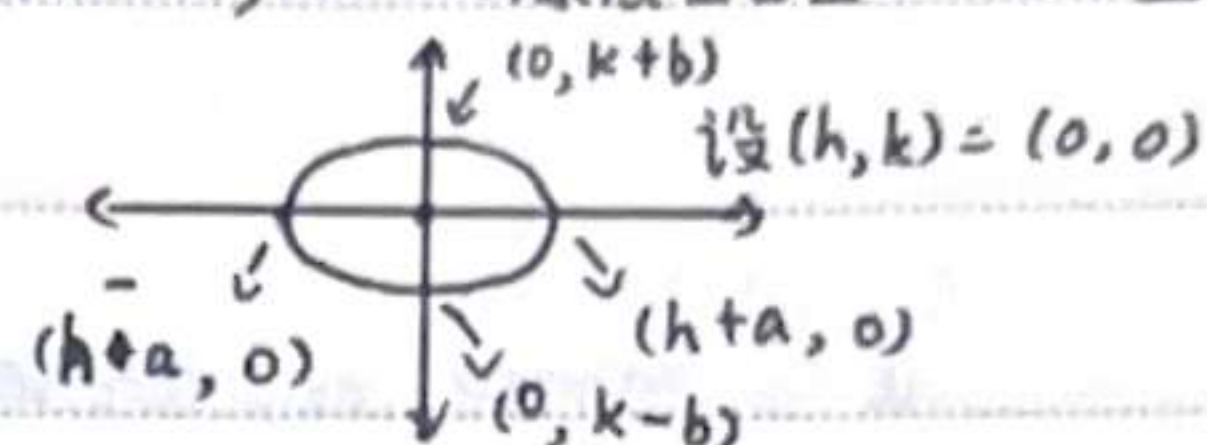
如: $y = \frac{1}{x^2 - 4}$, 如果得到一个正数: $\frac{1}{5}$, 则把图画 x -axes 上面
负数, 则画下面

11.3

The circle and the ellipse 圓和橢圓

Circle equation: $(x-h)^2 + (y-k)^2 = r^2$ centre (h, k) 半徑 = r

Ellipse equation: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$ centre (h, k) 从原点左右 $\pm a$ 上下 $\pm b$



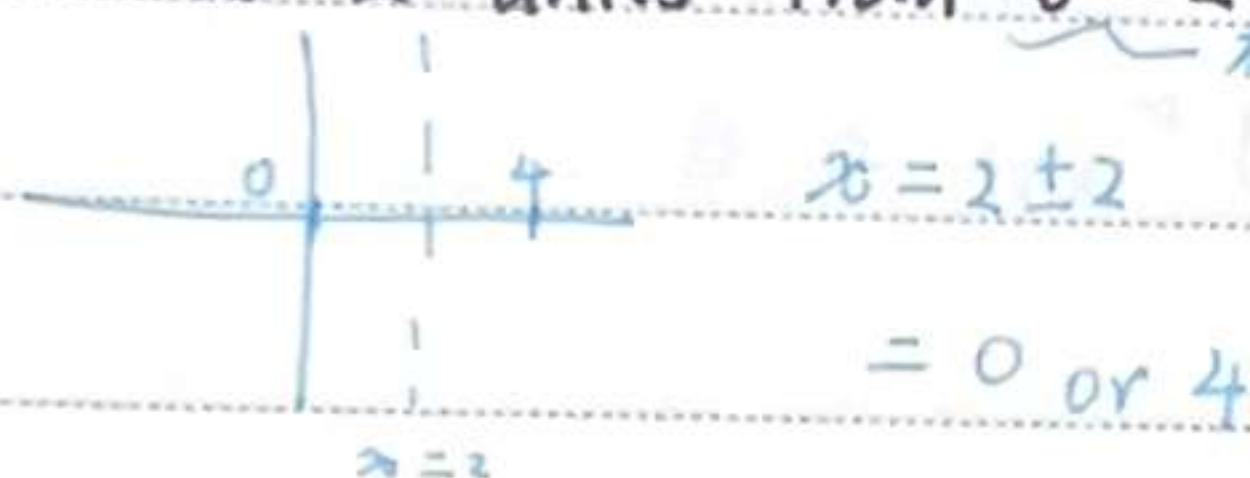
例题

State the equation of the locus of a point $P(x, y)$ such that:

(a) P is 3 units from the y -axis



(b) P is 2 units from $x=2$ 不是 from axis



(c) P is equidistant from R and S $R(0, 2)$ $S(4, 2)$ $R(1, 1)$ $S(3, 1)$

distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Distance $|PR| = \sqrt{(x-0)^2 + (y-2)^2}$

Distance $|PS| = \sqrt{(x-4)^2 + (y-2)^2}$

Since distance $|PR| = |PS|$

$\sqrt{(x-0)^2 + (y-2)^2} = \sqrt{(x-4)^2 + (y-2)^2}$

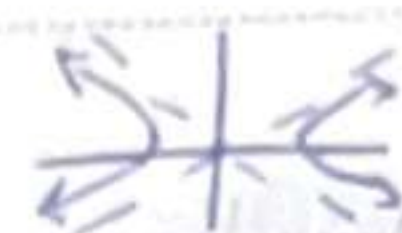
$(x-0)^2 + (y-2)^2 = (x-4)^2 + (y-2)^2$

$x = 2$

11.4

The hyperbola

Hyperbola equation: $\pm \frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$



or: $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$



$\pm$: centre or intersection of asymptote = (h, k)

vertices: $(-a+h, k)$ $(a+h, k)$

asymptote equation: $y = \frac{b}{a}(x-h) + k$

$y = -\frac{b}{a}(x-h) + k$

Example:

$$4x^2 - 9y^2 = 36$$

$$\frac{4x^2}{36} - \frac{9y^2}{36} = 1$$

$$\frac{x^2}{9} - \frac{y^2}{4} = 1$$

① $a=3$, $b=2$ centre $(0,0)$

② Vertices $(-3, 0)$ $(3, 0)$

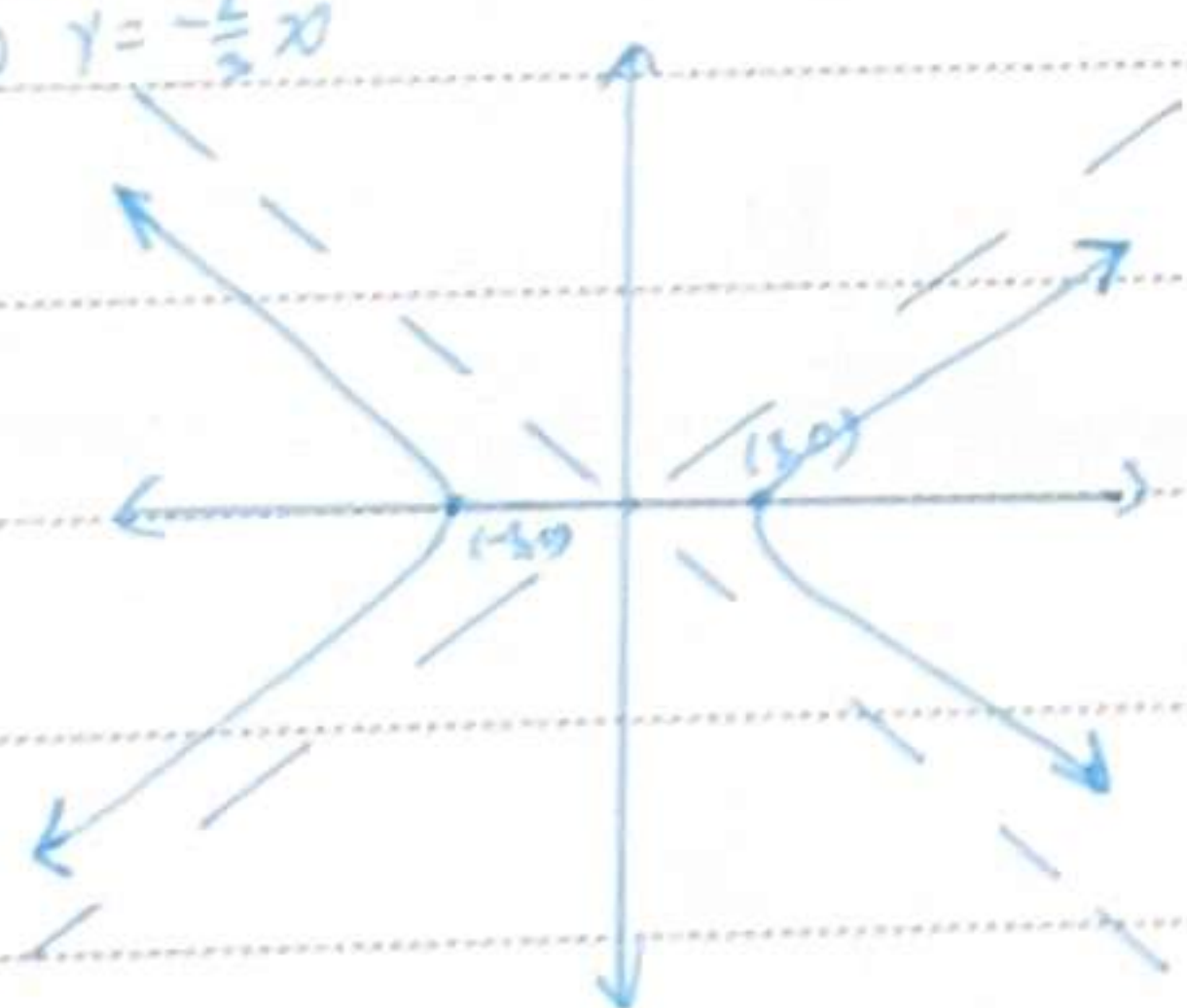
③ Asymptote $y = \pm \frac{2}{3}(x-h) + k$

$$= -\frac{2}{3}x, \frac{2}{3}x$$

a 是实数轴所截长度

④ $y = -\frac{2}{3}x$

$y = \frac{2}{3}x$



11.5

Polar coordinates, equations and graphs

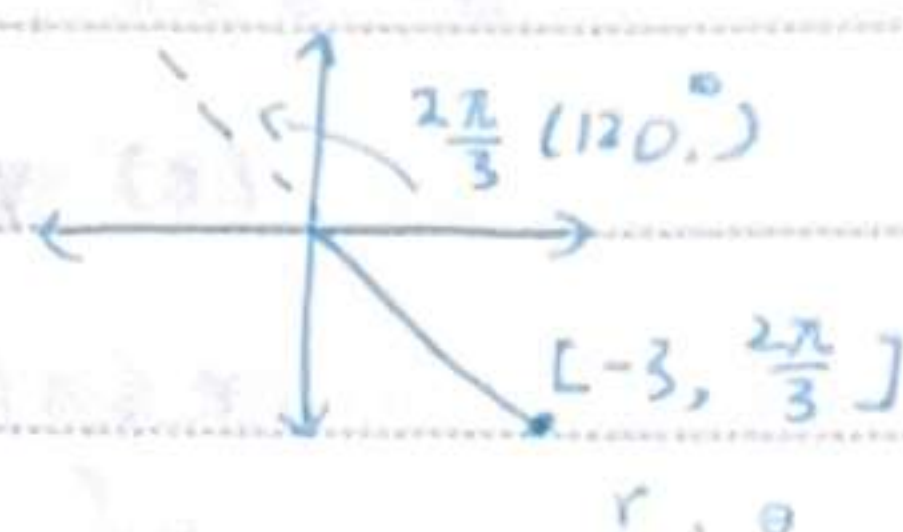
Aims: 把任意 polar coordinate $[r, \theta]$ ~~转换~~ 转换成 Cartesian 正常 (x, y) (或反转)

把任意 polar equation $r = \dots$ 转换成 Cartesian equation $y = \dots, x = \dots$ (或反转)

Plot polar coordinate

Plot polar coordinate $[-3, \frac{2\pi}{3}]$

是 r 不是 x
negative, so rotate $\frac{2\pi}{3}$, anticlockwise



Convert Polar $[r, \theta]$ to Cartesian (x, y)

$$x = r \cos(\theta)$$

$$y = r \sin(\theta)$$

Convert Cartesian (x, y) to Polar $[r, \theta]$

$$\tan(\theta) = \frac{y}{x}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right), \quad r = \sqrt{x^2 + y^2}$$

Convert Polar equation to Cartesian equation

例: (a) $r = 2$

$$r^2 = 4$$

$$r^2 = x^2 + y^2$$

$$x^2 + y^2 = 4$$

(b) $r = 6 \sin(\theta)$

$$r^2 = 6 r \sin(\theta) \quad \therefore y = r \sin(\theta)$$

$$r^2 = x^2 + y^2$$

$$x^2 + y^2 = 6y$$

$$x^2 + y^2 - 6y + 9 - 9 = 0$$

$$x^2 + (y - 3)^2 = 9$$

Convert Cartesian equation to polar equation

例题: (a) $x = 3$

(b) $x^2 + y^2 = 9$

$$x = r \cos(\theta)$$

$$r^2 \cos^2(\theta) + r^2 \sin^2(\theta) = 9$$

$$r \cos(\theta) = 3$$

$$r^2 = \frac{9}{\cos^2(\theta) + \sin^2(\theta)}$$

$$r = \frac{3}{\cos(\theta)}$$

$$r^2 = 9 \quad \text{因为 } \cos^2(\theta) + \sin^2(\theta) = 1$$

$$r = 3$$

(c) $y = x$

$$r \sin(\theta) = r \cos(\theta)$$

$$\frac{r}{r} = \frac{\cos(\theta)}{\sin(\theta)}$$

$$1 = \tan(\theta)$$

$$\theta = \tan^{-1}(1) = \frac{\pi}{4}$$

Aim: Use CAS to convert $[r, \theta]$ to (x, y) 反之亦然

例: (a) $[12, 210^\circ]$ to (x, y)

(b) $(5, 6)$ to $[r, \theta]$

Rad mode

Degree mode

Action/vector/to Rect(

Action/vector/to Pol(

Math 2 / []

Math 2 / []

Type [12] [(210)]
 $\uparrow$ $\uparrow$ $\uparrow$
 math 2 左 math 3 右 Trig 右

Type [5] [6]
 $\uparrow$
 math 2 左

$$= (-6\sqrt{3}, -6)$$

$$= [\sqrt{61} \quad 50^\circ 12']$$

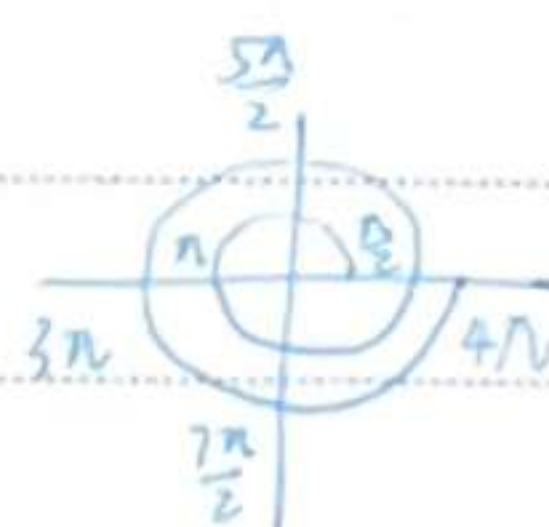
Aim: Use CAS sketch polar form

(a) sketch $r = \theta$, $0 \leq \theta \leq 4\pi$ rad mode

CAS Graph: Type (R = Type)

$$r = \theta \quad | \quad 0 \leq \theta \leq 4\pi$$

$\uparrow$
Trig



setting / view window 更改信息: min: 0, max: 4π , step: $\pi/6$

11.6

参数方程
Parametric equations

Aims: 把各类方程转化为以 x 和 y 作未知数的方程, 也就是 Cartesian

例: Find the Cartesian equation

(a) $x = t + 1, y = t^2 + 4$

① 先找 t , $t = x - 1$

② 套进去, $y = (x - 1)^2 + 4$

③ 简化, $y = x^2 - 2x + 5$

(b) $x = t^2 + t, y = t^2 - t$

$$x + y = 2t^2$$

$$x - y = 2t$$

$$(x - y)^2 = 4t^2$$

$$(x - y)^2 = 2(x + y)$$

(c) $x = \cos(2\theta), y = \sin(\theta)$

$$y^2 = \sin^2(\theta)$$

$$\cos(2\theta) = 1 - 2\sin^2(\theta)$$

$$x = 1 - 2y^2$$

(d) $x = \cos(2\theta), y = \sin(2\theta)$

$$x^2 = \cos^2(2\theta)$$

$$y^2 = \sin^2(2\theta)$$

$$\star \cos^2(2\theta) + \sin^2(2\theta) = x^2 + y^2$$

$$= 1$$

$$x^2 + y^2 = 1$$

The Absolute Value Function

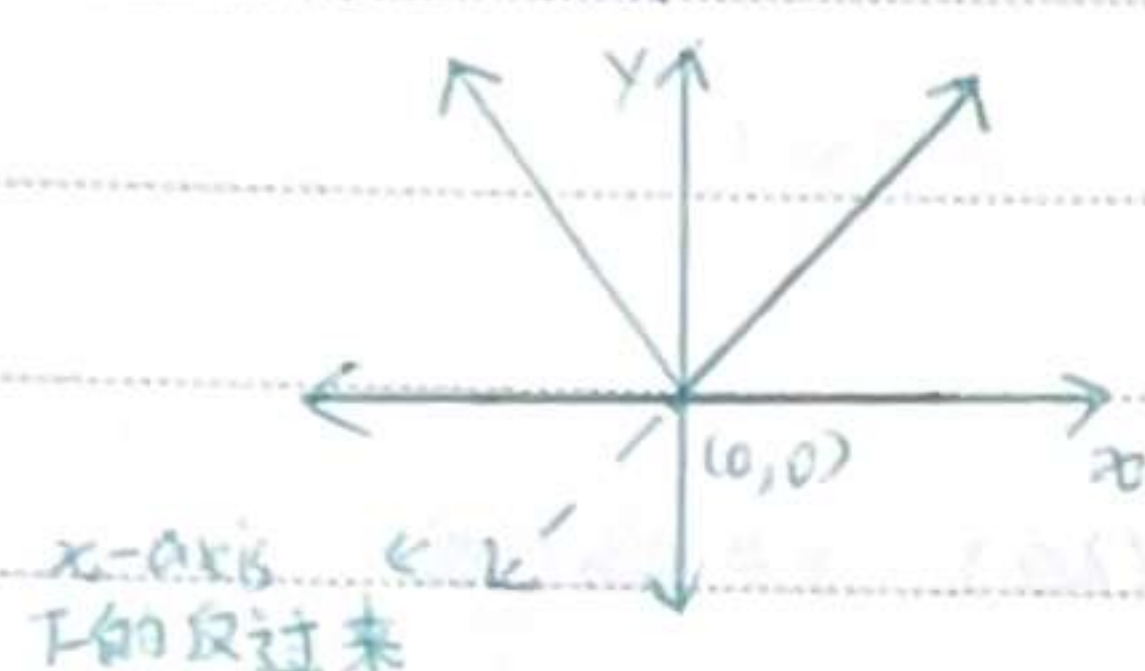
- To sketch the graph of $y = |f(x)|$

1. Sketch the Graph of $y = f(x)$

2. Refel Reflect the portion of the graph which is below the x -axis

Example Question

① $y = |2x|$



② $f(x) = |x^2 - 3x| + 2$

(a) Rewrite it as a hybrid function

$$f(x) = \begin{cases} 1 \text{ 乘以 } (x^2 - 3x) + 2, \\ -1 \text{ 乘以 } (x^2 - 3x) + 2, \end{cases}$$

$$= \begin{cases} x^2 - 3x + 2, & \text{Domain?} \\ -x^2 + 3x + 2, & \text{Domain?} \end{cases}$$

找 domain: 找 1 竖线里 equation 的 x -in

$$x^2 - 3x = x(x-3)$$



$$x\text{-in} = (0, 0) (3, 0)$$

$$f(x) = \begin{cases} x^2 - 3x + 2, & x \leq 0 \cup x \geq 3 \\ -x^2 + 3x + 2, & x \in (0, 3) \end{cases}$$

(b) find $f(-1)$ 和 $f(2)$

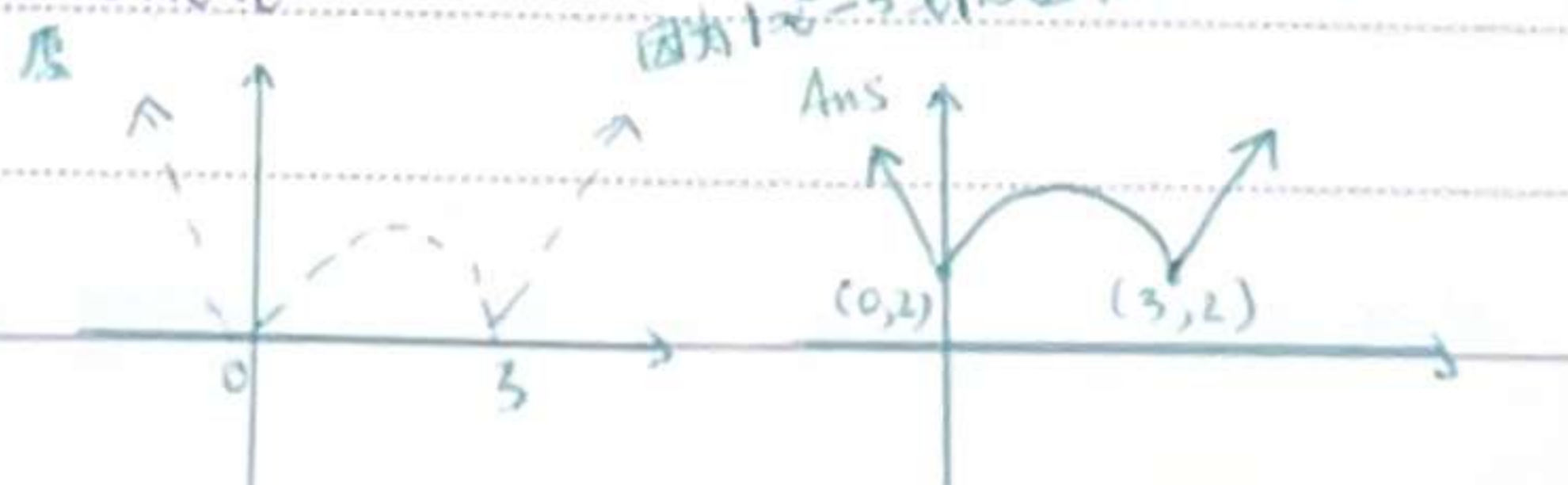
$$-1 \leq 0, \text{ 用 } x^2 - 3x + 2$$

$$1 + 3 + 2 = 6, \quad f(-1) = 6$$

$$2 \in (0, 3) \text{ 用 } -x^2 + 3x + 2$$

$$-4 + 6 + 2 = 4, \quad f(2) = 4$$

(c) Sketch



Basic Differentiation

Chain Rule: $y = a (f(x))^n$

$$\frac{dy}{dx} = a n (f(x))^{n-1} \times f'(x)$$

"e" Euler "oiler" number

spelling
sound

$$e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$$

Differentiate "e": $y = a e^{f(x)}$

$$\frac{dy}{dx} = a e^{f(x)} \times f'(x)$$

$$e^0 = 1$$

Example Question:

① Find equa of tangent y with

$$y = e^{2x} \text{ at } x = 0$$

$$f(0) = (0, 1) \leftarrow \begin{matrix} f(x) \text{ \& tangent} \\ \text{common point} \end{matrix}$$

$$f'(x) = 2e^{2x}$$

$$f'(0) = 2, \text{ Gradient} = 2$$

$$y = 2x + c$$

$$(0, 1) \quad 1 = 0 + c, c = 1$$

$$y = 2x + 1$$

② Differentiate $5e^{-4x} + 2e$

$$f'(x) = -20e^{-4x} + 2 \times 0 e^{1-0}$$

$$= -20e^{-4x}$$

Differentiation : Trigonometric Function

$$\frac{d}{dx} (\sin(f(x))) = \cos(f(x)) \cdot f'(x) \quad \text{in RADIAN}$$

$$\frac{d}{dx} (\cos(f(x))) = -\sin(f(x)) \cdot f'(x)$$

$$\frac{d}{dx} (\tan(f(x))) = \sec^2(f(x)) \cdot f'(x)$$

$$\hookrightarrow \sec(x) = \frac{1}{\cos(x)}$$

例: ① $\frac{d}{dx} (\sin(6x^\circ))$

$$\sin(6x^\circ) = \sin(6x \times \frac{\pi}{180})$$

$$= \sin(\frac{\pi}{30}x)$$

$$\frac{d}{dx} (\sin 6x^\circ) = \frac{d}{dx} (\sin(\frac{\pi}{30}x))$$

$$= \cos(\frac{\pi}{30}x) \cdot \frac{\pi}{30}$$

$$= \frac{\pi}{30} \cos(\frac{\pi}{30}x)$$

② Find equation of tangent to

$$y = \tan(2x) \text{ at } x = -\frac{\pi}{8}$$

$$f'(x) = 2 \sec^2(2x)$$

$$f'(-\frac{\pi}{8}) = \sec^2(-\frac{\pi}{4}) \times 2$$

$$\sec^2(-\frac{\pi}{4}) = \frac{1}{\cos^2(-\frac{\pi}{4})} \quad \begin{matrix} S \\ + \\ A \\ - \\ C \end{matrix} \quad \frac{\frac{\sqrt{2}}{2} \times \frac{\sqrt{2}}{2}}{1} = \frac{1}{\frac{1}{2}}$$

$$= 1 \div \frac{1}{2} = 2$$

$$f'(-\frac{\pi}{8}) = 2 \times 2$$

$$= 4 \quad y = 4x + c$$

$$f(-\frac{\pi}{8}) = -1 \quad (-\frac{\pi}{8}, -1)$$

$$-1 = 4 \times -\frac{\pi}{8} + c$$

$$c = -1 + \frac{\pi}{2}$$

$$y = 4x - 1 + \frac{\pi}{2}$$

Differentiation : Natural Logarithm

自然对数 (以e为底的对数)

- if $y = \log_e(ax+b)$ for $x > -\frac{b}{a}$, then $\frac{dy}{dx} = \frac{a}{ax+b}$
- if $h(x) = \log_e(f(x))$, then the chain rule gives $h'(x) = \frac{f'(x)}{f(x)}$

CAS : Alg Decimal Real Rad

- Main / keyboard / Math $\frac{1}{\square}$ / ~~第2行第3个~~ ln 第2行3、4个 ln, log $\square$
第2行第3个 log
- keyboard / abc / Math / 最右列 e

Example Question

① For $2 \log_e(x-2) = y$

Find value of a, given (a, 0)

$$0 = 2 \log_e(x-2)$$

$$\log_e(x-2) = 0$$

come from $e^0 = x-2$

$$* e^0 = 1$$

$$1 = x-2$$

$$x = 3, a = 3$$

6A

Chain, Product and Quotient rule

Chain rule: $y = g(x)^n$
 $\frac{dy}{dx} = n \cdot g(x)^{n-1} \cdot g'(x)$

Product rule: if $y = u \times v$
 then $\frac{dy}{dx} = u \times v' + v \times u'$

Quotient rule: if $y = \frac{u}{v}$
 then $\frac{dy}{dx} = \frac{v \cdot u' - u \cdot v'}{v^2}$

Example:

① $x^5 \sin(x)$

$u = x^5, v = \sin(x)$

$u' = 5x^4, v' = \cos(x)$

$\frac{dy}{dx} = 5x^4 \sin(x) + x^5 \cos(x)$

② $\frac{dy}{dx}, \sec^2(x)$

Chain rule: $\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$

$\frac{dy}{dx} = n g(x)^{n-1} \cdot g'(x)$

$= 2 \sec(x) \times \frac{d}{dx} \sec(x)$

To find $\frac{d}{dx}$ of $\sec(x)$:

$\frac{d}{dx} \left(\frac{1}{\cos(x)} \right)$

Quotient Rule:

$u = 1, v = \cos(x)$

$u' = 0, v' = -\sin(x)$

$\frac{0 - (-\sin(x))}{\cos^2(x)} = \frac{\sin(x)}{\cos^2(x)}$

Now, combine:

$\frac{dy}{dx} = 2 \sec(x) \times \frac{\sin(x)}{\cos^2(x)}$

$= 2 \tan(x) \sec^2(x)$

6 I

Implicit Differentiation

Implicit differentiation uses the Chain and Product rule

$$y = f(x)^n$$

$$\text{Derivative} = n f(x)^{n-1} \cdot f'(x)$$

For y^2 , $f(x) = y$

$$\text{Derivative} = 2y \cdot \frac{dy}{dx}$$

Example Question:

① Find $\frac{dy}{dx}$

(a) $x^3 + y^3 = 1$

$$\frac{d}{dx} [x^3 + y^3] = \frac{d}{dx} [1]$$

$$3x^2 + 3y^2 \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-3x^2}{3y^2}$$

$$= -\frac{x^2}{y^2}$$

(c) $\sin(y) = \cos^2(x)$

$$\frac{d}{dx} [\sin(y)] = \frac{d}{dx} [\cos^2(x)]$$

$$\cos(y) \cdot \frac{dy}{dx} = 2 \cos(x) \cdot (-\sin(x))$$

$$\frac{dy}{dx} = \frac{-2 \cos(x) \sin(x)}{\cos(y)}$$

$$= \frac{-\sin(2x)}{\cos(y)}$$

(b) $y = x^2 e^y$

$$\frac{d}{dx} [y] = \frac{d}{dx} [x^2 e^y]$$

$$\frac{dy}{dx} = \frac{d}{dx} [x^2 e^y]$$

Product rule: $u = x^2$ $v = e^y$

$$u' = 2x \quad v' = e^y \cdot \frac{dy}{dx}$$

$$\frac{d}{dx} = 2x e^y + x^2 e^y \cdot \frac{dy}{dx}$$

$$\frac{dy}{dx} = 2x e^y + x^2 e^y \cdot \frac{dy}{dx}$$

$$\frac{dy}{dx} (1 - x^2 e^y) = 2x e^y$$

$$\frac{dy}{dx} = \frac{2x e^y}{1 - x^2 e^y}$$

(d) $2x^2 - 2xy + y^2 = 5$

$$2x^2 - 2xy + y^2 = 5$$

↑
select all

Interactive/Calculation/ImpDiff

$$y' = \frac{2x - y}{x - y}$$

(ii) Find Gradient at (1, 3)

$$\frac{2x - y}{x - y} \mid x = 1 \mid y = 3$$

↑
copy from last line
Math3 最右列

Basic Integration

Basic Antiderivatives

	$f(x)$	$\int f(x) dx$
①	x^n	$\frac{x^{n+1}}{n+1} + C, n \neq -1$
	$(ax+b)^n$	$\frac{1}{a(n+1)} (ax+b)^{n+1} + C, n \neq -1$

★ Example Question 例题

① Find $f(x)$

(a) $f'(x) = \frac{3x}{2} - 4x^2 + 2x^3$

$$f(x) = \frac{3}{2} \cdot \frac{x^2}{2} - 4 \cdot \frac{x^3}{3} + 2 \cdot \frac{x^4}{4} + C$$

$$= \frac{3}{4} x^2 - \frac{4}{3} x^3 + \frac{1}{2} x^4 + C$$

(b) $f'(x) = (3x-1)^3$

$$f(x) = \frac{1}{3(3+1)} (3x-1)^4 + C$$

$$= \frac{1}{12} (3x-1)^4 + C$$

Antiderivatives of Exponential & Trigonometric functions

	$f(x)$	$\int f(x) dx$
②	$\frac{1}{ax+b}$	$\frac{1}{a} \log_e(ax+b) + C, \text{ for } ax+b > 0$
③	e^{ax+b}	$\frac{1}{a} e^{ax+b} + C$
④	$\sin(ax+b)$	$-\frac{1}{a} \cos(ax+b) + C$
⑤	$\cos(ax+b)$	$\frac{1}{a} \sin(ax+b) + C$
⑥	$\sec^2(ax+b)$	$\frac{1}{a} \tan(ax+b) + C$

★ Example Question 例题

(a) Find $\int 2 \sin(5x) + 3 \cos\left(\frac{x}{3}\right) dx$

$$= 2 \cdot -\frac{1}{5} \cos(5x) + 3 \cdot \frac{1}{1/3} \cdot \sin\left(\frac{x}{3}\right) + C$$

$$= -\frac{2}{5} \cos(5x) + 9 \sin\left(\frac{x}{3}\right) + C$$

Antiderivatives of $f(x) = \frac{1}{x}$

As we know, $y = \log_e(x)$ 其 $\frac{dy}{dx} = \frac{1}{x}$. Thus:

$$\begin{array}{c} f(x) \\ \frac{1}{x} \\ \frac{1}{ax+b} \end{array}$$

$$\int f(x) dx$$

$$\log_e(x) + C, x > 0$$

$$\frac{1}{a} \log_e(ax+b) + C, x > -\frac{b}{a}$$

★ Example Question 例题

(a) Find $\int \frac{3}{4x+7} dx$

$$= 3 \cdot \frac{1}{4} \cdot \log_e(4x+7) + C$$

$$= \frac{3}{4} \log_e(4x+7) + C, x > -\frac{7}{4}$$

$$4x+7 > 0$$

$$x > -\frac{7}{4}$$

70

Integration by Substitution

Integration by substitution / change of variable rule.

$$\int f(u) \cdot \frac{du}{dx} dx = \int f(u) du$$

Example

① Find $\int 2x (x^2+1)^3 dx$

$u = x^2 + 1, \frac{du}{dx} = 2x$

$\int u^3 \frac{du}{dx} dx$

$= \frac{1}{4} u^4 + C$

$= \frac{1}{4} (x^2+1)^4 + C$

② Find $\int \frac{\cos(x)}{\sin^2(x)} dx$

$= \int \cos(x) \cdot \sin^{-2}(x) dx$

 u 选有 index 的: $u = \sin(x)$

$\frac{du}{dx} = \cos(x)$

$\int u^{-2} \frac{du}{dx} dx$

$= \frac{1}{(-2+1)} u^{-1} + C$

$= -\frac{1}{u} + C$

$= -\frac{1}{\sin(x)} + C$

③ Find $\int \frac{3x}{x^2+1} dx$

$\int 3x \cdot (x^2+1)^{-1} dx$

$u = x^2 + 1, \frac{du}{dx} = 2x, \frac{3x}{2x} = \frac{3}{2}$

$\frac{3}{2} \int 2x \cdot (x^2+1)^{-1} dx$

$= \frac{3}{2} \int u^{-1} \cdot \frac{du}{dx} dx$

$= \frac{3}{2} \cdot \log_e(u) + C$

$\therefore * \int u^{-1} = \log_e(u)$

$= \frac{3}{2} \log_e(\cancel{2x} x^2 + 1) + C$

④ Find $\int \frac{2x-1}{(x-1)^2} dx$

$u = x-1, \frac{du}{dx} = 1, x = u+1$

$\int (2(u+1)-1) \cdot u^{-2} \frac{du}{dx} dx$

$= \int (2u+1) \cdot u^{-2} du$

$= \int 2u^{-1} + u^{-2} du$

$= 2 \cdot \log_e(u) + \frac{1}{-1} \cdot u^{-1} + C$

$= 2 \log_e(u) - \frac{1}{u} + C$

$= 2 \log_e(x-1) - \frac{1}{x-1} + C$

 $* \downarrow$ 别忘了把 u 套回去

7E

USE of Trigonometric Identities for Integration

- Antiderivatives involving Trigonometric Identities:

$$\sin^2(ax) = \frac{1}{2}(1 - \cos(2ax))$$

$$\cos^2(ax) = \frac{1}{2}(1 + \cos(2ax))$$

$$\sin(ax) \cos(ax) = \frac{1}{2} \sin(2ax)$$

$$\cos^3(ax) = \cos^2(ax) \cos(ax)$$

$$= (1 - \sin^2(ax)) \cos(ax), \quad u = \sin(ax)$$

$$\sec^2(ax) = 1 + \tan^2(ax)$$

- Integrals of the form $\int \sin^m(x) \cos^n(x) dx$ where m and n are 正数CASE A: the power of $\sin(x)$ is odd. 例: $\sin^3(x)$ if m is odd, write $m = 2k + 1$. Then

$$\sin^{2+1}(x) = (\sin^2(x)) \sin(x)$$

$$= (1 - \cos^2(x)) \sin(x)$$

$$\text{then } u = \cos(x) \quad \frac{du}{dx} = -\sin(x)$$

CASE B: the power of $\cos(x)$ is odd. 例: $\cos^5(x)$ if n is odd, and m is even, write $n = 2k + 1$

$$n = 2 \times 2 + 1, \text{ Then}$$

$$\cos^{2k+1}(x) = (\cos^2(x))^k \cos(x) = (1 - \sin^2(x))^k \cos(x)$$

$$\cos^{4+1}(x) = (\cos^2(x))^2 \cos(x) = (1 - \sin^2(x))^2 \cos(x)$$

$$\text{then } u = \sin(x) \quad \frac{du}{dx} = \cos(x)$$

$$\int (1 - u^2)^2 du$$

- Example Question:

$$\textcircled{1} \int 2 \sin(3x) \cos(3x) dx$$

$$= 2 \int \frac{1}{2} \sin(2 \cdot 3x) dx$$

$$= \int \sin(6x) dx$$

$$= -\frac{1}{6} \cos(6x) + C$$

$$\begin{aligned} \textcircled{2} \int \cos^3(2x) dx \\ &= \int \cos^2(2x) \cos(2x) dx \\ &= \int (1 - \sin^2(2x)) \cos(2x) dx \end{aligned}$$

$$u = \sin(2x) \quad \frac{du}{dx} = ?$$

$$\frac{d}{dx} (\sin(2x))$$

$$* \sin(2x) = 2 \sin(x) \cos(x)$$

$$* \text{Product rule for } \frac{dy}{dx}$$

$$2 \frac{d}{dx} (\sin(x) \cos(x))$$

$$u = \sin(x) \quad v = \cos(x)$$

$$u' = \cos(x) \quad v' = -\sin(x)$$

$$= 2 (\cos^2(x) - \sin^2(x))$$

$$* \cos(2x) = \cos^2(x) - \sin^2(x)$$

$$\rightarrow = 2 \cos(2x)$$

$$\frac{du}{dx} = 2 \cos(2x), \quad \frac{1}{2} = \frac{1}{2}$$

$$= \frac{1}{2} \int (1 - u^2) \frac{du}{dx} dx$$

$$= \frac{1}{2} \left[u - \frac{u^3}{3} \right] + C$$

$$= \frac{1}{2} \sin(2x) - \frac{1}{6} \sin^3(2x) + C$$

$$\textcircled{4} \int \sin^3\left(\frac{\pi}{4}\right)$$

$$\int \sin^2\left(\frac{\pi}{4}\right) \sin\left(\frac{\pi}{4}\right) dx$$

$$\int (1 - \cos^2\left(\frac{\pi}{4}\right)) \sin\left(\frac{\pi}{4}\right) dx$$

$$u = \cos\left(\frac{\pi}{4}\right), \quad \frac{du}{dx} = -\sin\left(\frac{1}{4}x\right) \cdot \frac{1}{4}, \quad \frac{1}{-1/4} = 1 \times -\frac{4}{1} = -4$$

$$-4 \int (1 - u^2) du$$

$$= -4 \left[u - \frac{1}{3} u^3 \right] + C$$

$$= -4 \cos\left(\frac{\pi}{4}\right) + \frac{4}{3} \cos^3\left(\frac{\pi}{4}\right) + C$$

$$\textcircled{3} \int \tan^3(x) dx$$

$$= \int \frac{\sin^3(x)}{\cos^3(x)} dx$$

$$= \int \frac{\sin^2(x) \sin(x)}{\cos^3(x)} dx$$

$$= \int \frac{(1 - \cos^2(x)) \sin(x)}{\cos^3(x)} dx$$

$$u = \cos(x) \quad \frac{du}{dx} = -\sin(x)$$

$$= - \int \frac{1 - u^2}{u^3} du$$

$$= - \int u^{-3} - u^{-1} du$$

$$= - \left[-\frac{1}{2} u^{-2} - \log_e(u) \right] + C$$

$$= \frac{1}{2u^2} + \log_e(u) + C$$

$$= \frac{1}{2 \cos^2(x)} + \log_e(\cos(x)) + C$$

7G

Partial Fraction 部分分數

$$f(x) = \frac{4x+2}{x^2-1}$$

$$\begin{cases} g(x) = 4x+2 \\ h(x) = x^2-1 \end{cases} \text{ in this case}$$

□ if the degree of $g(x)$ is less than the degree of $h(x)$, the $f(x)$ is a proper fraction 真分數

□ if the degree of $g(x)$ is greater or equal to the degree of $h(x)$, then $f(x)$ is a improper fraction 假分數

Proper Fractions

①. For every linear factor $ax+b$ in the $h(x)$ 分母, there will be a partial fraction of the form $\frac{A}{ax+b}$.

②. For every repeated linear factor $(cx+d)^2$ in the denominator, there will be partial partial fraction of the form $\frac{B}{cx+d}$ and $\frac{C}{(cx+d)^2}$.

③. For every irreducible quadratic factor ax^2+bx+c in the denominator, it will have a partial fraction of $\frac{Dx+E}{ax^2+bx+c}$.

Example Questions

① Express $\frac{5x+1}{(x-1)(x+2)}$ in partial

② $\frac{2x+3}{(x-3)^2}$ in PFraction

Fraction. 用 ①

$$\frac{5x+1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$$

★ 不可省略的一步: $5x+1 = A(x+2) + B(x-1)$

when $x=1$, $6 = 3A + 0$, $A=2$

when $x=-2$, $-9 = -3B$, $B=3$

→ Hence, $\frac{5x+1}{(x-1)(x+2)} = \frac{2}{x-1} + \frac{3}{x+2}$

用 ②

$$\frac{2x+3}{(x-3)^2} = \frac{A}{x-3} + \frac{B}{(x-3)^2}$$

$2x+3 = A(x-3) + B$

when $x=3$, $B=9$

when $x=0$, $3 = -3A + 9$, $A=2$

Hence, $\frac{2x+3}{(x-3)^2} = \frac{2}{x-3} + \frac{9}{(x-3)^2}$

$$\textcircled{3} \quad \frac{3x^2 + 2x + 5}{(x^2 + 2)(x + 1)}$$

遇到不可 factorise 的分母 factor 时用

用 (1) 和 (2)

$$= \frac{A}{(x+1)} + \frac{Bx+C}{(x^2+2)} \leftarrow Bx+C \text{ 必须放在不可 factorise 的 factor 上}$$

$$3x^2 + 2x + 5 = A(x^2 + 2) + (Bx + C)(x + 1)$$

$$x = -1, \quad 6 = 3A, \quad A = 2$$

$$\text{套 A 进: } 3x^2 + 2x + 5 = 2x^2 + 4 + Bx^2 + Bx + Cx + C$$

$$= x^2(2+B) + x(B+C) + C + 4$$

Equate coefficients 相等 so:

$$3 = 2 + B, \quad B = 1$$

$$2 = B + C, \quad C = 1$$

$$\text{Hence, } \frac{3x^2 + 2x + 5}{(x^2 + 2)(x + 1)} = \frac{2}{(x+1)} + \frac{x+1}{(x^2+2)}$$

$$\textcircled{4} \quad \frac{3x^2 - 4x - 2}{(x-1)(x-2)}$$

$$= \frac{3x^2 - 4x - 2}{x^2 - 3x + 2}$$

since 分子与分母有相同 power, this is an improper fraction 假分数

$$\begin{array}{r} 3 \\ x^2 - 3x + 2 \overline{) 3x^2 - 4x - 2} \\ \underline{-(3x^2 - 9x + 6)} \\ 5x - 8 \end{array}$$

$$\text{Hence, } \frac{3x^2 - 4x - 2}{(x-1)(x-2)} = 3 + \frac{5x-8}{x^2-3x+2} \quad \text{真分数}$$

$$\begin{aligned} \frac{5x-8}{x^2-3x+2} &= \frac{5x-8}{(x-1)(x+2)} \\ &= \frac{A}{(x-1)} + \frac{B}{(x+2)} \end{aligned}$$

$$5x - 8 = A(x + 2) + B(x - 1)$$

$$\text{when } x = 1, \quad -3 = -A, \quad A = 3$$

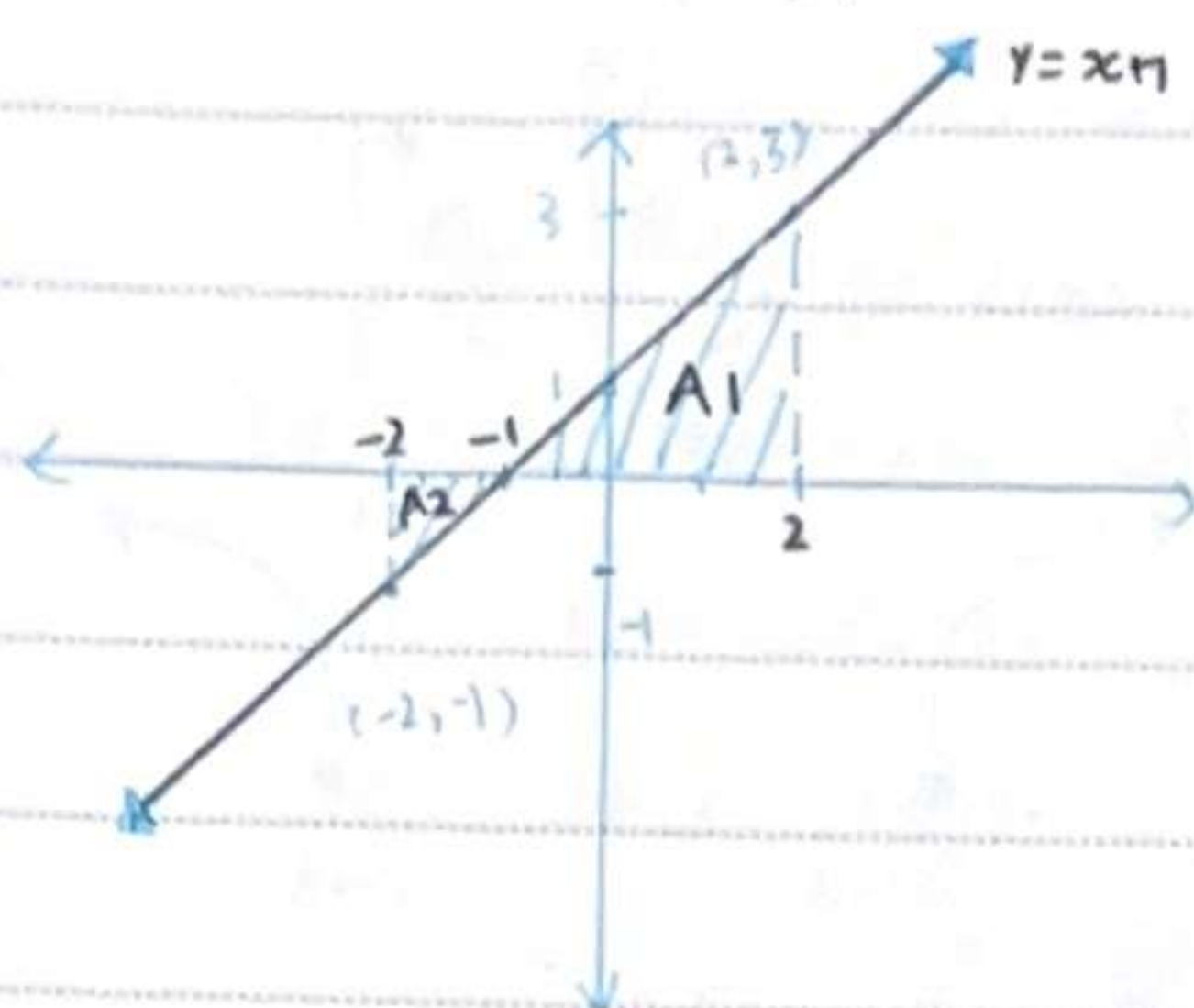
$$\text{when } x = 2, \quad 2 = 0 + B, \quad B = 2$$

$$\text{Hence, } \frac{3x^2 - 4x - 2}{(x-1)(x-2)} = 3 + \frac{3}{(x-1)} + \frac{2}{(x-2)}$$

~~Mouse Score 3~~

8A

Application of Integration - Area



符号区域

Signed area is $A_1 - A_2$ Total area is $A_1 + A_2$

$$A_2 = \int_{-2}^{-1} x + 1 \, dx$$

$$= \frac{1}{2} \text{ square unit}$$

$$A_1 = \int_{-1}^2 x + 1 \, dx$$

$$= 4\frac{1}{2} \text{ square units}$$

* Regions above x-axis have positive signed area

Regions below x-axis have negative signed area

$\int_a^b f(x) \, dx$ is called the definite integral of $f(x)$ from $x = a$ to $x = b$.

⊙ if $f(x) \leq 0$ for all $x \in [a, b]$ the area between $x \in [a, b]$ is: $-\int_a^b f(x) \, dx$

⊙ Properties of the definite integral

$$\int_a^c f(x) \, dx = \int_a^b f(x) \, dx + \int_b^c f(x) \, dx$$

$$\int_a^a f(x) \, dx = 0$$

$$\int_a^b k f(x) \, dx = k \int_a^b f(x) \, dx$$

$$\int_a^b f(x) \, dx = -\int_b^a f(x) \, dx$$

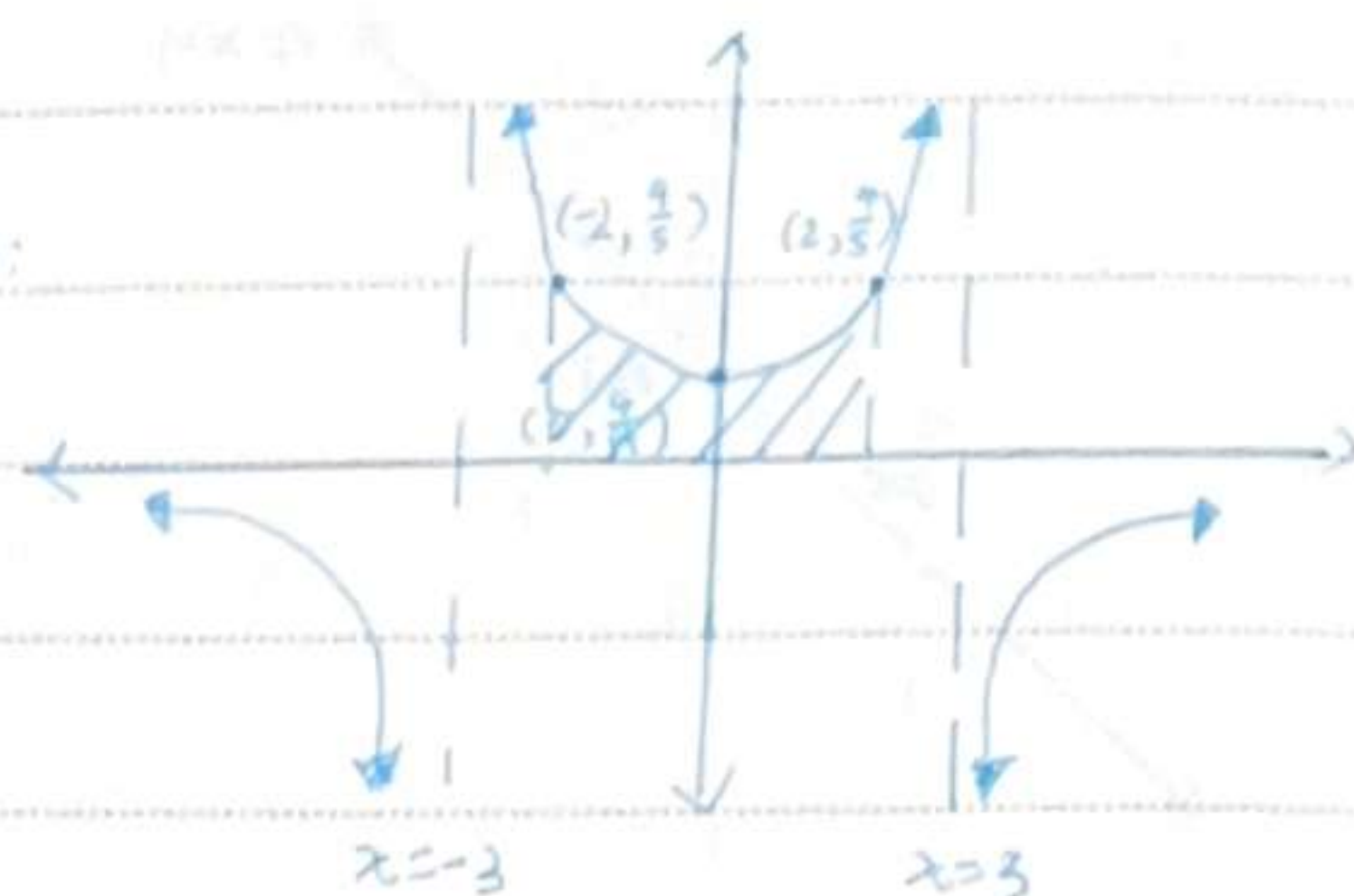
Example Question: 前方高能

① Sketch $g(x) = \frac{4}{9-x^2}$ and find the area of region with $-2 \leq x \leq 2$ Sketch: $9-x^2 = 0$ $x = \pm 3$ ← Vertical asymptote:

$x = -2, y = \frac{4}{5}$

$x = 2, y = \frac{4}{5}$

$x = 0, y = \frac{4}{9}$

Find area: ① Simplify $g(x)$:

$$\frac{4}{9-x^2} = \frac{4}{(3+x)(3-x)} = \frac{a}{3+x} + \frac{b}{3-x}$$

$$4 = a(3-x) + b(3+x)$$

$$x=3, 4 = 0 + 6b, b = \frac{2}{3}$$

$$x=-3, 4 = 6a, a = \frac{2}{3}$$

$$\text{Thus } \frac{4}{9-x^2} = \frac{2}{3(3+x)} + \frac{2}{3(3-x)}$$

$$\frac{2}{9+3x} + \frac{2}{9-3x}$$

② Anti-differentiate $g(x)$

$$\int_{-2}^2 \frac{2}{3} \cdot (x+3)^{-1} - \frac{2}{3} \cdot (x-3)^{-1}$$

这里中间变成减号是因为(3-x)变成(x-3)了

$$= \left[\frac{2}{3} \cdot (\log_e(x+3)) - \frac{2}{3} \cdot (\log_e(x-3)) \right]_{-2}^2$$

$$= \left[\frac{2}{3} \log_e(5) - \frac{2}{3} \log_e(-1) \right] - \left[\frac{2}{3} \log_e(1) - \frac{2}{3} \log_e(-5) \right]$$

$$= \frac{2}{3} \log_e(5) - \frac{2}{3} \log_e(1) - \frac{2}{3} \log_e(-1) + \frac{2}{3} \log_e(-5)$$

$$= \frac{2}{3} \log_e(5) - \frac{2}{3} \log_e\left(\frac{-1}{-5}\right)$$

$$= \frac{2}{3} \log_e\left(\frac{5}{1/5}\right)$$

$$= \frac{2}{3} \log_e(25)$$

$$= \frac{2}{3} \log_e(5^2)$$

$$= 2 \cdot \frac{2}{3} \cdot \log_e(5) = \frac{4}{3} \log_e(5) \text{ square units}$$

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② Find $\int_0^{\frac{\pi}{2}} \sin^3(2x) dx$

① 给 $\sin^3(2x)$ 变形:

$$\begin{aligned} & \int \sin^2(2x) \sin(2x) dx \\ &= \int (1 - \cos^2(2x)) \sin(2x) dx \end{aligned}$$

② Antidifferentiate $\sin^3(2x)$

$u = \cos^2(2x) \quad \frac{du}{dx} = -2 \sin(2x)$ ← 这一步是 differentiate u 哦!
 不是 differentiate anti

$$-\frac{1}{2} \int (1 - u^2) \cdot -2 \sin(2x) dx$$

$$-\frac{1}{2} \int (1 - u^2) \frac{du}{dx} dx$$

$$= -\frac{1}{2} \left[u - \frac{1}{3} u^3 \right]$$

$$= -\frac{1}{2} \cos(2x) + \frac{1}{6} \cos^3(2x)$$

③ 套进去找 Area

$$\left[-\frac{1}{2} \cos(2x) + \frac{1}{6} \cos^3(2x) \right]_0^{\frac{\pi}{2}} \quad \cos(\pi) = -1 \quad \cos(0) = 1$$

$$= \left[-\frac{1}{2} (-1) + \frac{1}{6} (-1) \right] - \left[-\frac{1}{2} + \frac{1}{6} \right]$$

$$= \frac{1}{2} - \frac{1}{6} + \frac{1}{2} - \frac{1}{6}$$

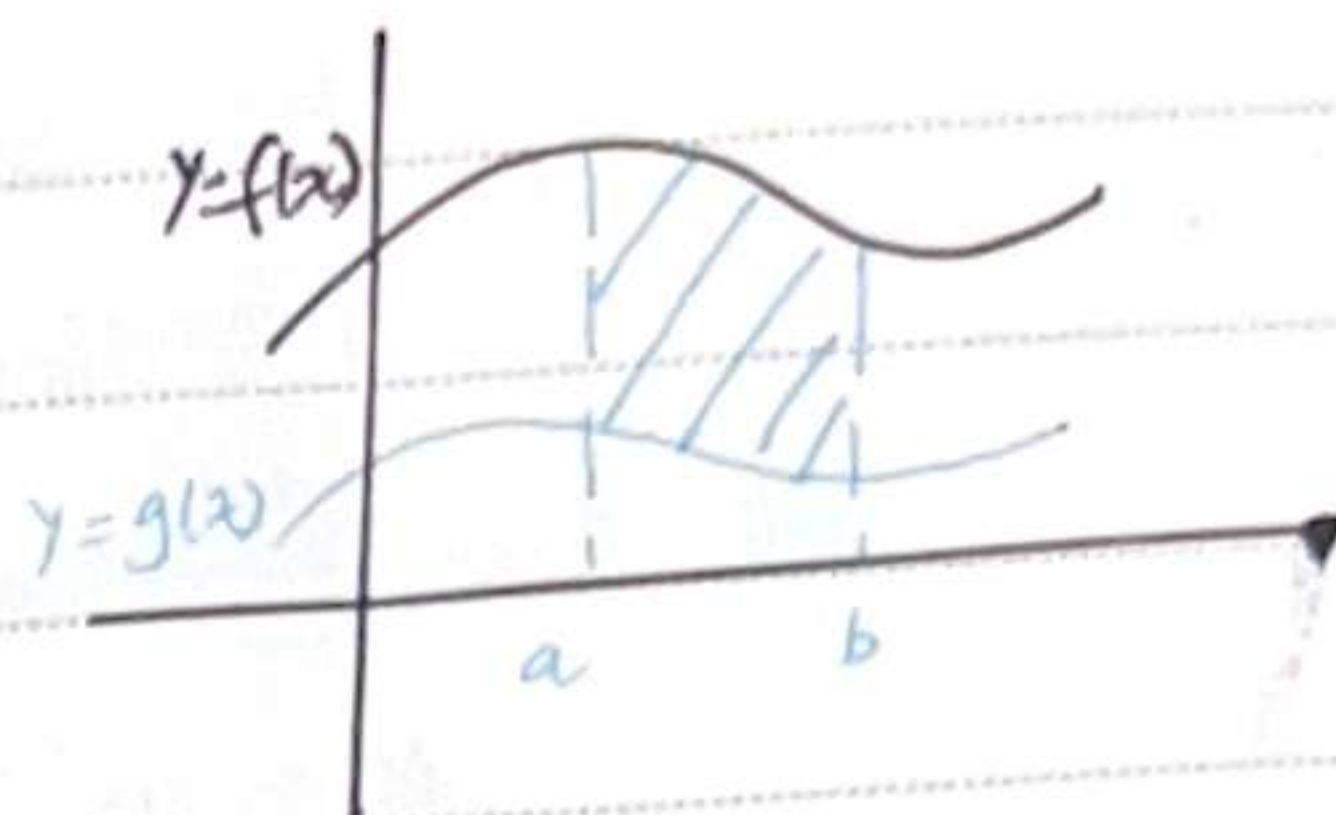
$$= \frac{2}{3} \text{ square units}$$

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Application of Integration - Area between two curves

前提: $f(x) \geq g(x)$ for all $x \in [a, b]$

- The area bounded by the two curves and the lines $x=a$ and $x=b$ can be found by:



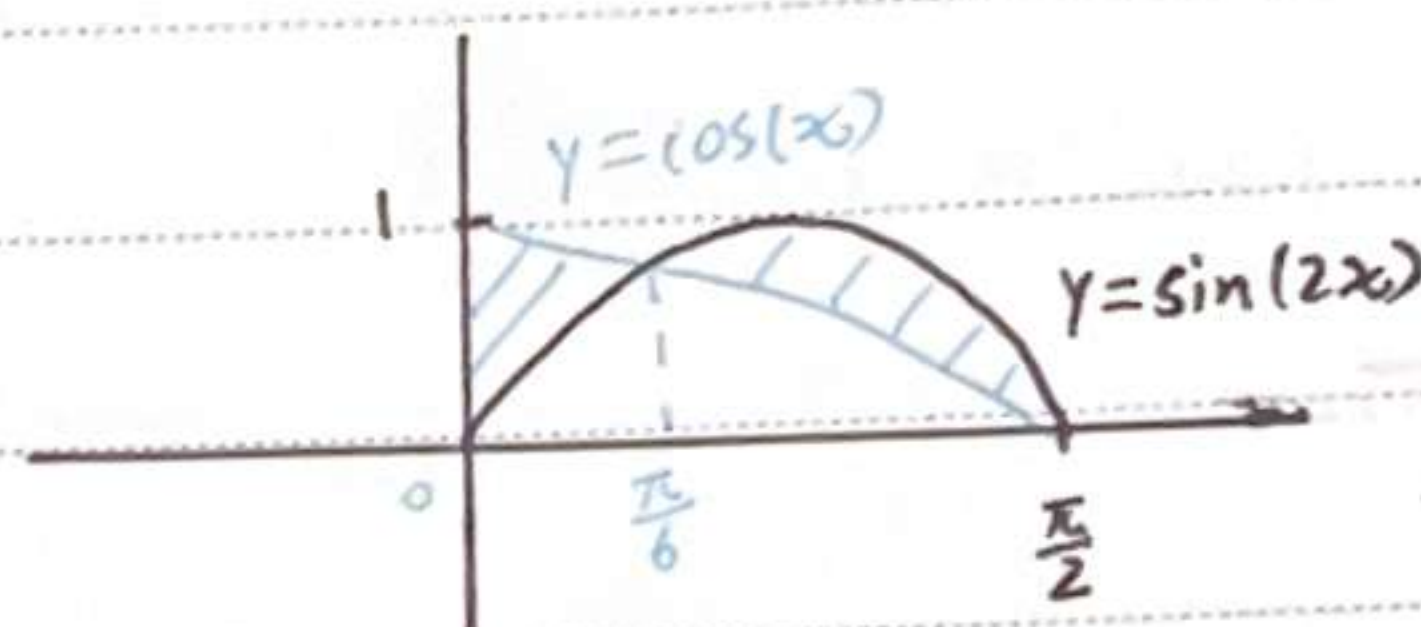
$$\int_a^b f(x) dx - \int_a^b g(x) dx$$

$$= \int_a^b f(x) - g(x) dx$$

如果有 cross 的 area, 也就是 $f(x)$ 不总是等于或大于 $g(x)$ 时 如 
 则需分别测量并将测量好的部分 area 加起来。

Example Question

① Find shaded area.



① 找 intersection

$$\cos(x) = \sin(2x)$$

$$\cos(x) = 2 \sin(x) \cos(x)$$

$$0 = \cos(x) \cdot (2 \sin(x) - 1)$$

$$\cos(x) = 0 \text{ or } \sin(x) = \frac{1}{2}$$

$$x = \frac{\pi}{2} \text{ or } \frac{\pi}{6} \text{ for } x \in [0, \frac{\pi}{2}]$$

② 找 Area

$$\text{Area} = \int_0^{\frac{\pi}{6}} \cos(x) - \sin(2x) dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \sin(2x) - \cos(x) dx$$

$$= \left[\sin(x) + \frac{1}{2} \cos(2x) \right]_0^{\frac{\pi}{6}} + \left[-\frac{1}{2} \cos(2x) - \sin(x) \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= \frac{1}{4} - \frac{1}{2} + \frac{1}{4} + \frac{1}{2}$$

$$= \frac{1}{2} \text{ square units}$$

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② Sketch graph and find area enclosed by:

$$y = \sqrt{x}, \quad y = 6 - x \quad \text{and} \quad y = 1$$

$$y = \sqrt{x}$$

$$y = -x + 6, \quad \text{Y-int } (0, 6)$$

$$x\text{-int } (6, 0)$$



① Find intersection

$$\sqrt{x} = 6 - x$$

$$x = (6 - x)^2$$

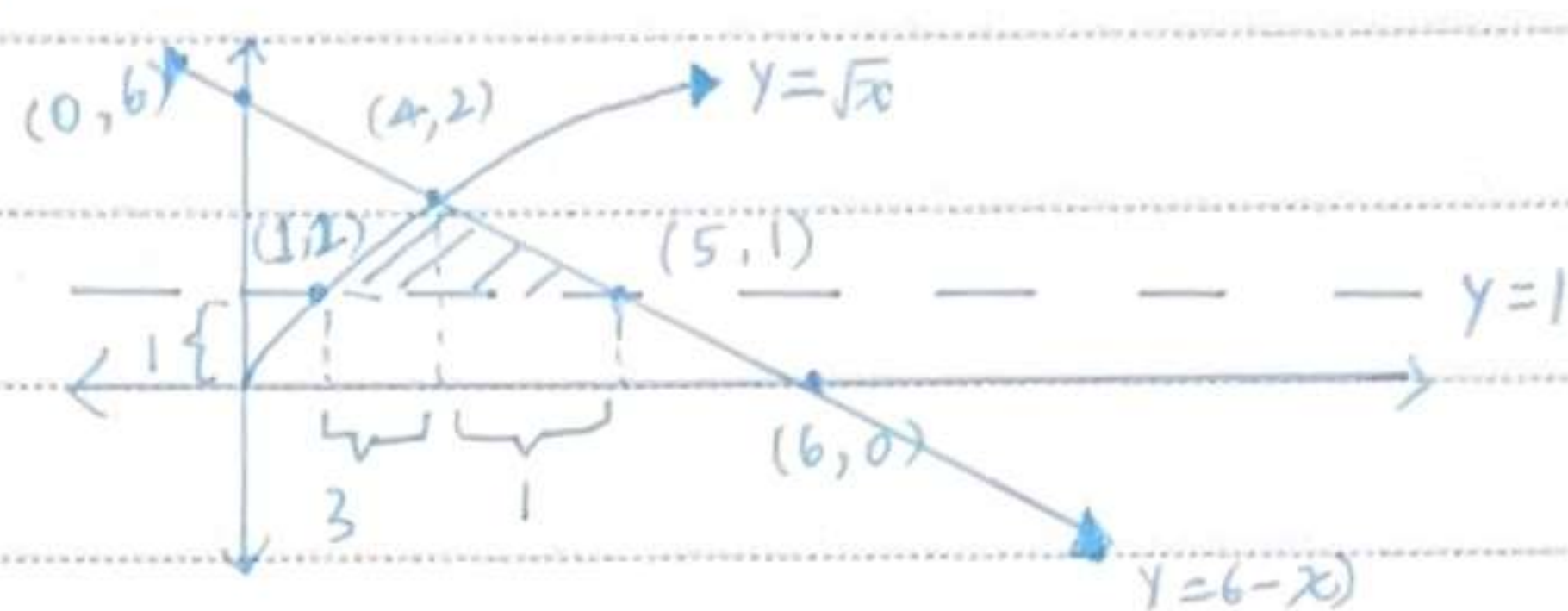
$$\sqrt{x} = 1, \quad x = 1, \quad (1, 1)$$

$$6 - x = 1, \quad x = 5, \quad (5, 1)$$

$$x^2 - 13x + 36 = 0$$

$$(x - 4)(x - 9) = 0 \quad \text{intersect at } x = 4, 9 \quad (4, 2) \quad (9, -3) \quad \text{ignore}$$

② Sketch



③ Find Area

$$\text{求左半边: } \int_1^4 \sqrt{x} \, dx - 3 \times 1$$

$$\text{求右半边: } \int_4^5 (6 - x) \, dx - 1 \times 1$$

$$\text{再将左右加起来: 用 CAS 求得 } \frac{13}{6} \text{ square units.}$$

Measurement

圆面积 $S = \pi r^2$

周长 $C = \pi d$

Area of a trapezium $\frac{1}{2}(a+b)h$

curved surface area of a cylinder $\frac{1}{2}\pi rh$ 

volume of a cylinder $\pi r^2 h$ 体积

volume of a cone $\frac{1}{3}\pi r^2 h$

Δ volume of a pyramid $\frac{1}{3} \times \text{base area} \times \text{height}$

$\oplus$ Area of a sphere $\frac{4}{3}\pi r^3$

Area of a triangle $\frac{1}{2}bc \sin(A)$

Sine Rule $\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$

Consine rule $a^2 = b^2 + c^2 - 2bc \cos(A)$

Algebra (complex numbers)

$Z = x + iy = r(\cos(\theta) + i \sin(\theta)) = \underline{r \text{ cis } (\theta)}$

$|Z| = \sqrt{x^2 + y^2} = r$

$Z_1 Z_2 = r_1 r_2 \text{ cis } (\theta_1 + \theta_2)$

$\frac{Z_1}{Z_2} = \frac{r_1}{r_2} \text{ cis } (\theta_1 - \theta_2)$

$Z^n = r^n \text{ cis } (n\theta)$ de Moivre's theorem

$x = r \cos \theta$

$y = r \sin \theta$

$Z = r \cos \theta + r \sin \theta i$

Probability and statistic

distribution of sample mean $\bar{X}$

Complex 用 rad mode

Tri 用 deg mode.

正式书信

尊敬的领导：

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□□此致

敬礼

学生：何颖仪

2019年5月31日

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{Midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

		30°	45°	60°	90°	180°
	①	6	4	3	2	π
S	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0
C	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
T	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	∞	0

$$\text{Factorise : } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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